Spring 2026: Topics in Algebra

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1 Cohomology of Varieties in Mixed Characteristics

1.1 De Rham Cohomology

Let X be a smooth variety over field F. Consider the complex (of sheaves of F-vector spaces):

$$\mathcal{O}_X \xrightarrow{d} \Omega^1_{X/F} \xrightarrow{d} \Omega^2_{X/F} \xrightarrow{d} \cdots$$

explicitly, $d(fdg) = df \wedge dg$, etc. We define: $H^n_{\mathrm{dR}}(X/F)$ to be the hypercohomology of this complex.

Example. *If X is affine,*

$$H^n_{\mathrm{dR}} = H^n(\mathcal{O}(X) \to \Omega^1(X) \to \cdots)$$

In the case $X/\mathbb{C}$ smooth variety.

Theorem 1.1 (Grothendieck)**.** $H^n_{\mathrm{dR}}(X/\mathbb{C}) \simeq H^n(X(\mathbb{C}), \mathbb{C})$

Example. *If X is affine, we have:*

$$\mathcal{O}_X(X) \xrightarrow{d} \Omega^1(X) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^{\dim X}(X)$$

Tensor with $\mathcal{O}_X^{\mathrm{an}}$, I'm getting

$$\mathcal{O}_X^{\mathrm{an}}(X(\mathbb{C})) \xrightarrow{d} \Omega^{1,\mathrm{an}}(X(\mathbb{C})) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^{\dim X,\mathrm{an}}(X(\mathbb{C}))$$

This is a quasi-isomorphism. One can even further extend the scalar to smooth differential forms, and it remains to be a quasi-isomorphism.

Example. *Let $X = \mathbb{G}_m/\mathbb{C}$. So the de Rham complex is given by:*

$$\mathbb{C}[t, t^{-1}] \xrightarrow{d} \mathbb{C}[t, t^{-1}]dt$$

One sees:

$$H^i_{\mathrm{dR}}(X) = \begin{cases} \mathbb{C} \cdot 1, & \text{if } i = 0; \\ \mathbb{C}[\dfrac{dt}{t}], & \text{if } i = 1; \end{cases}$$

Remark. *Grothendieck's theorem implies that for an affine variety X, we have a special case for Artin vanishing theorem:*

$$H^n(X(\mathbb{C}), \mathbb{C}) = 0, \quad n > \dim_{\mathbb{C}} X$$

Remark. *$H^n_{\mathrm{dR}}(X/\mathbb{C})$ is finite dimensional for each n.*

Example. *Let $X/\mathbb{Q}$ be smooth.*

$$H^n_{\mathrm{dR}}(X/\mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C} \simeq H^n_{\mathrm{dR}}(X_{\mathbb{C}}, \mathbb{C}) \simeq H^n_{\mathrm{sing}}(X(\mathbb{C}), \mathbb{C}) \simeq H^n_{\mathrm{sing}}(X(\mathbb{C}), \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{C}$$

Let's turn to positive characteristic. Pick p prime, k perfect field of characteristic p, X/k smooth variety. A fact for perfect field is that:

$$\Omega^1_{X/k} \simeq \Omega^1_{X/\mathbb{F}_p}$$

Example. *Let $X = \mathbb{A}^1_k$. I have de Rham complex:*

$$k[t] \xrightarrow{d} k[t]dt$$

Note that in this case:

$$H^0_{\mathrm{dR}}(\mathbb{A}^1) = \ker d = \bigoplus_{i \geq 0} k \cdot t^{pi}$$

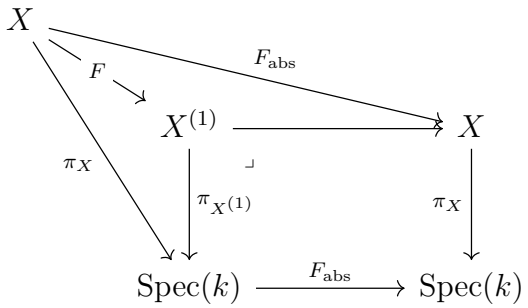
$$H^1_{\mathrm{dR}}(\mathbb{A}^1) = \operatorname{coker} d = \bigoplus_{i \geq 0} k \cdot [t^{pi+p-1}dt]$$

Look at the complex:

$$\mathcal{O}_X \xrightarrow{d} \Omega^1(X) \xrightarrow{d} \cdots$$

Consider the subsheaf $\mathcal{O}^p_X = \langle f^p \mid f \in \mathcal{O}_X \rangle \subset \mathcal{O}_X$. A basic observation is that this complex is $\mathcal{O}^p_X$-linear.

Let $F_{\mathrm{abs}} : X \to X$ be the absolute Frobenius map, and F be the relative Frobenius.



Now I have a complex of quasi-coherent sheaves on $X^{(1)}$

$$F_*\mathcal{O}_X \xrightarrow{d} F_*\Omega^1_{X/k} \xrightarrow{d} \cdots$$

where: $\mathcal{O}_{X^{(1)}} = (F_*\mathcal{O}_X)^p \hookrightarrow F_*\mathcal{O}_X$. The differentials are $\mathcal{O}_{X^{(1)}}$-linear.

Example. *Let $X = \operatorname{Spec} k[t], X^{(1)} = \operatorname{Spec} k[t^p]$. The complex:*

$$k[t] \xrightarrow{d} k[t]dt$$

is $k[t^p]$-linear, so the de Rham cohomology are $k[t^p]$-modules.

Remark. *The relative Frobenius $F : X \to X^{(1)}$ is finite flat of degree $p^{\dim X}$*

Remark.

$$H^n_{\mathrm{dR}}(X/k) = \mathbb{H}^n_{\mathrm{Zar}}(X^{(1)}, F_*\Omega^\bullet_X : F_*\mathcal{O}_X \to F_*\Omega^1_X \to \cdots)$$

Theorem 1.2 (Cartier isomorphism)**.** *Let X/k, we let $\mathsf{H}^i(F_*\Omega^\bullet_X)$ be the cohomology sheaf, in fact a coherent $\mathcal{O}_{X^{(1)}}$-module. In fact we have:*

$$\mathsf{H}^i(F_*\Omega^\bullet_X) \simeq \Omega^i_{X^{(1)}}$$

Example. *Let $X = \mathbb{A}^1_k$, my complex is again*

$$k[t] \to k[t]dt$$

So that $\mathsf{H}^1 = k[t]dt/d(k[t]) \simeq k[t^p][t^{p-1}dt]$, we wish to identify this with:

$$\Omega^1_{X^{(1)}}(X^{(1)}) = k[s]ds$$

I thus must map

$$[t^{p-1}dt] \mapsto ds$$

1.2 Crystalline Cohomology

From now on, for a variety X/k, we write $R\Gamma_{\mathrm{dR}}(X/k) \in D^b(\mathrm{Vect}_k)$.

Definition 1.3. Let k be a perfect field. The **Witt vector** $W(k)$ be the unique ring such that:

1. it's p-torsion free.
2. its p-adically complete.
3. $W(k)/p \simeq k$

We will write $W_n(k) = W(k)/p^n$. We call X a **p-adic formal scheme** over $W(k)$ if it is a collection:

$$\{X_n/W_n(k)\}_n$$

such that we have compatible structure $X_{n+1} \times_{W_{n+1}(k)} W_n(k) \simeq X_n$. Then the inverse limit of the structure sheaves $\mathcal{O}_{X_n}$ is a sheaf on $|X|$.

Example. $W(\mathbb{F}_p) = \mathbb{Z}_p$.

Example. *If $Y/W(k)$ is a scheme, I can form a formal scheme $\hat{Y}$ with:*

$$(\hat{Y})_n = Y \times_{W(k)} W_n(k)$$

Remark. *If R is a p-torsion free ring,*

$$W(R) \simeq \prod_{n \geq 0} R$$

by putting a ring structure by "ghost coordinates":

$$w_n \coloneqq a_0^p + p a_1^{p^{n-1}} + \cdots p^n a_n$$

And the isomorphism is mapping:

$$(a_0, a_1, a_2, \cdots) \mapsto (w_0, w_1, \cdots)$$

We wish to construct **Crystalline Cohomology Functor**

$$R\Gamma_{\text{cris}} : \{\text{smooth varieties}/k\} \to D(W(k)\text{-mod})$$

such that:

1. $R\Gamma_{\text{cris}}(X/W(k)) \otimes^{\mathbb{L}}_{W(k)} k \simeq R\Gamma_{\text{dR}}(X/k)$
2. If $\tilde{X}$ is a smooth formal scheme over $W(k)$. Note that the base change $\tilde{X}_n/W_n(k)$ is smooth. Now if $\tilde{X}_1 = \tilde{X} \times_W k = X$, then:

   $$R\Gamma_{\text{cris}}(X/W(k)) \simeq R\Gamma_{\text{dR}}(\tilde{X}/W(k)) \coloneqq R\Gamma_{\text{Zar}}(X, \hat{\mathcal{O}}_{\tilde{X}} \xrightarrow{d} \hat{\Omega}^1_{\tilde{X}} \to \cdots)$$

   where $\hat{\Omega}^i_{\tilde{X}} = \varprojlim \Omega^i_{X_n}$
3. If $X_n/W_n(k)$ is smooth, such that $X_n \times_{W_n(k)} k \simeq X$, then:

   $$R\Gamma_{\text{cris}}(X/W(k)) \otimes^{\mathbb{L}}_{W(k)} W_n(k) \simeq R\Gamma_{\text{dR}}(X_n/W_n(k))$$

   We will denote LHS as $R\Gamma_{\text{cris}}(X/W_n(k))$.

Example. *If $\tilde{X}$ is affine, so that $(\tilde{X})_n = \text{Spec } A_n$, so that:*

$$R\Gamma_{\text{dR}}(\tilde{X}/W(k)) = \mathcal{O}(\tilde{X}) \xrightarrow{d} \Omega^1(\tilde{X}) \xrightarrow{d} \cdots$$

Example. *Let $X = \mathbb{A}^1_k$, $\tilde{X} = \hat{\mathbb{A}}^1_{W(k)}$. We can explicitly calulate the above complex:*

$$\mathcal{O}(\tilde{X}) = W\langle t \rangle = \{\sum a_n t^n \mid v_p(a_n) \to \infty, n \to \infty\} \xrightarrow{d} W\langle t \rangle dt \to \cdots$$

So that we can compute:

$$H^0_{\text{dR}}(\tilde{\mathbb{A}}^1_{W(k)}) = \ker d = W \cdot 1$$

Remark. *If $Y/W(k)$ is a smooth scheme, then:*

$$R\Gamma_{\text{dR}}(\hat{Y}/W(k)) = \text{derived } p\text{-adic completion of } R\Gamma_{\text{dR}}(Y/W(k))$$

For the later, we just p-adically complete the complex termwise. In particular, if Y is smooth proper, then $R\Gamma_{\text{dR}}(Y/W(k))$ is a "perfect complex", so that the completion is doing nothing, and we have:

$$R\Gamma_{\text{dR}}(\hat{Y}/W(k)) = R\Gamma_{\text{dR}}(Y/W(k))$$

Remark. *Consider a more classical setting $\mathbb{C}[[t]] \to \mathbb{C}$. Let $\tilde{X}$ be a smooth t-adic formal scheme over $\mathbb{C}[[t]]$, then we have a classical result:*

$$R\Gamma_{\mathrm{dR}}(\tilde{X}/\mathbb{C}[[t]]) \simeq R\Gamma_{\mathrm{dR}}(\tilde{X}_{t=0}/\mathbb{C}) \otimes_{\mathbb{C}} \mathbb{C}[[t]]$$

One ingredient for proving this is $\mathbb{C}[[t]] \to \mathbb{C}$ admits a splitting, but this is not available in our case.

Corollary 1.4. *Let $\tilde{X}/W(k)$ be smooth formal scheme, and f and automorphism on $\tilde{X}$ such that $f = \mathrm{id}$ modulo p, then $f^* = \mathrm{id}$ on $H^n_{\mathrm{dR}}(\tilde{X}/W(k))$.*

Proof. (Without assuming the existence of crystalline cohomology). Let f acts on $\mathcal{O}_{\tilde{X}} = \varprojlim \mathcal{O}_{\tilde{X}_n}$. Let's consider:

$$\log f \coloneqq \sum_{n\geq 1} (-1)^{n+1} \frac{(f-\mathrm{id})^{\circ n}}{n}$$

This makes sense since $n \mid p^n$ in $\mathbb{Z}_p$ and $v_p(p^n/n) \to \infty$ as $n \to \infty$. One may check $\log f$ is a derivation of $\mathcal{O}_{\tilde{X}}$.

Let's assume $p > 2$, then we have $f = \exp(\log f)$, this is because in this case $p \mid \log f$. We write $\log f = p \cdot D$, so $f = \sum_{n\geq 0} \frac{p^n}{n!} D^{\circ n}$. Let's view the derivation D as a vector field in $T_{\tilde{X}/W}$, then f^* acts on $\Omega^i_{\tilde{X}}$ as $\sum_{n\geq 0} \frac{p^n}{n!} L_D^{\circ n}$, where L_D is the Lie derivative (in algebraic geometry sense). So that we can choose $f^* = \mathrm{id} + L_D \circ G$ for some operator G. Now comes the main lemma:

Lemma. *For any derivation D, L_D on $\Omega^\bullet_{\tilde{X}}$ is homotopic to 0.*

Proof. This is a direct computation:

$$L_D = d\iota_D + \iota_D d$$

for ι_D the contraction with vector field D. □

Now the result follows. □

Proof. (of Cartier isomorphism) For $i = 0$, note we have an obvious map:

$$\mathcal{O}_{X^{(1)}} \to \ker(F_*\mathcal{O}_X \xrightarrow{d} F_*\Omega^1_X), \quad f \mapsto f^p$$

Locally, at every point I have an open neighborhood étale over $\mathbb{A}^n_k$. So it suffices to check the map is isomorphism for $\mathbb{A}^n_k$ by functoriality, by explicit computation.

In general, let's consider the affine case $X = \mathrm{Spec}\ A$. The goal is to show:

$$\mathsf{H}^i_{\mathrm{dR}}(A/k) \simeq \Omega^i_{A^{(1)}/k}$$

I have a distinguished trangle:

$$R\Gamma_{\mathrm{dR}}(\mathrm{Spec}\ A/k) \to R\Gamma_{\mathrm{cris}}(\mathrm{Spec}\ A/W_2(k)) \to R\Gamma_{\mathrm{dR}}(\mathrm{Spec}\ A/k) \to$$

This gives me long exact sequence for cohomology:

$$\to H^i_{\mathrm{dR}}(\mathrm{Spec}\ \tilde{A}/W_2(k)) \simeq H^i_{\mathrm{cris}}(\mathrm{Spec}\ A/W_2(k)) \to H^i_{\mathrm{dR}}(\mathrm{Spec}\ A/k) \xrightarrow{\beta_i} H^{i+1}_{\mathrm{dR}}(\mathrm{Spec}\ A/k) \to$$

Lemma. *1. $\oplus\beta_i$ on $\bigoplus_{i\geq 0} H^i_{\mathrm{dR}}(A/k)$ is a derivation.*

2. $\beta_{i+1}\circ\beta_i = 0$.

Then it is a formal consequence (using ring structure) that there exists map $\Omega^i_{A^{(1)}} \to H^i_{\mathrm{dR}}(A/k)$. Then we check étale locally this is an isomorphism. □

1.3 $\mathcal{D}$-modules and Crystals

Let's now introduce $\mathcal{D}$-modules to this picture. Let X be a smooth scheme over R, let $\mathcal{D}_{X/R}$ be the sheaf of differential operators on X. Very explicitly:

$$\mathcal{D}_{X/R} = \mathcal{O}_X\{\partial_v \mid v \in T_X\}/(\partial_{fv_1+v_2} = f\partial_{v_1} + \partial_{v_2}, [\partial_v, f] = L_v(f), [\partial_v, \partial_w] = \partial_{[v,w]})$$

Remark. *If $R \supset \mathbb{Q}$, one have an injection:*

$$\mathcal{D}_{X/R} \to \mathrm{End}_R(\mathcal{O}_X)$$

But this is not necessarily true in characteristi p. For example, consider $X = \mathbb{A}^1_{\mathbb{F}_p}$, so that $\mathcal{D}_X = \mathbb{F}_p[t, \partial_t]/([\partial_t, t] = 1)$. But note that ∂_t^p acts as 0 on $\mathcal{O}_X$. Nevertheless, we still have the standard filtration by degree:

$$\mathcal{D}^{\leq n}_{X/R} \subset \mathcal{D}_{X/R}$$

and $\mathrm{gr}\mathcal{D}_{X/R} \simeq \mathrm{Sym}^\bullet T_{X/R}$

Lemma 1.5. *For any smooth scheme X/R, then $\Omega^\bullet_{X/R}$ is isomorphic to*

$$\mathcal{RH}om_{\mathcal{D}_{X/R}}(\mathcal{O}_X, \mathcal{O}_X)$$

Proof. Let's write a locally projective resolution for $\mathcal{O}_X$ as a $\mathcal{D}$-module:

$$\cdots \to \mathcal{D}_{X/R} \otimes_{\mathcal{O}_X} \wedge^2 T_{X/R} \to \mathcal{D}_{X/R} \otimes_{\mathcal{O}_X} T_{X/R} \xrightarrow{\delta\otimes v \mapsto \delta\cdot\partial_v} \mathcal{D}_{X/R} \xrightarrow{\delta\mapsto\delta(1)} \mathcal{O}_X \to 0$$

We then thus compute Hom termwise. □

The next goal is to construct crystalline cohomology using $\mathcal{D}$-modules.

Theorem 1.6. *If $p > 2, n \geq 1$ and X/k be a smooth variety. We can output certain abelian (tensor) category* $\mathrm{Cris}(X/W_n(k))$ *called* **crystals** *with a distinguished element $\mathcal{O}^{\mathrm{cris}}_X$ such that: for any $X_n/W_n(k)$ with $X_n(k) \times_{W_n(k)} k \simeq X$ such that:*

$$\mathrm{Cris}(X/W_n(k)) \simeq \mathrm{Mod}(\mathcal{D}_{X_n/W_n(k)})$$

$$\mathcal{O}^{\mathrm{cris}}_X \mapsto \mathcal{O}_{X_n}$$

With this theorem, one can just define:

$$R\Gamma_{\mathrm{cris}}(X/W_n(k)) \coloneqq R\mathcal{H}om_{\mathrm{Cris}(X/W_n(k))}(\mathcal{O}^{\mathrm{cris}}_X, \mathcal{O}^{\mathrm{cris}}_X) \in D(W_n(k))$$

Remark. $\mathsf{QCoh}(X_n)$ *as a tensor category recovers X_n.*

Remark. *Sheaf of algebras $\mathcal{D}_{X_n/W_n}$ genuinely depends on X_n.*

Theorem 1.7 (Deformation theory)**.** *Let $A \to A/I$ be a square 0 thickenining (meaning $I^2 = 0$). Let $Y/(A/I)$ smooth scheme.*

1. *There exists $\mathrm{ob}_{Y,A} \in H^2(Y, T_{Y/(A/I)} \otimes_{A/I} I)$ such that:*

$$\mathrm{ob}_{Y,A} = 0 \iff \exists \tilde{Y}/A \text{ smooth with } \rho : \tilde{Y} \times_A A/I \simeq Y$$

   In particular, every curve has a lift.

2. *If there exists one such lift $(\tilde{Y}, \rho)$, then there is a bijection:*

$$\{(\tilde{Y}', \rho')\} \simeq H^1(Y, T_{Y/(A/I)} \otimes_{A/I} I)$$

3. *Given a lift (Y, ρ), we have a bijection:*

$$\{f : \tilde{Y} \to \tilde{Y} \mid f = \mathrm{id} (\mathrm{mod}\ I)\} \simeq H^0(Y, T_{Y/(A/I)} \otimes_{A/I} I)$$

Let X/k be smooth. We cover X by affine opens $U_i, i \in I$. Using the theorem to the setting $W_{n+1}(k) \to W_n(k)$, we see there exists unique $\tilde{U}_i/W_n(k)$ smooth, with $\tilde{U}_i \times k \simeq U_i$. By uniqueness of lift, for each $i, j \in I$ there eixsts an isomorphism:

$$f_{ij} : \mathcal{O}_{\tilde{U}_i}|_{U_i \cap U_j} \simeq \mathcal{O}_{\tilde{U}_j}|_{U_i \cap U_j}$$

What I must have is the cocycle condition modulo p:

$$f_{mi} \circ f_{jm} \circ f_{ij} \equiv 1 (\mathrm{mod}\ p)$$

Lemma 1.8. *Let $Y_n/W_n(k)$ be a smooth affine scheme with $p > 2$. Let $f : Y_n \to Y_n$ with $f \equiv 1 (\mathrm{mod}\ p)$. Consider $f^* \in \mathrm{Aut}(\mathcal{D}_{Y_n/W_n(k)}(Y_n))$, there exists $D_f \in D_{Y_n/W_n}(Y_n)^\times$ with $D_f \equiv 1 \bmod p$ such that:*

$$f^*(\delta) = D_f \delta D_f^{-1}$$

Proof. As before , let:

$$p \cdot \Delta_f = \log f^* = \sum_{r \geq 1} (-1)^{r+1} \frac{(f^* - \mathrm{id})^{\circ r}}{r} \in \mathrm{End}(\mathcal{O}(Y_n))$$

For some derivation $\Delta_f \in \mathrm{Der}(\mathcal{O}(Y_{n-1}))$, then we will pick:

$$D_f \coloneqq \exp(p \cdot \partial_{\Delta_f}) \in D(Y_n)$$

□

Corollary 1.9. *For M any D_{Y_n}-module, there is a natural isomorphism $M \simeq (f^*)_* M$*

Proof. Consider $m \mapsto D_f \cdot m$. □

Let's now construct the category $\mathrm{Cris}(X/W_n(k))$. The idea is to mimic the way to define $\mathcal{D}$-modules. We consider:

$$\{\text{modules } M_i/\mathcal{D}_{\tilde{U}_i}(\tilde{U}_i), \alpha_{ij} : f_{ij}^*(M_i|_{\tilde{U}_i\cap\tilde{U}_j}) \simeq M_j|_{\tilde{U}_i\cap\tilde{U}_j}\}$$

satisfying the cocycle conditon replacing equality by the natural isomorphism as in the previous corollary.

Given $f : X \to Y$ map of smooth varieties over k. There is an induced map:

$$f^* : R\Gamma_{\mathrm{cris}}(Y/W(k)) \to R\Gamma_{\mathrm{cris}}(X/W(k))$$

Let's pick $f = F : X \to X^{(1)}$ be the relative Frobenius. Then I have:

$$F^* : R\Gamma_{\mathrm{cris}}(X^{(1)}/W(k)) \to R\Gamma_{\mathrm{cris}}(X/W(k))$$

Assume $k = \mathbb{F}_q$ with $q = p^r$. Then:

$$(F^*)^r : R\Gamma_{\mathrm{cris}}(X/W(k)) \to R\Gamma_{\mathrm{cris}}(X/W(k))$$

Theorem 1.10. *Let $X/\mathbb{F}_q$ be a smooth proper scheme. We know $H^n_{\mathrm{cris}}(X/W(\mathbb{F}_q))$ is finitely generated $W(\mathbb{F}_q)$-module. Assume that there exists an algebraic scheme lift $\tilde{X}/W(\mathbb{F}_q)$, then:*

$$H^n_{\mathrm{cris}}(X/W(\mathbb{F}_q)) \simeq H^n_{\mathrm{dR}}(\tilde{X}/W(\mathbb{F}_q))$$

Then we still have Lefschetz Trace Formula:

$$\#X(\mathbb{F}_q) = \sum_{i\geq 0}(-1)^i \mathrm{Tr}((F^*)^r \mid H^i_{\mathrm{cris}}(X/W(\mathbb{F}_q))[\frac{1}{p}])$$

Remark. *This enhances Atiyah-Bott fixed point formula: for $X/\mathbb{F}_q$ smooth proper, we have:*

$$\#X(\mathbb{F}_q) = \sum_{n\geq 0}(-1)^n \mathrm{Tr}((F^*)^r; H^n(X, \mathcal{O}_X)) \bmod p$$

Corollary 1.11. *If $X/\mathbb{F}_q$ is smooth proper, $H^i(X, \mathcal{O}_X) = 0, \forall i > 0$. Then:*

$$\#X(\mathbb{F}_q) \cong 1 (\mathrm{mod}\ q)$$

Proof. It suffices to show that for each $n > 0$:

$$\mathrm{Tr}((F^*)^r; H^n_{\mathrm{cris}}(X)[\frac{1}{p}]) = 0 \bmod q$$

Then $\#X(\mathbb{F}_q) \equiv \mathrm{Tr}((F^*)^r; H^0_{\mathrm{cris}}(X)[\frac{1}{p}]) \equiv 1 \bmod q$ Given any M a $W(\mathbb{F}_q)$-module, let M_{tf} the maximal torsion free quotient. It suffices to show:

$$F^* : H^n_{\mathrm{cris}}(X^{(1)}/W(\mathbb{F}_q))_{\mathrm{tf}} \to H^n_{\mathrm{cris}}(X/W(\mathbb{F}_q))_{\mathrm{tf}}$$

is divisible by p. But note:

$$R\Gamma_{\mathrm{cris}}(X/W(\mathbb{F}_q)) \otimes^{\mathbb{L}}_{W(\mathbb{F}_q)} \mathbb{F}_q \simeq R\Gamma_{\mathrm{dR}}(X/\mathbb{F}_q)$$

we get:

$$0 \to H^n_{\text{cris}}(X/W(\mathbb{F}_q))/p \to H^n_{\text{dR}}(X/\mathbb{F}_q) \to H^{n+1}_{\text{cris}}(X/W(\mathbb{F}_q))[p] \to 0$$

Therefore it suffices to prove:

$$F^* : H^n_{\text{dR}}(X^{(1)}/\mathbb{F}_q) \to H^n_{\text{dR}}(X/\mathbb{F}_q)$$

is zero. Note that the map between de Rham complex looks

$$\begin{array}{ccccccc}
\mathcal{O}_X & \longrightarrow & \Omega^1_X & \longrightarrow & \Omega^2_X & \longrightarrow & \cdots \\
{\scriptstyle f\mapsto f^p}\uparrow & & \uparrow{\scriptstyle 0} & & \uparrow{\scriptstyle 0} & & \\
F^{-1}\mathcal{O}_{X^{(1)}} & \longrightarrow & F^{-1}\Omega^1_{X^{(1)}} & \longrightarrow & F^{-1}\Omega^2_{X^{(1)}} & \longrightarrow & \cdots
\end{array}$$

So the above map factors through $H^n(X^{(1)}, \mathcal{O}_{X^{(1)}}) = 0$. □

Proposition 1.12. *Let k be a perfect field of characteristic p. Suppose X/k smooth scheme, $\tilde{X}/W(k)$ a smooth formal scheme such that $\tilde{X} \times_{W(k)} k \simeq X$. Suppose there is $\tilde{F} : \tilde{X} \to \tilde{X}^{(1)} = \tilde{X} \times_{W(k)} W(k)^{(1)}$ such that $\tilde{F} \bmod p = F$. Then:*

$$\mathcal{O}_{\tilde{X}} \xrightarrow{d} \Omega^1_{\tilde{X}} \xrightarrow{d} \cdots$$

is quasi isomorphic to

$$\mathcal{O}_{\tilde{X}^{(1)}} \xrightarrow{dp} \Omega^1_{\tilde{X}^{(1)}} \xrightarrow{dp} \cdots$$

Corollary 1.13. *If X/k is smooth admitting a lift $(\tilde{X}, \tilde{F})$. Then the de Rham complex is quasi isomorphic to:*

$$\mathcal{O}_{X^{(1)}} \xrightarrow{0} \Omega^1_{X^{(1)}} \xrightarrow{0} \cdots$$

In particular:

$$H^n_{\text{dR}}(X/k) \simeq \bigoplus_{i+j=n} H^i(X^{(1)}, \Omega^j_{X^{(1)}})$$

Proof. (of proposition).

$$\begin{array}{ccccccc}
\mathcal{O}_{\tilde{X}} & \xrightarrow{d} & \Omega^1_{\tilde{X}} & \xrightarrow{d} & \Omega^2_{\tilde{X}} & \longrightarrow & \cdots \\
{\scriptstyle \tilde{F}^*}\uparrow & & \uparrow{\scriptstyle \tilde{F}^*/p} & & \uparrow{\scriptstyle \tilde{F}^*/p^2} & & \\
\mathcal{O}_{\tilde{X}^{(1)}} & \xrightarrow{d} & \Omega^1_{\tilde{X}^{(1)}} & \xrightarrow{d} & \Omega^2_{\tilde{X}^{(1)}} & \longrightarrow & \cdots
\end{array}$$

It suffices to check that this is a quasi isomorphism modulo p. □

Theorem 1.14 (Berthelot-Ogus)**.** *Let X/k be a smooth scheme, with a lift $\tilde{X}/W(k)$ smooth formal scheme with* $\dim X < p$*. Then:*

$$\mathcal{O}_{\tilde{X}} \xrightarrow{d} \Omega^1_{\tilde{X}} \xrightarrow{d} \cdots$$

is quasi isomorphic to

$$\mathcal{O}_{\tilde{X}^{(1)}} \xrightarrow{dp} \Omega^1_{\tilde{X}^{(1)}} \xrightarrow{dp} \cdots$$

In particular:

$$H^n_{\mathrm{dR}}(X/k) \simeq \bigoplus_{i+j=n} H^i(X^{(1)}, \Omega^j_{X^{(1)}})$$

Let X/k for k perfect field of characteristic p. We can think of $\mathcal{D}_X$-modules as $\mathsf{QCoh}(\widehat{X^{\mathrm{dR}}})$ for some stack $\widehat{X^{\mathrm{dR}}}$. For now assume $k = F$ has characteristic 0.

Lemma 1.15. *Let M be a $\mathcal{D}_X$-module on X. Given $x \in X(F)$, there is a natural isomorphism:*

$$\varprojlim_n (M \otimes_{\mathcal{O}_X} \mathcal{O}_{X,x}/\mathfrak{m}_x^n) = M|_{\widehat{\mathcal{O}_{X,x}}} \simeq M_x \widehat{\otimes}_F \widehat{\mathcal{O}_{X,x}}$$

Proof. We will construct:

$$M_x \xrightarrow{\sim} (M|_{\widehat{\mathcal{O}_{X,x}}})^{\nabla=0}$$

We choose a local coordinate:

$$\widehat{\mathcal{O}_{X,x}} \simeq F[[t_1, \cdots, t_d]]$$

For $s \in M_x$, it has a lift $\tilde{s} \in M(U)$ for some $x \in U$, we then send:

$$s \mapsto \sum_{i_1,\cdots,i_d \geq 0} (-1)^{i_1+\cdots+i_d} \frac{t_1^{i_1} \cdots t_d^{i_d}}{i_1! \cdots i_d!} \nabla^{i_1}_{\partial t_1} \cdots \nabla^{i_d}_{\partial t_d}(\tilde{s})$$

To its Taylor expansion. This will be independent of $\tilde{s}$ from the nature of stalks. □

1.4 A Modern Approach

Consider $\Delta \hookrightarrow X \times_F X$, with $p_1, p_2 : X \times X \to X$ be the two projection maps. let's consider:

$$(X \times_F X)^{\widehat{}}_{\Delta}$$

the union of closed subscheme of $X \times X$ cut out by $\mathcal{I}^n_\Delta$.

Theorem 1.16 (Grothendieck)**.** *Let X/F be a smooth scheme over field of characteristic* 0*. The data of a $\mathcal{D}_X$-module is the same data of a quasi coherent sheaf M on X with isomorphism:*

$$\alpha : p_1^* M \xrightarrow{\sim} p_2^* M$$

on $(X \times X)^{\widehat{}}_{\Delta}$ satisfying certain cocycle condition.

Corollary 1.17. *Consider:*

$$\text{Spec } R_{\text{red}} \hookrightarrow \text{Spec } R \xrightarrow{s} X$$

Then s^*M only depends on $s|_{\text{Spec }(R_{\text{red}})}$.

Proof. Given two s, s' agrees on the reduced part. It induces a map:

$$\text{Spec } R \to (X \times X)^{\widehat{}}_{\Delta}$$

□

Example. *Let $X = \mathbb{A}^1_F$ with* char $F = 0$. *Then $\mathcal{D}_{\mathbb{A}^1}$-modules is the same data as a quasi coherent sheaf $\mathbb{A}^1$ iwth $\widehat{\mathbb{G}_a}$-equivariant structure. Since we can identify:*

$$(\mathbb{A}^1 \times \mathbb{A}^1)^{\widehat{}}_{\Delta} \simeq \mathbb{A}^1 \times \widehat{\mathbb{G}_a}$$

$$(x, y) \mapsto (x, x - y)$$

Therefore it's the same data as $\mathsf{QCoh}(\mathbb{A}^1/\widehat{\mathbb{G}_a})$

Theorem 1.18 (Drinfeld,Bhatt-Lurie)**.** *Let X/k be smooth variety, for k perfect of characteristic p. If $\tilde{X}/W_2(k)$ is a smooth lift of X, there is a natural $\mathbb{Z}/p$-grading:*

$$H^n_{\text{dR}}(X/k) \simeq \bigoplus_{i \in \mathbb{Z}/p} H^n_{\text{dR}}(X/k)_i$$

such that for $n < p$:

$$H^n_{\text{dR}}(X/k)_i \simeq H^{n-i}(X^{(1)}, \Omega^i_{X^{(1)}})$$

Let R be a $\mathbb{Q}$-algebra, recall that $\mathcal{D}_{\mathbb{A}^1_R}$-module is equivalent to $\mathsf{QCoh}(\mathbb{A}^1_R/\widehat{\mathbb{G}_a})$. Very explicilty, we just send:

$$M \mapsto M/R[t]$$

with a coaction map $M \to M\widehat{\otimes}\mathcal{O}(\widehat{\mathbb{G}_a}) \simeq M[[t]]$, $m \mapsto \sum_{n \geq 0} \partial_t^n(m) \otimes \frac{t^n}{n!}$.

Now let $\mathbb{G}^{\#}_{a,\mathbb{F}_p} = \text{Spec } \mathbb{F}_p[t, \frac{t^n}{n!} \mid n > 1]$, where we define:

$$\mathbb{F}_p[t, \frac{t^n}{n!} \mid n > 1] = \mathbb{Z}[t, \frac{t^n}{n!} \mid n \geq 1] \otimes_{\mathbb{Z}} \mathbb{F}_p$$

One can check that this ring is isomorphic to $\mathbb{F}_p[t_0, t_1, t_2, \cdots]/(t_0^p, t_1^p, \cdots)$ by "choosing a coordinate".

We also define:

$$\widehat{\mathbb{G}^{\#}_{a,\mathbb{F}_p}} = \bigcup \text{Spec } \mathcal{O}(\mathbb{G}^{\#}_{a,\mathbb{F}_p})/(\frac{t^m}{m!} \mid m \geq n) = \text{Spf } \mathbb{F}_p[t][[\frac{t^n}{n!} \mid n \geq 1]]$$

We can endow both with natural (formal) group scheme structure, making:

$$\widehat{\mathbb{G}^{\#}_{a,\mathbb{F}_p}} \to \mathbb{G}^{\#}_{a,\mathbb{F}_p}$$

a map of formal group schemes.

Remark. $\mathbb{G}^{\#}_{a,\mathbb{Z}}$ *has a unique group structure such that* $\mathbb{G}^{\#}_{a,\mathbb{Z}} \to \mathbb{G}_{a,\mathbb{Z}}$ *is a group scheme map.*

Lemma 1.19. *Let* R *be any ring. Then* $\mathcal{D}(\mathbb{A}^1_R)$*-module is equivalent to* $\mathsf{QCoh}(\widehat{\mathbb{A}^1_R/\mathbb{G}^{\#}_{a,R}})$

Remark. *Let* $R = \mathbb{F}_p$*, consider:*

$$f : \mathbb{G}^{\#}_{a,\mathbb{F}_p} \to \mathbb{G}_{a,\mathbb{F}_p}$$

this is given by ring map:

$$\mathbb{F}_p[t] \to \mathbb{F}_p[t, \frac{t^p}{p}, \cdots]$$

Then $\ker f \simeq \operatorname{Spec} \mathcal{O}(\mathbb{G}^{\#}_a)/t$*. Similarly,*

$$\ker(\widehat{f} : \widehat{\mathbb{G}^{\#}_a} \to \mathbb{G}_a) \simeq \operatorname{Spf} \mathcal{O}(\widehat{\mathbb{G}^{\#}_a})/t$$

In particular, for all $\mathcal{D}$*-module* M*, then* $\ker \widehat{f}$ *acts via an autormophism of* M*.*

Let X/k with k of characteristic p. Recall we have:

$$\mathcal{D}_X \xrightarrow{\alpha} \mathcal{E}nd_R(\mathcal{O}_X)$$

Locally, given each vector field $v \in T_X(U) = \operatorname{Der}_k(\mathcal{O}_X, \mathcal{O}_X)(U)$, we know $v^{[p]} = v^{\circ p}$ is again a derivation.

Example. *Let* $X = \mathbb{A}^1_{\mathbb{F}_p} = \operatorname{Spec} \mathbb{F}_p[t]$*. Then:*

$$v = \partial_t \implies v^{[p]} = 0$$

$$v = t\partial_t \implies v^{[p]} = t\partial_t$$

One checks $\partial_v^p - \partial_{v^{[p]}} \in \mathcal{D}_X(U)$ is in the kernel of α.

Proposition 1.20. *1.* $Z(F_*\mathcal{D}_X) \simeq \operatorname{Sym}_{\mathcal{O}_{X^{(1)}}} T_{X^{(1)}}$

$$\partial_v^p - \partial_{v^{[p]}} \leftarrow v \in T_{X^{(1)}}$$

2. $\ker \alpha$ *is the ideal generated by* $\operatorname{Sym}^{\geq 1}_{\mathcal{O}^{(1)}_X} T_{X^{(1)}}$

Definition 1.21. Let X/k with M a $\mathcal{D}_X$-module. The *p-curvature* of M along v is

$$\psi_v = \partial_v^{\circ p} - \partial_{v^{[p]}} \in \operatorname{End}_{\mathcal{D}}(M)$$

Definition 1.22. Let X/k be smooth, we call a $\mathcal{D}_X$-module M is *quasi-nilpotent* (with nilpotent p-curvature) if for all $U \subset X$ and all $v \in T_X(U)$, ψ_v acts on $M(U)$ is locally nilpotent.

Example. *For any* X*,* $\mathcal{O}_X$ *as a* $\mathcal{D}_X$*-module is quasi-nilpotent. But* $\mathcal{D}_X$ *as a* $\mathcal{D}_X$*-module is not.*

Quasi-nilpotent $\mathcal{D}_{\mathbb{A}^1_{\mathbb{F}_p}}$-modules is equivalent to the category $\mathsf{QCoh}(\mathbb{A}^1_{\mathbb{F}_p}/\mathbb{G}^{\#}_{a,\mathbb{F}_p})$.

Lemma 1.23. *Let X/k be smooth.*

$$\Omega^{\bullet}_{X/k} \simeq R\mathcal{H}om_{\text{qnil } \mathcal{D}\text{-mod}}(\mathcal{O}, \mathcal{O}_X)$$

Definition 1.24. Let X/R be smooth for a $\mathbb{Z}/p^n$-algebra R. We call $\mathcal{D}_{X/R}$-module M quasinilpotent if M/p is quasi-nilpotent on $X \times_R R/p$.

Lemma 1.25. *Quasi-nilpotent $\mathcal{D}_{\mathbb{A}^1_R}$-modules is equivalent to the category $\mathsf{QCoh}(\mathbb{A}^1_R/\mathbb{G}^{\#}_{a,R})$.*

Let $Y \hookrightarrow X/k$ be smooth embedding. We define the *divided power envelope*:

$$X^{\#}_Y \coloneqq \operatorname{Spec}_X \mathcal{O}_X[\frac{f^n}{n!} \mid n \geq 1, f \in \mathcal{I}_Y]$$

Theorem 1.26. *Let X/k be a smooth variety. Quasi-nilpotent $\mathcal{D}_X$-modules are equivalent to quasi-coherent sheaves on M/X, with $p_1^* M \simeq p_2^* M$ on:*

$$(X \times X)^{\#}_{\Delta} \xrightarrow{p_i} X$$

satisfying the cocycle conditions.

Definition 1.27. A quotient stack is a pair (G, X), denoted as $[X/G]$. A strong equivalence $(G, X) \sim (G', X')$ if there exists $f : G' \to G$ a flat surjection and $X' \to X$ is a $\ker f$-torsor making the obvious diagram commutes. Then we call $[X/G]$ and $[X'/G']$ equivalent if they can be connected by a chain of strong equivalences. We thus denote:

$$\mathsf{QCoh}([X/G]) \coloneqq G\text{-equivariant sheaves on } X$$

Definition 1.28. Let X/k be affine scheme, $X \hookrightarrow \mathbb{A}^n_k$. We define:

$$X^{\mathrm{dR}} \coloneqq [(\mathbb{A}^n)^{\#}_X / \mathbb{G}_a^{\#\times n}]$$

This will be independent of the embedding.

Remark. *A more intrinsic definition will be:*

$$X^{\mathrm{dR}} = X/(X \times X)^{\#}_{\Delta}$$

Proposition 1.29. $\mathsf{QCoh}(X^{\mathrm{dR}}) \simeq (\mathcal{D}_X\text{-mod})^{\mathrm{qnil}}$.

The ingredient for the proof is by observing: $X \mapsto X^{\mathrm{dR}}$ for every X is recovered from the k-algebra stack $\mathbb{A}^{1,\mathrm{dR}}_k$.

Definition 1.30.

$$\mathbb{G}^{\#}_m = (\mathbb{G}_m)^{\#}_1 = \operatorname{Spec} k[t^{\pm 1}, \frac{(t-1)^n}{n!} \mid n \in \mathbb{N}]$$

This is a group scheme.

Theorem 1.31. *Suppose X/k has a lift $\tilde{X}/W_2(k)$, then we have a $\mathbb{G}^{\#}_m$ action on X^{dR}.*

Remark. *This theorem will imply the original theorem. The action will induce $\mathbb{G}_m^{\#}$-action on $\mathrm{Ext}^n_{\mathsf{QCoh}(X^{\mathrm{dR}})}(\mathcal{O}_{X^{\mathrm{dR}}}, \mathcal{O}_{X^{\mathrm{dR}}}) = H^n_{\mathrm{dR}}(X/k)$. Just like $\mathbb{G}_m$ action will induce a $\mathbb{Z}$-grading, the $\mathbb{G}_m^{\#}$-action will induce a $\mathbb{Z}/p$-grading via:*

$$\mathbb{G}_m^{\#} \xrightarrow{\log} \mathbb{G}_a$$

But we can compute $\ker \log = \mu_p$. *In particular, this implies we have a μ_p-action on $H^n_{\mathrm{dR}}(X/k)$,giving a $\mathbb{Z}/p$-grading as desired.*

Remark.

$$\mathbb{G}^{\#}_{m,k} \simeq \mu_p \times \mathbb{G}_a^{\#}$$

Proposition 1.32. *If X/k admits $\tilde{X}/W_2(k)$ and*

$$\tilde{F} : \tilde{X} \to \tilde{X}^{(1)}$$

lifting $F : X \to X^{(1)}$. Then there exists a natural action $\mathbb{G}_m$-action on $X^n_{\mathrm{dR}}(X/k)$.

Remark. *For $X = \mathbb{A}^1$, it admits a unique lift $(\tilde{X}, \tilde{F})$. But this lift is not compatible with the ring scheme structure.*

Lemma 1.33.

$$\ker(\mathbb{G}_a^{\#} \to \mathbb{G}_a) \simeq \mathbb{G}_a^{\#,(1)} = \mathrm{Spec}\ k[\frac{t^p}{p}, \frac{(\frac{t^p}{p})^p}{p}, \cdots]$$

$$\mathrm{coker}\ (\mathbb{G}_a^{\#} \to \mathbb{G}_a) \simeq \mathbb{G}_a^{(1)}$$

There will be a natural action of $\mathbb{G}_m^{\#}$ on $\ker(\mathbb{G}_a^{\#} \to \mathbb{G}_a)$.

Lemma 1.34. *There exists unique flat $\mathbb{Z}_p$-algebra $\mathcal{O}(W)$ with $F : \mathcal{O}(W) \to \mathcal{O}(W)$ such that $F(y) \equiv y^p \bmod p$ which is universal in the sense for every flat R and a morphism $\mathbb{Z}_p[x] \to R$, there exists unique lift $f : \mathcal{O}(W) \to R$ such that $f \circ F = F_R \circ f$.*

Let $W = \mathrm{Spec}\ \mathcal{O}(W) \to \mathbb{A}^1$.

Proposition 1.35. *1. W has a unique ring scheme structure such that $W \to \mathbb{A}^1$ is a ring map.*

2. $\mathcal{O}(W) \simeq \mathbb{Z}_p[x = a_0, a_1, \cdots]$ such that

$$F^n(x) = a_0^{p^n} + pa_1^{p^{n-1}} + \cdots p^n a_n$$

W is called Witt vector scheme: for every $\mathbb{Z}_p$-algebra $W(R)$ is a ring. As a set:

$$W \simeq \prod_{n \geq 0} R$$

We insist the ghost components:

$$w_n : W(R) \to R$$
$$(a_0, a_1, \cdots) \mapsto a_0^{p^n} + \cdots p^n a_n$$

to be a ring homomorphism. Now that $F : W \to W$ is a Frobenius lift. This is a ring scheme map.

Lemma 1.36.

$$\mathbb{A}^{1,\mathrm{dR}} \simeq [W/_{\bullet p}W]$$

where the action is given by $a \in W$ acts by pa.

With this presentation, we can define a $\mathbb{G}_m^{\#}$ action on $\mathbb{A}^{1,\mathrm{dR}}$. This is because we can identify:

$$\mathbb{G}_m^{\#} \simeq W^{\times,F=1}$$

$$t \mapsto \mathrm{can}(t-1)+1$$

where $\mathrm{can} : \mathbb{G}_a^{\#} \hookrightarrow W$ is the canonical embedding. Now that the map:

$$W \xrightarrow{p} W$$

is $\mathbb{G}_m^{\#}$-equivariant, for the first action by rescaling and the second action being trivial.

2 Hitchin Systems

2.1 Moduli Stack of Principal G-bundle

Let k be a field, X smooth irreducible projective curve over k. Let G be a split connected reductive group. Most of time, we will work for $k = \mathbb{C}$, $G = \mathrm{GL}_n, \mathrm{PGL}_n, \mathrm{SL}_n$.

Definition 2.1. A principal G-bundle on X is a G-variety P/k with $\pi : P \to X$ right G-equivariant, such that locally π looks like:

$$X \times G \to X$$

with G acting by right multiplication.

Remark. *In general, need to etale locally.*

So concretely, a principal G-bundle is defined by an atlas of charts U_i on X and clutching maps

$$g_{ij} : U_i \cap U_j \to G$$

satisfying cocycle conditions. Given P, we can find U_i such that $P|_{U_i}$ is trivial, and get these g_{ij}; conversely, given these g_{ij} we can glue to get P. Therefore the principal G-bundles on X are classified by:

$$H^1_{\text{ét}}(X, \mathcal{O}_{X,G})$$

where $\mathcal{O}_{X,G}$ is the sheaf of G-valued regular functions on X. This is because g_{ij} defines a Cech 1-cocycle valued in $\mathcal{O}_{X,G}$. If $\tilde{g_{ij}} = h_i g_{ij} h_j^{-1}$ for $h_i : U_i \to G$, then $g_{ij}, \tilde{g_{ij}}$ define isomorphic bundles.

Remark. $H^1_{\text{ét}}(X, \mathcal{O}_{X,G})$ *is only a set, unless G is abelian.*

For every $x \in X(F)$, $\pi^{-1}(x)$ is a principal homogeneous G-space.

Let's talk about moduli of G-bundles. There exists a stack

$$\mathrm{Bun}_G(X)$$

which classifies G-bundles on X. Namely, for S a test scheme, we have:

$$\mathrm{Maps}(S, \mathrm{Bun}_G(X)) = \{G\text{-bundles on } X \times S\}$$

It turns out this is a smooth algebraic stack.

Example. *Quotient stack is an algebraic stack. Let Y be a scheme with an algberaic group H action, we get a quotient stack Y/H given by:*

$$\mathrm{Maps}(S, Y/H) = \{(E, \varphi) \mid E \in \mathrm{Prin}_H(S), \varphi : E \to Y \text{ is } H\text{-equivariant}\}$$

We will see $\mathrm{Bun}_G(X)$ is a nested union of quotient stacks for smooth Y_n/H_n for Y_n smooth.

Example. *Let $BG = */G$. We know:* $\mathrm{Hom}(S, BG)$ *classifies principal G-bundles on S. In fact:*

$$\underline{\mathrm{Hom}}(X, BG) = \mathrm{Bun}_G(X)$$

Example. *Let $G = \mathrm{GL}_1 = \mathbb{G}_m$, and X be a smooth curve. We wish to describe* $\mathrm{Bun}_{\mathbb{G}_m}(X)$. *This is closely related to the Picard variety* $\mathrm{Pic}(X)$. *The only difference is that in* $\mathrm{Bun}_{\mathbb{G}_m}(X)$ *we need to remember the inner symmetry of each line bundle. Explicitly:*

$$\mathrm{Bun}_{\mathbb{G}_m}(X) \simeq \mathrm{Pic}(X) \times B\mathbb{G}_m$$

as stacks.

Example. *Let $G = \mathrm{GL}_2$, we wish to study* $\mathrm{Bun}_G(\mathbb{P}^1)$. *Let E be a rank 2 vector bundle on $\mathbb{P}^1$, then it has a vector subbundle:*

$$\mathcal{O}(m) \hookrightarrow E$$

for sufficiently negative m. So we have a short exact sequence:

$$0 \to \mathcal{O}(m) \to E \to \mathcal{O}(n) \to 0$$

Recall that $\mathcal{O}(m)$ has clutching function z^m. Thus E has clutching matrix:

$$\begin{pmatrix} z^m & f(z) \\ & z^n \end{pmatrix}$$

We can change trivialization so that $f(z)$ has only monomials z^s for $m < s < n$ (note if $m \geq n-1$ then $f = 0$ and we have a direct sum). This agrees with the fact that:

$$\mathrm{Ext}^1(\mathcal{O}(n), \mathcal{O}(m)) = H^1(\mathbb{P}^1, \mathcal{O}(m-n)) = H^0(\mathbb{P}^1, \mathcal{O}(n-m-2))$$

has dimension $n-m-1$ for $n > m$, and 0 otherwise.

Let's find largest r such that there exists a nonzero map: $\mathcal{O}(r) \to E$. Namely, we need to have a nonzero section:

$$v \in \Gamma(\mathbb{P}^1, \mathcal{O}(-r) \otimes E)$$

Assume on U_∞, $v = \begin{pmatrix} x_\infty(z^{-1}) \\ y_\infty(z^{-1}) \end{pmatrix}$; on U_0, $v = \begin{pmatrix} x_0(z) \\ y_0(z) \end{pmatrix}$. The the gluing condition reads:

$$\begin{pmatrix} x_0(z) \\ y_0(z) \end{pmatrix} = z^{-r} \begin{pmatrix} z^m & f(z) \\ & z^n \end{pmatrix} \begin{pmatrix} x_\infty(z^{-1}) \\ y_\infty(z^{-1}) \end{pmatrix}$$

After doing somealgebra one can see the maximal $r \geq \frac{m+n}{2}$. Hence $r' \coloneqq m+n-r \leq r$. Also $\mathcal{O}(r) \to E$ must be an inclusion of line bundle, otherwise r can be raised. So we get:

$$0 \to \mathcal{O}(r) \to E \to \mathcal{O}(r') \to 0$$

and $\mathrm{Ext}^1(\mathcal{O}(r'), \mathcal{O}(r)) = 0$. We thus know every rank 2 vector bundle $\mathbb{P}^1$ splits.

Connected components of $\mathrm{Bun}_{\mathrm{GL}_2}(\mathbb{P}^1)$ are labelled by degree $m+n$, but "degreee of degeneracy" is $n-m$. So points are $P_0, P_1, P_2, \cdots$ (ordered by $n-m$) and closure of P_i consists of P_j with $j \geq i$. So open sets are $\{P_0, P_1, \cdots, P_{j-1}\}$ and the whole space. (This is different from Zariski topology in which points are closed).

More generally, we have:

Theorem 2.2 (Grothendieck-Birkhoff)**.** *A vector bundle on $\mathbb{P}^1$ of rank N is uniquely represented as:*

$$\mathcal{O}(m_1) \oplus \mathcal{O}(m_2) \cdots \oplus \mathcal{O}(m_N)$$

with $m_1 \leq \cdots \leq m_N$

Proof. Induction. Choose m_0 maximal such that there exists:

$$0 \to \mathcal{O}(m_0) \to E \to \mathcal{O}(m_1) \oplus \cdots \mathcal{O}(m_N) \to 0$$

by induction hypothesis. Then we have $E' \subset E$ where:

$$0 \to \mathcal{O}(m_0) \to E' \to \mathcal{O}(m_i) \to 0$$

SO that $m_0 \geq \frac{m_0+m_i}{2} \implies m_0 \geq m_i$ for each i. But this also shows each extension is trivial, so that our first SES splits. □

Theorem 2.3. *If G is a connected reductive group. Then for all G-bundle on $\mathbb{P}^1$, there is a T-structure, and we have an bijection given by induction:*

$$\mathrm{Bun}_T / W \simeq \mathrm{Bun}_G$$

So bundles are labelled by $\Lambda_+^\vee = X^\vee(T)_+$.

Let G be a semisimple group. In this case it is known that every G-bundle on an affine smooth curve is trivial. So we can glue from two charts: D_x (the formal disc around x), and $X\backslash x$. The clutching map is an element:

$$g \in G(K), \quad K = k[D_x^\times] \simeq k((t))$$

Now we can modify g by right multiplication by $G(\mathcal{O}), \quad \mathcal{O} = k[D_x] = k[[t]]$; we can modify g on the left by $G(R)$ for $R = k[X\backslash x]$. So we have:

$$\mathrm{Bun}_G(X) = G(R)\backslash G(K)/G(\mathcal{O})$$

as a set.

Definition 2.4. We call $G(K)/G(\mathcal{O}) \simeq \frac{G(k((t)))}{G(k[[t]])}$ the *affine Grassmannian*, denoted by Gr_G (it is an ind-variety). So $\mathrm{Bun}_G(X)$ is the set of $G(R)$-orbits on Gr_G.

We can do the same if we remove any finite subset of points $S \subset X$. So we get:

$$\mathrm{Bun}_G(X) = G(R)\backslash \prod_{x\in S} G(K_x)/\prod_{x\in S} G(\mathcal{O}_x)$$

where $R = k[X\backslash S]$. This only works for semisimple groups.

In fact, more generally we can take $S = X(k)$. Then we get:

$$\mathrm{Bun}_G(X)(k) = G(F)\backslash G(\mathbb{A})/G(\mathcal{O}_{\mathbb{A}})$$

where $F = k(X), \mathbb{A} = \prod'_{x\in X(k)} K_x, \mathcal{O}_{\mathbb{A}} = \prod_{x\in X(k)} \mathcal{O}_x$. This is valid for any group.

Let E_γ be the bundle defined by clutching map $\gamma \in G((t))$ (in the semisimple case). Let $\mathfrak{g} = \mathrm{Lie}(G)$. By definition:

$$T_{E_\gamma}\mathrm{Bun}_G(X) = \mathfrak{g}((t))/(\mathrm{Ad}\gamma^{-1}(\mathfrak{g}(X\backslash x_0)) + \mathfrak{g}[[t]])$$

Definition 2.5. Let $E \in \mathrm{Bun}_G(X)$, its *associated adjoint bundle* is the vector bundle

$$\mathrm{ad}E = E \times^G \mathfrak{g} \to X$$

One can show:

$$T_E\mathrm{Bun}_G(X) \simeq H^1(X, \mathrm{ad}E) \simeq H^0(X, K_X \otimes \mathrm{ad}E)^*$$

where the second isomorphism is from Serre duality and by fixing an invariant inner product on E to identify $\mathrm{ad}E$ with $(\mathrm{ad}E)^*$.

Definition 2.6. We call:

$$T^*_E\mathrm{Bun}_G(X) = H^0(X, K_X \otimes \mathrm{ad}E)$$

the **space of Higgs fields**.

Assume that a genric bundle has automorphism group $Z(G)$ (this will be the case if G is semisimple and $g > 1$). Let $\mathrm{Bun}_G^0(X)$ be the open set of regularly stable bundles. For $E \in \mathrm{Bun}_G^0(X)$, we know:

$$H^0(X, \mathrm{ad}E) = \mathrm{Lie}(\mathrm{Aut}E) = 0 \implies H^1(X, K_X \otimes \mathrm{ad}E) = 0$$

So

$$\dim H^0(X, K_X \otimes \mathrm{ad}E) = \chi(K_X \otimes \mathrm{ad}E) = \chi(K_X) \dim \mathfrak{g} = \dim \mathfrak{g} \cdot (g-1)$$

as $\mathrm{ad}E$ is a vector bundle of degree 0. Thus $\mathrm{Bun}_G^0(X)$ is a smooth variety of dimension $\dim \mathfrak{g} \cdot (g-1)$. Moreover, this vareity is known to be irreducible, if G is simply connected.

If G is reductive:

$$\dim \mathrm{Bun}_G^0(X) = (g-1) \dim \mathfrak{g} + \dim Z(\mathfrak{g})$$

Example. *For $G = \mathrm{GL}_n$, we have:*

$$\dim \mathrm{Bun}_n^0(X) = n^2(g-1) + 1$$

The symplectic variety $T^*\mathrm{Bun}_G^0(X)$ is an open subset in the **Hitchin stack**:

$$\mathcal{M}_G = \{(E, \varphi) \mid E\text{: principal } G\text{-bundle}, \varphi\text{: Higgs field}\}$$

2.2 Hithcin Integrable Sysmtem

Suppose a group H acts on a variety Y. Then we have a Hamiltonian H-action on T^*Y , and we have **moment map**:

$$\mu : T^*Y \to \mathfrak{h}^*$$

dual to

$$\mathfrak{h} \to \mathrm{Vect}(Y)$$

If H acts freely, we have **Hamiltonian reduction**:

$$\mu^{-1}(0)/H \simeq T^*(Y/H)$$

Moreover, every H-stable function on T^*Y, it descends to $T^*(Y/H)$; and any two with $\{f_1, f_2\} = 0$ descends to ones with $\{f_1, f_2\} = 0$. Thus $T^*\mathrm{Bun}_G(X)$ is the Hamiltonian reduction of $T^*G(K)$ by $G(R) \times G(\mathcal{O})$.

Let X be an irreducible smooth projective curve $\mathbb{C}$, with genus $g \geq 2$. Let G be connected reductve group over $/\mathbb{C}$. Recall we have:

$$\mathrm{Bun}_G(X) \supset \mathrm{Bun}_G^0(X)$$

the piece of regular stable bundle as a variety. For every $E \in \mathrm{Bun}_G^0(X)$, $\mathrm{Aut}(E) = Z(G)$. For $G = \mathrm{GL}_n, \mathrm{PGL}_n, \mathrm{SL}_n$, we have $\mathrm{Bun}^0(X)$ parametrizes stable bundles: $\mu(E) = \frac{\deg(E)}{\mathrm{rank}(E)}$ and for every $F \subset E, \mu(F) < \mu(E)$.

Consider the case $G = \mathrm{PGL}_n$.

Definition 2.7. Hitchin base of X for PGL_n is the vector space $\mathcal{B} = \bigoplus_{i=1}^{n-1} H^0(X, K_X^{\otimes i+1})$. Note:

$$\dim H^0(X, K_X^{\otimes m}) = (2m-1)(g-1)$$

for $m \geq 2$ by Riemann-Roch. Hence:

$$\dim \mathcal{B} = \sum_{m=2}^{n-1} (2m-1)(g-1) = (n^2-1)(g-1) = \dim \mathrm{Bun}_G^0(X)$$

Definition 2.8. The **Hitchin map** is the map:

$$p : T^*\mathrm{Bun}_G^0(X) \to \mathcal{B}$$

$$p(E, \varphi) = (\mathrm{Tr}\Lambda^2\varphi, -\mathrm{Tr}\Lambda^3\varphi, \cdots, (-1)^n \mathrm{Tr}\Lambda^n\Phi)$$

Theorem 2.9 (Hitchin)**.** *p is generically a Lagrangian fibration (i.e. generic fibres are Lagrangian). Concretely, this means the coordinate functions on $\mathcal{B}$ pulled back are algebraically independent (dominant) and Poisson commute (i.e. form an integrable system).*

Choose some b_j for $\mathcal{B}$ compatible with direct sum decomposition. Then

$$p(E, \varphi) = \sum_{j=1}^{\dim \mathcal{B}} H_j(E, \varphi) b_j$$

for $H_j : T^*\mathrm{Bun}_G^0(X) \to \mathbb{C}$, with $\{H_i, H_k\} = 0$.

Now consider the case $G = \mathrm{GL}_n$, we get the same Hitchin map but adding $\mathrm{Tr}\varphi$:

$$p(E, \varphi) = (-\mathrm{Tr}\varphi, \mathrm{Tr}\Lambda^2\varphi, \cdots)$$

and:

$$\mathcal{B} = \bigoplus_{i=0}^{n-1} H^0(X, K_X^{\otimes i+1})$$

has an additional summand $H^0(X, K_X)$ of dimension g. Then one can compute $\dim \mathcal{B} = n^2(g-1) + 1 = \dim \mathrm{Bun}_n^0(X)$. We have same statement of Hitchin system.

For general G, recall we have Chevalley theorem:

$$\mathbb{C}[\mathfrak{g}]^G = \mathbb{C}[Q_1, \cdots, Q_r]$$

for Q_i homogeneous of degree d_i, where d_i are degrees of G.

Example. *For $G = \mathrm{GL}_n$, it has degrees $1, 2, \cdots, n$, $Q_i = (-1)^i \mathrm{Tr}\Lambda^i\varphi$*

Now we decompose $\mathfrak{g}$ as a module over the principal $\mathfrak{sl}_2$:

$$\mathfrak{g} = \bigoplus_{i=1}^{r} L_{2(d_i-1)}$$

We thus can define:

$$\mathcal{B} = \bigoplus_{i=1}^{r} H^0(X, K_X^{\otimes d_i})$$

Then

$$\dim \mathcal{B} = \sum (2d_i - 1)(g-1) + \#\{i \mid d_i = 1\} = \dim \mathfrak{g}(g-1) + \dim Z(G) = \dim \mathrm{Bun}_G^0(X)$$

We can thus define the Hitchin map:

$$p : T^*\mathrm{Bun}_G^0(X) \to \mathcal{B}$$

$$p(E, \varphi) = (Q_1(\varphi), \cdots, Q_r(\varphi))$$

Then the Hitchin's theorem holds for any G.

Proof. (of Poisson commutativity). The input will be Hamiltonian reduction. Let G be semisimple for simplicity,

$$\mathrm{Bun}_G(X) = G(R)\backslash G((t))/G[[t]]$$

by choosing a point $x \in X$ for t local coordinate, and $R = \mathbb{C}[X\backslash x]$. Let $G((t))^0$ be the preimage of $\mathrm{Bun}_G^0(X)$. So:

$$H = G(R) \times G((t))/Z(G)$$

acts freely on $G((t))^0 = Y$. Now $\mathrm{Bun}_G^0(X) = Y/H$, then:

$$T^*(Y/H) = \mu^{-1}(0)/H = T^*Y//H$$

If $\{F_i\}$ are H-invariant Poisson commuting functions on T^*Y, then I can descend them to Poisson commuting functions $\overline{F_i}$ on $T^*(Y/H) = T^*Y//H$.

Now we need to construct Poisson commuting functions on $T^*G((t)) = \mathfrak{g}((t))^* \times G((t))$. But we can canonically identify:

$$\mathfrak{g}((t))^* = \mathfrak{g} \otimes \mathbb{C}((t))dt$$

using the residue pairing. Therefore a point on $T^*G((t))$ is a pair (g, φ) for $\varphi \in \mathfrak{g} \otimes \mathbb{C}((t))dt$-Higgs field on D_x^*. I can assign

$$\varphi \mapsto (Q_1(\varphi), \cdots Q_r(\varphi)), \quad Q_i(\varphi) \in \mathbb{C}((t))(dt)^{\otimes d_i}$$

Write $Q_{i,n}(\varphi)$ for the Laurent coefficients of $Q_i(\varphi)$. Note that $Q_{i,n}(\varphi)$ are 2-sided invariant functions on $T^*G((t))$. Therefore it descend to $Y/H = \mathrm{Bun}_G^0(X)$. What we get is precisely the Hitchin map:

$$\begin{array}{ccc} T^*G((t))^0 & \xrightarrow{(Q_1,\cdots,Q_r)} & \bigoplus_{i=1}^r \mathbb{C}((t))dt^{\otimes d_i} \\ \uparrow & & \\ \mu^{-1}(0) & \text{expand at } x \text{ in coordinate } t & \uparrow \\ \downarrow & & \\ T^*\mathrm{Bun}_G^0(X) & \xrightarrow{p} & \bigoplus_{i=1}^r H^0(X, K_X^{\otimes d_i}) \end{array}$$

So p^*b_j are obtained by descending linear combinations of $Q_{i,n}$, hence they are Poisson commute because $Q_{i,n}$s are Poisson commute. □

Remark. *Suppose A is a Lie group, with* $\mathrm{Lie}(A) = \mathfrak{a}$. *Then recall $\mathfrak{a}^*$ is Poisson, with Poisson bracket coming from the Lie bracket on $\mathfrak{a}$. It follows that:*

$$f \in \mathbb{C}[\mathfrak{a}^*]$$

we have:

$$\{a_j, f\} = \sum c_{ij}^k a_k \frac{\partial f}{a_i} = 0$$

by choosing a basis for $\mathfrak{a}$ and computing the structure constants.

Now let's start proving the dominance of p for $G = \mathrm{GL}_n$ ($n \geq 2$). We need the notion of **Spectral Curve**. Let $b = (b_1, \cdots, b_n) \in \mathcal{B}$. Consider the polynomial:

$$\lambda^n + b_1\lambda^{n-1} + \cdots + b_n = \prod(\lambda - \lambda_i)$$

For each $x \in X$, we have $\lambda_i \in T_x^*X$. Then these λ_i trace out a curve $C_b \subset T^*X$ equipped with a degree n map $\pi : C_b \to X$. The C_b is called the spectral curve. Note that if $b = p(E, \varphi)$, then C_b represents the spectrum of φ.

Theorem 2.10 (Hitchin)**.** *For generic $b \in \mathcal{B}$, C_b is smooth and irreducible.*

Proof. Since smooth and irreducible are open conditions, so enough to show there exists such b. We need:

Lemma. *1. If L is a line bundle on X with degree $\geq 2g$, then a generic section has only simple zeros.*

2. A generic section of $K_X^{\otimes n}$ has simple zeroes.

Proof. (of Lemma) (1) $\implies$ (2) since $\deg K_X^{\otimes n} = (2g-2)n \geq 2g$. To prove (1), consider $Y \subset H^0(X, L)$ variety of section which has a double zero. Assume $\deg L = d \geq 2g$. Riemann-Roch tells if M is any line bundle $M \neq K_X$ of degree $m \geq 2g - 2$, then

$$\dim H^0(X, M) = m - g + 1$$

If $s \in Y$ has double zero at z, then s is a section of $L \otimes \mathcal{O}(-2z)$. But now $\deg(L \otimes \mathcal{O}(-2z)) \geq 2g - 2$, so if $L \otimes \mathcal{O}(-2z) \neq K_X$, then:

$$\dim H^0(L \otimes \mathcal{O}(-2z)) \leq d - g - 1$$

It take all possible z, we get dimension $\dim \leq d - g$. Moreover $L \otimes \mathcal{O}(-2z) \simeq K_X$ only for finitely many z, so this piece has contribution $\dim \leq d - g \implies \dim Y \leq d - g$. So that $Y \neq H^0(Y, L)$ since the later has dimension $d - g + 1$. □

We take $b = (0, 0, \cdots, s)$ for $s \in H^0(X, K_X^{\otimes n})$. It has simple zeros by the lemma. So spectral curve is then defined by $\lambda^n = s(x)$ is is smooth and irreducible. We also need:

Theorem (Hitchin)**.** *The genus for C_b is $n^2(g-1)+1$ the dimension of $\mathrm{Bun}_n^0(X)$, when it is smooth.*

Proof. Check at b as above, use Hurwitz formula. s has $2n(g-1)$ simple zeroes. It follows:

$$\chi(C_b) = n\chi(X) - (n-1)2n(g-1) = -n^2(2g-2) \implies g(C_b) = n^2(g-1)+1$$

□

□

If $b = p(E,\varphi)$, then on C_b we have an "eigenbundle" L for φ. Geometrically, φ has distinct eigenvalues, and L_z for $z \in C_{p(E,\varphi)}$ is the eigenspace of φ on E_x. Moreover, we can reconstruct E from C, L by $E = \pi_* L$. Also on $\pi_* L$ we have a canonical Higgs field $\pi_* L \to \pi_* L \otimes K_X$ which acts on each summand by the corrresponding eigenvalue, and under the identification $\pi_* L \simeq E$ it goes to φ. Therefore the data of Higgs pair (E,φ) is the same data as (C, L). So:

$$p^{-1}(b) \subset \mathrm{Pic}_d(C) \simeq \mathrm{Jac}(C)$$

because C only depends on b. Therefore $\dim p^{-1}(b) \leq n^2(g-1)+1$. This proves p is dominant.

2.3 Bundles with Parabolic Structures

Let E be a G-bundle on X, $x \in X$, $H \subset G$ subgroup.

Definition 2.11. An H-structure on E at x is an H-orbit in E_x. Hence possible H-structures are parametrized (non-canonically) by G/H.

Example. *Take H to be parabolic, do it at several points:*

$$t_1, \cdots, t_N \subset X; \quad P_1, \cdots, P_N \text{ parabolic subgroups}$$

Then we define:

$$\mathrm{Bun}_G(X, t_1, \cdots, t_N, P_1, \cdots, P_N)$$

to be the moduli stack of G-bundles on X with parabolic structures at $t_1, \cdots, t_N$ of types $P_1, \cdots, P_N$.

Example. *Let $G = \mathrm{GL}_N, P = P(N_1, \cdots, N_s)$ with $\sum N_i = N$. Then recall a GL_N-bundle is just a rank n vector bundle; a P-structure at x is the same data as a partial flag:*

$$0 = V_0 \subset V_1 \subset \cdots V_s = E_x \simeq \mathbb{C}^N, \quad \dim V_i/V_{i-1} = N_i$$

We have a natural projection:

$$\mathrm{Bun}_G(X, t_1, \cdots, t_N, P_1, \cdots, P_N) \to \mathrm{Bun}_G(X)$$

This is a fibration with fibre $G/P_1 \times \cdots G/P_N$. Open set $\mathrm{Bun}_G^0(X) \subset \mathrm{Bun}_G(X)$

Example. *Let $X = \mathbb{P}^1$, consider trivial G-bundles. The stack of such bundles is:*

$$G \setminus_{\text{diag}} \prod_{i=1}^{N} G/P_i$$

Let's now take $G = \mathrm{PGL}_2, P_i = B$. In this case we get:

$$\mathrm{PGL}_2 \setminus (\mathbb{P}^1)^N$$

It will have an open piece isomorphic to $(\mathbb{P}^1)^{N-3}$.

We can generalize the Hitchin system to the parabolic case. Note that:

$$T^*_E \mathrm{Bun}^0_G(X, t_1, \cdots, t_N, P_1, \cdots, P_N) = \{\varphi : E \to E \otimes K_X\}$$

where φ has at most simple poles at t_i with residue α strictly preserving the flag at t_i. Then:

$$p(E, \varphi) = (Q_1(\varphi), \cdots, Q_r(\varphi)) \in \mathcal{B}$$

Let's compute the Hitchin system for $X = \mathbb{P}^1$, with parabolic structures at $t_1, \cdots, t_N \in \mathbb{A}^1 \subset \mathbb{P}^1$. We take trivial bundles, then the parabolic structure $s_i = [y_1 : 1] \in \mathbb{P}^1$. The Higgs field will be:

$$\varphi = \sum_{i=1}^{N} \frac{A_i}{z - t_i} dz$$

This have no pole at ∞, this translates to $\sum_i A_i = 0$. Now if

$$A_i = \begin{pmatrix} a_i & b_i \\ c_i & -a_i \end{pmatrix}$$

we need

$$\begin{pmatrix} a_i & b_i \\ c_i & -a_i \end{pmatrix} \cdot \begin{pmatrix} y_i \\ 1 \end{pmatrix} = 0 \implies A_i = p_i \begin{pmatrix} y_i & -y_i^2 \\ 1 & -y_i \end{pmatrix}$$

$$H = \frac{1}{2}\mathrm{Tr}\varphi^2 = \frac{1}{2}\mathrm{Tr} \sum_{i,j} \frac{A_i A_j}{(z - t_i)(z - t_j)} (dz)^2 = \sum_{i<j} \frac{\mathrm{Tr} A_i A_j}{(z - t_i)(z - t_j)} (dz)^2$$

We can also compute:

$$\mathrm{Tr} A_i A_j = -p_i p_j (y_i - y_j)^2$$

We get:

$$H(z) = \sum_i \frac{G_i}{z - t_i} (dz)^2$$

for $G_i = \sum_{j \neq i} \frac{p_i p_j (y_i - y_j)^2}{t_j - t_i}$ the *Garnier Hamiltonians*. However recall $\sum A_i = 0$, this implies:

$$\sum G_i = \sum t_i G_i = \sum t_i^2 G_i = 0$$

In fact from these G_i I obtain $N - 3$ independent functions.

Our phase space is

$$\{(\underline{y},\underline{p})\in\mathbb{C}^{2N}\mid \sum p_i=\sum p_iy_i=\sum p_iy_i^2\}/\mathrm{PGL}_2$$

Let's now discuss the twisted version.

$$\mathrm{Bun}_G(X,t_1,\cdots,t_N)=G(R)\setminus G((z_1))\times\cdots\times G((z_N))/I_1\times\cdots I_N$$

for $R=\mathbb{C}[X\setminus\{t_1,\cdots,t_N\}]$, $I\subset G[[z]]$ consists of elements which at $z=0$ evaluate to element of B(i.e. $G[[z]]/I=G/B$).

Now given characters:

$$\lambda_i:\mathrm{Lie}I\to\mathbb{C}$$

We can form the *twisted Hamiltonian reduction*:

$$\mu^{-1}(\lambda)/G(R)\times I_1\times\cdots\times I_N$$

which is a twisted cotangent bundle of Bun_G^0.

In our example of PGL_2, $X=\mathbb{P}^1$. The twisted cotangent space at $E,s_1,\cdots,s_N$ is

$$\varphi:E\to E\otimes K_X$$

with simpe poles at t_i and residues A_i such that $A_i|_{s_i}=\lambda_i$. Then if

$$\varphi=\sum\frac{A_i}{z-t_i}$$

$$H=\frac{1}{2}\mathrm{Tr}\varphi^2=\sum\frac{\lambda_i^2}{(z-t_i)^2}dz^2+\text{something with simple poles}$$

From this we will obtain Hamiltonians:

$$G_i=\sum_{j\neq i}\frac{(y_i-y_j)^2p_ip_j2(\lambda_ip_j-\lambda_jp_i)(y_i-y_j)-2\lambda_i\lambda_j}{t_j-t_i}$$

This is an integrable systemon on $T^*_{\text{twist}}(\mathbb{P}^1)^{N-3}$. This is called twisted Garnier system.

Example. *Let X be an elliptic curve, consider:*

$$\mathrm{Bun}_G(X,0,P)_0$$

where $G=\mathrm{PGL}_n$, with n of type $(1,n-1)$. A twist corresponds to a choice $c\in\mathbb{C}$. A generic degree 0 rank n vector bundle on X is a direct sum of degree 0 line bundles of form $L_q=\mathcal{O}(q)\otimes\mathcal{O}(0)^{-1}$ for $q\in X$, which corresponds to function $\frac{\theta(z-q)}{\theta(z)}$. So $E=\bigoplus_{i=1}^n L_{q_i}$. Let's assume $q_i\neq q_j$. A Higgs field is given by:

$$\varphi:E\to E\otimes K_X$$

Note in this case K_X is trivial. So φ is given by a matrix (φ_{ij}), $\varphi_{ij}:L_{q_i}\to L_{q_j}$. In fact we can choose:

$$\varphi_{ij}=a_{ij}\frac{\theta'(0)\theta(z+q_j-q_i)}{\theta(z)\theta(q_j-q_i)}\in\Gamma(L_{q_j}\otimes L_{q_i}^{-1})$$

where θ is an entire function on $\mathbb{C}$, with transformation law:

$$\theta(z+1) = \theta(z), \theta(z+\tau) = -e^z\theta(z)$$

for $X = \mathbb{C}/\mathbb{Z}\langle 1, \tau\rangle$. On the other hand ,$\varphi_{ii} = p_1$ is forced to be a constant. One can show that:

$$\mathrm{res}_0\varphi = \begin{pmatrix} (n-1)c & * \\ & -c\cdot \mathrm{id}_{n-1} \end{pmatrix} \implies \mathrm{res}_0\varphi + c\cdot \mathrm{id}$$

has rank 1. This forces $a_{ij} = c$ for each $i \neq j$ (by appropriate rescaling). We can thus compute:

$$H_2(z) = \frac{1}{2}\mathrm{Tr}\varphi^2 = \frac{n(n-1)}{2}c^2\wp(z) + \frac{1}{2}(\sum_i p_i^2 - c^2\sum_{i\neq j}\wp(q_i - q_j))$$

where the second factor $F_2 = \sum_i p_i^2 - c^2\sum_{i\neq j}\wp(q_i - q_j)$ is called **elliptic Calogero-Moser hamiltonian**. *There are also "higher integrals" coming from $\mathrm{Tr}\varphi^m$, which is $F_m = \sum p_i^m +$* lower terms.

2.4 Quantization of Hitchin Systems

Classical integerable system lives on T^*Y. Let's first consider quantization of T^*Y. Recall that $\mathcal{O}(T^*Y)$ is locally generated by $\mathcal{O}(Y)$ and $p_i = dy_i$. Quantization in this setting is to replace p_i by $\hbar\partial_i$, so quantization of $\mathcal{O}(T^*Y)$ is

$$\mathcal{D}_\hbar(Y) = \sum_{j=0}^{\infty}\hbar^j\mathcal{D}_{\leq j}(Y)$$

Quantization of classical integrable system $H_1, \cdots, H_n$ is a collection $\widehat{H_1}, \cdots, \widehat{H_n} \in \mathcal{D}_\hbar(Y)$ such that they pairwise commute and their classical limits recover H_i.

Example. *Recall the Garnier system:*

$$G_i = \sum_{j\neq i}\frac{(x_i - x_j)^2p_ip_j - 2(x_i - x_j)(\lambda_ip_j - \lambda_jp_i)}{t_j - t_i}$$

The naive quantization is just to replace p_i by $\hbar\partial_i$ (there is an ambiguity that where should I put my ∂_is):

$$\widehat{G_i} = \sum_{j\neq i}\frac{\hbar^2(x_i - x_j)^2\partial_i\partial_j - 2(x_i - x_j)(\hbar\lambda_ip_j - \hbar\lambda_jp_i)}{t_j - t_i}$$

$$= \hbar^2\sum_{j\neq i}\frac{(x_i - x_j)^2\partial_i\partial_j - 2(x_i - x_j)(\Lambda_ip_j - \Lambda_jp_i)}{t_j - t_i}$$

for $\Lambda_i = \lambda_1/\hbar$. It turns out they commute, and the system we get is independent on ordering. This is the **Gaudin system**. *In fact, there is a conceptual construction of the Gaudin system. Let $V_1, \cdots, V_N$ be representations of $\mathfrak{g}$, let $\Omega \in (S^2\mathfrak{g})^G$ be the Casimir. We consider:*

$$\mathcal{H} = V_1 \otimes \cdots \otimes V_N$$

We have an operator:

$$\Omega_{ij} : \mathcal{H} \to \mathcal{H}$$

where the first component of Ω acts on V_i and the second component acts on V_j. Now we can set:

$$\widehat{G_i} = \sum_{j \neq i} \frac{\Omega_{ij}}{t_i - t_j} \in \mathrm{End}(\mathcal{H})$$

One can check that $[\widehat{G_i}, \widehat{G_j}] = 0$, so they also act on $\mathcal{H}^G$. Now take $\mathfrak{g} = \mathfrak{sl}_2$, $V_i = \mathbb{C}[x]$, and:

$$f \mapsto -\partial_x, h \mapsto 2x\partial_x, e \mapsto x^2\partial_x - 2\Lambda_i x$$

we construct the above Gaudin system.

Example. *If we consider Calogero-Moser hamiltonian:*

$$H_2 = \frac{1}{2}\sum_i p_i^2 - c^2 \sum_{i \neq j} \wp(q_i - q_j)$$

We have quantization:

$$\widehat{H}_2 = \frac{\hbar^2}{2}(\sum_i \partial_i^2 - C^2 \sum_{i \neq j} \wp(q_i - q_j))$$

we call the inner bracket L_2 the quantum CM Hamilotnian.

Theorem. *Quantum CM Hamiltonian is completely integrable.*

Now let's discuss a little bit on Beilinson-Drinfeld quantization of Hitchin system. Recall Hitchin system was constructed by Hamiltonian reduction:

$$T^*\mathrm{Bun}_G(X) = T^*G(K_x)//G(R) \times G(\mathcal{O}_x)$$

and the two-sided invariant $\widetilde{H_i} = Q_i(\varphi)$ descends to quotient. Now in the quantized world, we should replace $T^*\mathrm{Bun}_G(X)$ by $\mathcal{D}_\hbar(\mathrm{Bun}_G(X))$. So we should seek two-sided invariant elements in $\mathcal{D}_\hbar(G(K_x))$ and hope it will descend to the quotient.

Let's consider the following toy case. For K a Lie group, then:

$$\mathcal{D}(K)_{\text{left}} = U(\mathfrak{k})$$

therefore:

$$\mathcal{D}(K)_{\text{2-sided}} = U(\mathfrak{k})^K = Z(U(\mathfrak{k}))$$

In fact, we need to consider differential operators on half densities on Bun_G: $K_{\mathrm{Bun}}^{\frac{1}{2}}$. This is natural in the sense that we hope our quantization to live in a Hilbert space (hence half densities). Therefore $\mathcal{D}(G(K_x))$ should be replaced by differential operators on line bundles on $G(K)$. We have Kac-Moody group:

$$1 \to \mathbb{C}^* \to \widehat{G} \to G((t)) \to 1$$

the Kac-Moody group has Kac-Moody Lie algebra:

$$\widehat{g} = \mathfrak{g}((t)) \oplus \mathbb{C}K$$

$$[a(t), b(t)] = [ab](t) + \mathrm{res}_0(da, b)K$$

One can show that:

$$K_{\mathrm{Bun}_G(X)}^{\frac{1}{2}} = \mathcal{L}^{-h^\vee}$$

for $h^\vee$ the dual Coxeter number of $\mathfrak{g}$, $\mathcal{L}$ a line bundle on Bun_G.

3 Strata in Reductive Groups

First let $G/\mathbb{C}$ be a connected reductive group, $\mathfrak{g} = \mathrm{Lie}G$. Let:

$$D_G = \{\omega : \mathbb{C}^* \to G \mid \exists \widetilde{\omega} : \mathrm{SL}_2(\mathbb{C}) \to G, \omega(t) = \widetilde{\omega}(\begin{pmatrix} t & \\ & t^{-1} \end{pmatrix})\}$$

Given ω, we have a filtration:

$$G_0^\omega \supset G_1^\omega \supset G_2^\omega \supset \cdots$$

where:

$$\mathfrak{g}_n = \{x \in \mathfrak{g} \mid \omega(t) = t^n x, \forall t \in \mathbb{C}^*\}$$

and $\mathrm{Lie}G_n^\omega = \mathfrak{g}_n \oplus \mathfrak{g}_{n+1} \oplus \cdots$. Then one can show G_0^ω is a parabolic subgroup, G_1^ω is its unipotent radical, G_2^ω is a subset of unipotent elements of G. In fact I have:

$$\mathrm{Hom}(\mathrm{SL}_2(\mathbb{C}), G)/G \to D_G/G \to \text{unipotent elements}/G$$

all bijections.

Now let G/k over an algebraically closed field. Then I can still define D_G and

$$G_0^\omega \supset G_1^\omega \supset G_2^\omega \supset \cdots$$

For ω, we denote $\overline{\omega}$ be its G-orbit. We define:

$$\widetilde{H}^{\overline{\omega}} = \bigcup_{\omega \in \overline{\omega}} G_2^\omega$$

$$H^{\overline{\omega}} = \widetilde{H}^{\overline{\omega}} - \bigcup_{\overline{\omega'} < \overline{\omega} \in D_G/G} \widetilde{H}^{\overline{\omega'}}$$

form a partition of unipotent variety. Each $H^{\overline{\omega}}$ contains a unique dense unipotent class. Over $\mathbb{C}$, each $H^{\overline{\omega}}$ actually contains only one unipotent class, in other words D_G/G parametrizes the unipotent classes over $\mathbb{C}$

Therefore, we have a well-defined embedding from unipotent classes of $G/\mathbb{C}$ to unipotent classes of G/k in charactersitic p, by picking the unique dense unipotent class in $H^{\overline{\omega}}$.

Now assume $k = \mathbb{C}$ or has large characteristic. Let's discuss a little bit Springer correspondence. Let $\mathcal{B}$ be the flag variety, with Weyl group $W = G \setminus \mathcal{B} \times \mathcal{B}$. We have Springer resolution:

$$\pi : \widetilde{\mathcal{B}} \to G$$

We know π is a small map. Let G_{rs} be the subset of regular semisimple elements, with preimage $\widetilde{\mathcal{B}}_{\mathrm{rs}}$. Given $w \in W$, I have a map:

$$r_w : \widetilde{\mathcal{B}}_{\mathrm{rs}} \to \widetilde{\mathcal{B}}_{\mathrm{rs}}$$

$$(g, B) \mapsto (g, \mathcal{B}')$$

where (B, B') has relative position w. This defines a map:

$$W \to \mathrm{Aut}(\widetilde{\mathcal{B}}_{\mathrm{rs}})$$

One can define the group structure on W via this map.

Now π is small, so W acts on $H^*(\mathcal{B}_g, \overline{\mathbb{Q}_\ell})$, for any element $g \in G$. There is a W-equivariant restriction map:

$$H_*(\mathcal{B}_g, \overline{\mathbb{Q}_\ell}) \to H_*(\mathcal{B}, \overline{\mathbb{Q}_\ell})$$

Let V_g be the image of this map in degree $d(g) = 2 \dim \mathcal{B}_g$. This V_g will be irreducible representations of W (the Springer representations).

This defines a stratification on G: we call $g \sim g' \iff V_g = V_{g'}$.

Facts:

1. Any stratum is a union of conjugacy classes of a fixed dimension.
2. Any stratum contains at most one unipotent class.
3. $\mathrm{Irr}_*(W) = \{V_g\} \subset \mathrm{Irr}(W)$ is independent of the characteristic.
4. Any stratum contains a unipotent element in some characteristic.
5. There is a canonical bijection between the strata on a reductive group and its Langlands dual.

Example. *Let $G = \mathrm{GL}_3(\mathbb{C})$. Let $G_d = \{g \in G \mid \dim(\text{conj class of } g) = d\}$. This is nonempty for $d = 6, 4, 0$. These are exactly the strata. These G_d are actually smooth irreducible varieties.*

Example. *Let $G = \mathrm{Sp}_4(\mathbb{C})$. We define G_d the same way. This will be nonempty for $d = 8, 6, 4, 0$. But G_4 will split:*

$$G_4 = G_4' \coprod G_4''$$

where G_4' is the semisimple conjugacy class for $\mathrm{diag}(1, 1, -1, -1)$. *$G_8, G_6, G_4', G_4'', G_0$ are the strata. All these strata have smooth components. But here G_6 is not irreducible.*

Let k_p be algebraic closure of $\mathbb{F}_p$, $k_0 = \mathbb{C}$. Let G_p be connected reductive groups of k_p, U_p the set of unipotent elements of G_p, $\underline{U}_p$ the set of unipotent conjugacy classes of G_p (this is a finite set).

Note that $\underline{U}_0 \subset \underline{U}_p$. We can write U_p as disjoint union of pieces. For each piece $\mathcal{P}$ it has the property that if over $\mathbb{F}_q \subset k_p$ G_p has split $\mathbb{F}_q$ strata, then $|\mathcal{P}(\mathbb{F}_q)| = f(q)$ independent of p.

Example. *Consider* $\mathrm{Sp}_4(\mathbb{F}_q)$*, then the decomposition of pieces look like:*

$$q^8 = [1] + [(q^4-1)] + [\frac{q(q-1)(q^4-1)}{2}, \frac{q(q+1)(q^4-1)}{2}] + [2 \cdot \frac{q^2(q^2-1)(q^4-1)}{2}]$$

for $p \neq 2$*. For* $p = 2$*, then it looks like:*

$$[1] + [(q^4-1)] + [(q^2-1)(q^4-1) + q^4 - 1] + [2 \cdot \frac{q^2(q^2-1)(q^4-1)}{2}]$$

Let $\underline{U}_* = \bigcup_p \underline{U}_p / (x \in \underline{U}_0 \sim \text{image of } x \text{ in } \underline{U}_p)$. For classical groups, $\underline{U}_* = \underline{U}_2$; for type G_2, $\underline{U}_* = \underline{U}_3$; for type E_8, $\underline{U}_* = \frac{\underline{U}_2 \cup \underline{U}_3}{\sim}$, here $\underline{U}_0 = 70, \underline{U}_2 = 74, \underline{U}_3 = 71$.

The irreducible representations of B_n are in bijection with the collections:

$$(a_0 \leq a_2 \leq \cdots \leq a_{2m}; a_1 \leq \cdots \leq a_{2m-1} \mid \sum a_i = n)/\sim$$

where $\sim$ corresponds to adding 0 to the front. We define:

$$\mathrm{Sym}B_n = (b_0 < \cdots < b_{2m}; b_1 < \cdots < b_{2m-1} \mid \sum b_i = n + m^2)/\sim$$

where we identify the above to

$$(0 < b_0 + 1 < b_2 + 1 < \cdots b_{2m} + 1; 0 < b_1 + 1 < \cdots)$$

We also define:

$$\mathrm{Sym}'B_n = (c_0 \ll \cdots \ll c_{2n}; c_1 \ll c_3 \ll \cdots \ll c_{2m-1} \mid c_1 \geq 1, \sum c_i = n + 2m^2 + m)/\sim$$

$$\mathrm{Sym}''B_n = (d_0 \lll \cdots \lll d_{2m}; d_1 \lll \cdots d_{2m-1} \mid d_1 \geq 2, \sum d_i = n + 4m^2 + 2m)/\sim$$

All these sets are in bijection. We need these symbols for the following reasons.

1. The special representations of $W = B_n$ are parametrized by elements in $\mathrm{Sym}(B_n)$ with $b_0 \leq b_1 \leq b_2 \cdots$.
2. Irreducible representation of $W = B_n$ corresponding to unipotent classes of $G_p, p \neq 2$ are parametrized by elements in $\mathrm{Sym}'B_n$ such that: $c_0 \leq c_1 \leq c_2 \leq \cdots$.
3. Irreducible representations of $W = B_n$ corresponding to unipotent classes of G_p for $p = 2$ are those given by similar elements in $\mathrm{Sym}''B_n$.

Remark. *The set (1) is contained in set (2) is contained in set (3).*

There will be similar parametrizations for type D. Springer representation gives me an bijection to a subset of Irr(W):

$$\underline{U}_p \to \mathrm{Irr}_p(W) \subset \mathrm{Irr}(W)$$

It turns out:

$$\mathrm{Irr}_*(W) = \bigcup_p \mathrm{Irr}_p(W)$$

Under Springer correspondence, this translates back to

$$\underline{U}_* = \bigcup_p \underline{U}_p/\sim$$

In other word, we can use $\underline{U}_*$ to parametrize the Strata.

Fix a root system $(V, V^*, R, R^\vee)$, we can pick $e \in V, e' \in V^*$ and define:

$$W_{e,e'} \subset W$$

corresponding to root system $(V, V^*, R_{ee'}, R^\vee(ee'))$, where:

$$R_e = \{\alpha \in R \mid \langle e, \alpha^\vee \rangle = 0\}, R^\vee e = \{\alpha^\vee \in R^\vee \mid \alpha \in R_e\}$$

and similarly for $Re', R^\vee e'$, and set $R_{ee'} = R_e \cap R_{e'}; R^\vee_{ee'} = R^\vee_e \cap R^\vee_{e'}$

For each $E \in \mathrm{Irr}(W_{e,e'})$, we define n_E to be the minimal number such that it appearin the n_E-th symmetric power of the standard representation. For E a special representation of $W_{e,e'}$, we know $\mathrm{Ind}^W_{W_{e,e'}} E$ has has a unique irreducible component $\widetilde{E}$ such that $n_{\widetilde{E}} = n_E$, call it $j^W_{W_{e,e'}}(E)$. We thus call $\widetilde{E} \in \mathrm{Irr}(W)$ is 2-special if it is of form $j^W_{W_{e,e'}}(E)$. It turns out that:

$$\mathrm{Irr}_*(W) = \{\text{2-special representations of } W\}$$

In summary, we parametrized Strata by $\underline{U}_*$ or $\mathrm{Irr}_*(W)$. Now let's try a different approach. Let $\mathrm{cl}(W)$ be the set of conjugacy classes in W. For $w \in W$, γ a conjugacy class in G_p, we define:

$$\mathcal{B}_{\gamma,w} = \{(g, B) \in G_p \times \mathcal{B}_p \mid (B, gBg^{-1}) \text{ are in relative position } w, g \in \gamma\}$$

Theorem 3.1. *Assume w has minimal length in the conjugacy class. Then there exists unique $\gamma \in \underline{U}_p$ such that $\mathcal{B}_{\gamma,w} \neq 0$ of minimal dimension. γ depends only on the conjugacy class of w.*

This enables us to define:

$$\varphi_p : \mathrm{cl}(W) \to \underline{U}_p$$

This map is surjective. The goal is to define:

$$\varphi_* : \mathrm{cl}(W) \to \underline{U}_*$$

For $C \in \mathrm{cl}(W)$, there are two possibilities:

1. If $\varphi_p(C)$ is independent of p, then we define $\varphi_*(C) = \varphi_p(C)$ for any p.
2. Otherwise, there will be an exceptional prime p_0, and we just set $\varphi_*(C) = \varphi_{p_0}(C)$.

Hence we get a map:

$$\mathrm{cl}(W) \to \mathrm{Irr}(W)$$

with image $\mathrm{Irr}_*(W)$. Each fibre of this map has at most one elliptic element.

Example. *In type E_8, we look at the representation 15_{56} (the 15-th dimensional representation first appearing in 56-th tensor power of reflection representation), its fibre under the above map has 5 elements (should think these of type A_5). The representation 700_{28} has fibre of type A_4; $1400_{29}, 2016_{19}$ has fibre of type $\widetilde{A}_3$; there are 5 representations has fibre of type A_3; 14 of them has fibre of type A_4; 52 of thme has fibre type A_1.*

Example. *In type C_n, then $\underline{U}_* = \underline{U}_2$. This set is given by partitions:*

$$v_1 \geq v_2 \geq \cdots$$

of $2n$ such that any odd part appears even number of times, and any even part which appears even number of times has an associated $\varepsilon = 0, 1$. Now the Weyl group elements are parametrized by permutations of $1, 2, \cdots, 2n$ which commutes with $(1, 2n)(2, 2n-1)\cdots(n, n+1)$. Now $w \in W$ is a product of cycles, then we just read off the sizes of the cycles and these are precisely our v_i's.

In type B_n, recall the representations can be parametrized by symbols $\mathrm{Sym}''B_n$:

$$d_0 \lll \cdots \lll d_{2m}; d_1 \lll \cdots \lll d_{2m-1}$$

The **elliptic representations** are precisely those with property:

$$d_0 \ll d_1 \ll d_2 \ll \cdots \ll d_{2m}$$

Recall we constructed a map:

$$W \to \mathrm{cl}(W) \to \mathrm{Str}(G)$$

Recall we have distinguishe conjugacy classes $\mathrm{cl}(W)_{\mathrm{ell}}$, whose image in the strata are called $\mathrm{Str}_{\mathrm{ell}}$. As we've discussed: $\mathrm{cl}(W)_{\mathrm{ell}} \simeq \mathrm{Str}_{\mathrm{ell}}$.

Now assume G semsimple Let's choose $C \in \mathrm{cl}(W)_{\mathrm{ell}}$, and $w \in C$ with minimal length d_C. It's image under the above map is an elliptic stratum Σ_C. The miracle is that for $g \in \Sigma_C$, we have:

$$d_C = \dim Z(g)$$

If G is almost simple in characteristic 0, and Σ is an elliptic stratum, then there exists $g \in \Sigma$ semisimple. There will be only one exception when G is of type E_8 for one particular stratum.

Let's now discuss the following theorem:

Theorem 3.2. *Any stratum is locally closed.*

Recall we have introduced

$$G_m = \{g \in G \mid \text{conj class of } g \text{ has dimension } m\}$$

this is coarser than the strata. But it is known that G_m is locally closed.

Recall that we wrote G as disjoint union of "pieces". More precisely, given any $g \in G$, we can consider Jordan decomposition:

$$g = g_s \cdot g_u$$

Now let $H_G(g)$ be the centralizer in G of connected center of $Z_G^\circ(g_s)$.

Now fix (L, S_1) where L is Levi subgroup of a parabolic, and $S_1 \subset L$ is the inverse image of an isolated class in $L/Z^\circ(L)$. Let $Y = Y_{L,S_1}$ be the union of all conjugacy class in G which meets $g \in S_1, H_G(g) = L$. We now claim:

1. There are only finitely many (L, S_1) up to conjugacy.
2. Y_{L,S_1} form a partition of G (these are called pieces of G)
3. (Lusztig 1987) The closure of a piece is a union of pieces.
4. Any Straum is a union of pieces.

Assume that J is a piece inside a stratum $\Sigma \subset G_m$, then $\overline{J} \cap G_m \subset \Sigma$. Now let's look at maximal element in $\{\overline{J} \cap G_m\}$, then the union of all these maximal elements will precisely give me Σ. This helps me deduce the theorem.

Recall we've defined:

$$\mathrm{cl}(W) \to \mathrm{Str}(G) \to \mathrm{Irr}(W)$$

We now want to define:

$$\mathrm{Irr}(W) \to \mathrm{Str}(G)$$

Recall Springer correspondence give me a map:

$$\mathrm{Irr}(W) \xrightarrow{\varepsilon_p} \underline{U}_p \xrightarrow{\sim} \mathrm{Irr}_p(W) \subset \mathrm{Irr}(W)$$

the first map is just forgetting the local system. Now take any $E \in \mathrm{Irr}(W)$, we consider $\varepsilon_p(E) \in \underline{U}_*$. As before there are two cases:

1. If $\varepsilon_p(E)$ is independent of p, we set $\varepsilon_*(E)$ to be their common value;
2. Otherwise there is an exceptional prime p_0, we take $\varepsilon_*(E) = \varepsilon_{p_0}(E)$

Remark. *The composition:*

$$\mathrm{Irr}(W) \xrightarrow{\varepsilon_*} \mathrm{Str}(G) \to \mathrm{Irr}(W)$$

is identity on $\mathrm{Irr}_*(W)$.

Let's now introduce *superstratum*. Recall that we can decompose:

$$\mathrm{Irr}(W) = \bigcup_c \mathrm{Irr}_c(W)$$

where c runs through all 2-sided cells. As an example, for type B_n, we call two representations lie in the same two-sided cell if their symbols involve the same numbers. Now for each 2-sided cell c, consider:

$$\bigcup_{\Sigma \text{ stratum } \sigma(\Sigma)\in c} \Sigma$$

where $\sigma : \mathrm{Str}(G) \to \mathrm{Irr}_*(W)$, this are called the superstratum. It is conjectured that superstraa should be locally closed.

Example. *In the case of* Sp_4*, then the strata of dimension* $6, 4, 4$ *will glue to a single super stratum.*

It is also conjectured that any irreducible component of a super stratum is rationally smooth (over $\mathbb{C}$).

4 Brauer-Manin Obstruction

4.1 Introduction

Let k be a global field, k_v the completion of k at v, $\mathcal{O}_v$ the valuation ring of k_v (let's take $\mathcal{O}_v = k_v$ if v is archimedean). We let $\mathbb{A}$ be the adele of k, X a smooth projective geometrically integral k-variety. The main question to ask is that whether X has a k-point.

Theorem 4.1 (Hasse-Minkowski, 1924)**.** *For* $X : q(x_0, \cdots, x_n) = 0$ *in* $\mathbb{P}^n$ *for* q *degree* 2 *homogeneous, then:* X *has a* k*-point* $\iff$ X *has a* k_v*-point for every* v*. In other words, "local-global principle" holds.*

However local gloabl principle is not true in general:

1. Hasse (1931): this can fail for more complicated X (norms from $\mathbb{Q}(\sqrt{-3}, \sqrt{13})$)
2. Lind, Geichardt (1940s): failure in the case of genus 1 curves (e.g. smooth projective model of $2Y^2 = 1 - 17X^4$, $3X^3 + 4Y^3 + 5Z^3 = 0$ in $\mathbb{P}^2_{\mathbb{Q}}$).
3. Swinnerton-Dyer (1962): failure for cubic surfaces (e.g. $5X^3 + 9Y^3 + 10Z^3 + 12W^3 = 0$ in $\mathbb{P}^3_{\mathbb{Q}}$)
4. Manin(1970) gives a systematic way to explain all these examples.

4.2 Brauer Group Over a Field

Let k be any field, k_s a separable closure, $\overline{k}$ an algebraic closure, $G_k = \mathrm{Gal}(k_s/k)$, A a finite dimensional associate k-algebra.

Proposition 4.2. *Under the above assumptions, TFAE:*

1. $A \otimes_k k_s \simeq \mathrm{Mat}_n(k_s)$ *for some* $n \geq 1$.
2. $A \otimes_k \overline{k} \simeq \mathrm{Mat}_n(\overline{k})$ *for some* $n \geq 1$.
3. A *is a central simple algebra over* k.
4. $A \simeq \mathrm{Mat}_r(D)$ *for some* $r \geq 1$ *and finite dimensional central division ring* D/k *(*r, D *are determined by* A*).*

Example. *The CSA over* $\mathbb{R}$ *are* $\mathrm{Mat}_n(\mathbb{R})$, $\mathrm{Mat}_n(\mathbb{H})$.

Example. *Assume* $\mathrm{char}(k) \neq 2$*, and* $a, b \in k^\times$*. Let's consider:*

$$H_{a,b} = k\langle i, j\rangle/(i^2 = a, j^2 = b, ij = -ji)$$

is four-dimensional CSA over k*. These are the quaternion algebras.*

Definition 4.3. Let A, B be CSAs over k. We call $A \sim B$ if there exists D and $r, s \geq 1$ such that $A \simeq \mathrm{Mat}_r(D), B \simeq \mathrm{Mat}_s(D) \iff \exists r, s \geq 1$, $\mathrm{Mat}_s(A) \simeq \mathrm{Mat}_r(B)$.

Definition 4.4. The **Brauer group** $\mathrm{Br}k \coloneqq \{\text{CSAs / k}\}/\sim$. The group structure is using $\otimes_k$, with identity k, and the inverse of A is A^{op}.

Remark. *In fact we have a functor* $\mathrm{Br} : \{\text{fields}\} \to \{\text{abelian groups}\}$*. If* $k \hookrightarrow L$*, we get* $\mathrm{Br}k \to \mathrm{Br}L, [A] \mapsto [A \otimes_k L]$.

Theorem 4.5. $\mathrm{Br}(k) \simeq H^2(G_k, k_s^\times)$.

Proof. (Sketched) The central simple algebras over k of dimension n^2 are parametrized by $H^1(G_k, \mathrm{Aut}\ M_n(k_s)) = H^1(G_k, \mathrm{PGL}_n(k_s))$. Now we have:

$$1 \to \mathbb{G}_m \to \mathrm{GL}_n \to \mathrm{PGL}_n \to 1$$

Hence we obtain a map:

$$1 \to H^1(\mathrm{PGL}_n(k_s)) \to H^2(k_s^\times)$$

Note that $H^1(\mathrm{PGL}_n(k_s))$ is not a group. However, one can see that the image of the above map lies in the n-torsion of $H^2(k_s^\times)$. Now let's union over all n, we get:

$$\{\mathrm{CSAs}/k\} \to H^2(k_s^\times)$$

We must check this descends to an isomorphism as claimed. □

Corollary 4.6. $\mathrm{Br}(k)$ *is a torsion abelian group.*

Proposition 4.7. *If $char(k) \nmid n$, then*

$$H^1(\mu_n) = k^\times / k^{\times n}$$

$$H^2(\mu_n) = \mathrm{Br}(k)[n]$$

Proof. Use

$$1 \to \mu_n \to k_s^\times \xrightarrow{n} k_s^\times \to 1$$

we get (by Hilbert 90 again):

$$0 \to k^\times \xrightarrow{n} k^\times \to H^1(\mu_n) \to 0 \to 0 \to H^2(\mu_n) \to \mathrm{Br}(k) \xrightarrow{n} \mathrm{Br}(k)$$

□

If $\mathrm{char}(k) \neq 2$, we have:

$$H^1(\mu_2) \times H^1(\mathbb{Z}/2\mathbb{Z}) \xrightarrow{\cup} H^2(\mu_2)$$

$$\iff k^\times/k^{\times 2} \times k^\times/k^{\times 2} \to (\mathrm{Br}k)[2]$$

$$(a, b) \mapsto [H_{a,b}]$$

Fact: Fix a. The map:

$$k^\times \to \mathrm{Br}(k), \quad b \mapsto [H_{a,b}]$$

has kernel $N_{k(\sqrt{a})/k} k(\sqrt{a})^\times$

Example. *If k is separably closed, then $\mathrm{Br}(k)$ is trivial.*

Example. *A theorem of Wedderburn:* $\mathrm{Br}(\mathbb{F}_q) = 0$

Example. *Tsen's theorem:* $\mathrm{Br}\mathbb{C}(t) = 0$. *This is in fact true for any k of transcendence degree 1 over an algebraically closed field.*

Example. *For local fields:*

1. $\mathrm{Br}(\mathbb{C}) = 0$, $\mathrm{Br}(\mathbb{R}) = \frac{1}{2}\mathbb{Z}/\mathbb{Z}$.
2. $\mathrm{Br}(\mathbb{Q}_p) \simeq \mathbb{Q}/\mathbb{Z}$. *More generally,* $\mathrm{Br}(k_v) \simeq \mathbb{Q}/\mathbb{Z}$ *for any global field k and nonarchimedean v.*

Example. *If k is a global field, we have an exact sequence:*

$$0 \to \mathrm{Br}(k) \to \bigoplus_v \mathrm{Br}(k_v) \xrightarrow{\Sigma} \mathbb{Q}/\mathbb{Z} \to 0$$

This is a by-product of the cohomological approach to the class field theory. Sometimes this is called "Brauer reciprocity".

4.3 Étale Cohomology

Let $f : Y \to X$ with $y \in Y$, and $k(y)$ the residue field at y.

Definition 4.8. The notion f is étale at y is characterized by the following:

1. (special case) Suppose I have $f : \operatorname{Spec} A[t_1, \cdots, t_n]/(g_1, \cdots, g_n) \to \operatorname{Spec} A$, then f is étale at $y \iff$ the the matrix $\frac{\partial g_i}{\partial t_j}(y) \in \operatorname{Mat}_n(k(y))$ is invertible.

2. (general case) f is étale at y iff and only if there exists open neighborhood Y' of Y of y, and X' of $f(y) \in X$, such that $f : Y' \to X'$ looks like the special case.

We call $f : Y \to X$ if it is étale at each $y \in Y$. Let Ét_X be the category of étale spaces above X.

Definition 4.9. An étale cover of V is a collection of étale morphisms $f_i : U_i \to V$ such that $\bigcup f_i(U_i) = V$.

Definition 4.10. The small étale site is defined to be:

$$X_{\text{ét}} \coloneqq (\text{Ét}_X, \{\text{étale covers of objects}\})$$

A presheaf on $X_{\text{ét}}$ is a functor:

$$\mathcal{F} : \text{Ét}_X^{\text{op}} \to \mathsf{Sets}$$

A sheaf on $X_{\text{ét}}$ is a presheaf satisfying the glue condition, for each étale cover of each V.

Example. *The sheaf $\mathbb{G}_m : V \to \mathcal{O}(V)^\times$ is an étale sheaf.*

We have a global section functor:

$$\text{Shv}_{\text{ét}}(X) \to \mathsf{Ab}$$

So that we have étale cohomology group:

$$H^i_{\text{ét}}(X, -)$$

Example. *Let $X = \operatorname{Spec} k$. An object in $\text{Ét}_{\operatorname{Spec} k}$ is just $\coprod_{i \in I} \operatorname{Spec} L_i$ of finite separable extensions of k. A sheaf $\mathcal{F} \in \text{Shv}(X_{\text{ét}})$ is given by the data $\mathcal{F}(L) \coloneqq \mathcal{F}(\operatorname{Spec} L)$. To record this, we can consider:*

$$\mathcal{F}(k_s) \coloneqq \varinjlim_{L/k} \mathcal{F}(L)$$

This is a Galois module. Then:

$$H^i_{\text{ét}}(X, \mathcal{F}) = H^i(G_k, \mathcal{F}(k_s))$$

4.4 Brauer Groups of Schemes

Let X be a scheme.

Definition 4.11. An **Azumaya algebra** on X is an $\mathcal{O}_X$-algebra $\mathcal{A}$ such that:

1. $\mathcal{A}$ is coherent as an $\mathcal{O}_X$-module (analogue of finite dimensional).
2. $\mathcal{A}$ is étale locally a matrix algebra. Explicitly, there exists an étale cover $(U_i \to X)_{i\in I}$ such that for each i, there exists $r_i \geq 1$ and an isomorphism:

$$\mathcal{A}|_{U_i} \simeq \mathrm{Mat}_{r_i}(\mathcal{O}_{U_i})$$

Remark. *An equivalent definition is to require $\mathcal{A}$ to be locally free of positive rank, and that:*

$$\mathcal{A} \otimes_{\mathcal{O}_X} \mathcal{A}^{\mathrm{op}} \to \mathcal{E}nd\mathcal{A}$$

to be an isomorphism.

Definition 4.12. Let $\mathcal{A}, \mathcal{A}'$ be Azumaya algebras. We call $\mathcal{A} \sim \mathcal{A}' \iff \mathcal{A} \otimes_{\mathcal{O}_X} \mathcal{E}nd_{\mathcal{O}_X}\mathcal{E} \simeq \mathcal{A}' \otimes_{\mathcal{O}_X} \mathcal{E}nd_{\mathcal{O}_X}\mathcal{E}'$ for some locally free coherent $\mathcal{O}_X$-modules $\mathcal{E}, \mathcal{E}'$ of positive rank at each $x \in X$.

Definition 4.13. We define:

$$\mathrm{Br}_{\mathrm{Az}}X \coloneqq \{\text{Azumaya algebras on } X\}/\sim$$

Note this is a functor:

$$\mathrm{Br}_{\mathrm{Az}} : \mathsf{Sch}^{\mathrm{op}} \to \mathsf{Ab}$$

Example. *Assume $2 \in \mathcal{O}_X^\times$. Let's take $f, g \in \mathcal{O}_X^\times$. Then:*

$$\mathcal{O}_X\langle i, j\rangle/(i^2 = f, j^2 = g, ij = -ji)$$

This is an Azumaya algebra on X.

Definition 4.14. The cohomological Brauer group is:

$$X \coloneqq H^2_{\text{ét}}(X, \mathbb{G}_m)$$

We have a natural transformation:

$$\mathrm{Br}_{\mathrm{Az}}(X) \to \mathrm{Br}X$$

Theorem 4.15 (Gabber, de Jong)**.** *Let X be quasi-projective over an affine scheme, then:*

$$\mathrm{Br}_{\mathrm{Az}}X \simeq (\mathrm{Br}X)_{\mathrm{tors}}$$

Proposition 4.16. *Let X be a regular integral noetherian scheme, with function field $k(X)$. Then:*

1. $\mathrm{Br}X \to \mathrm{Br}k(X)$ *is injective.*

2. $\mathrm{Br}X$ *is therefore torsion.*

Corollary 4.17. *If X is regular quasi-projective vareity over a field, then:*

$$\mathrm{Br}_{\mathrm{Az}}X \simeq (\mathrm{Br}X)_{\mathrm{tors}} \simeq \mathrm{Br}X$$

Corollary 4.18. *If $k = \overline{k}$, and X is a nice curve over k. Then* $\mathrm{Br}X = 0$

Proof. $\mathrm{Br}k(X) = 0$ (this is Tsen's theorem). □

Proposition 4.19. *If R is a complete DVR, with residue field k. Then*

$$\mathrm{Br}R \to \mathrm{Br}k$$

is an isomorphism.

Proof. It's an instance of Hensel's lemma. □

Corollary 4.20. *Let k be a global field.*

1. *If v is nonarchimedean, then* $\mathrm{Br}\mathcal{O}_v = 0$.

2. *The image of:*
$$\mathrm{Br}(k) \to \prod \mathrm{Br}(k_v)$$
lies in the direct sum: $\bigoplus \mathrm{Br}(k_v)$.

Proof. (of second part) For a finite set S of places, let $\mathcal{O}_{k,S} \coloneqq \{x \in k \mid v(x) \geq 0, \forall v \notin S\}$. Then:

$$\mathcal{O}_{k,S} \to \prod_{v \in S} k_v \times \prod_{v \notin S} \mathcal{O}_v$$

So:

$$\mathrm{Br}\mathcal{O}_{k,S} \to \prod_{v \in S} \mathrm{Br}k_v$$

Take direct limit:

$$\mathrm{Br}k \to \bigoplus_v \mathrm{Br}k_v$$

□

Now let's introduce Hochschild spectral sequence. Let X be a nice k-variety. Let $X_s \coloneqq X \times_k k_s$. Let $G = \mathrm{Gal}(k_s/k)$. Then we have a spectral sequence:

$$E_2^{pq} \coloneqq H^p(G, H^q_{\text{ét}}(X_s, \mathbb{G}_m)) \Longrightarrow H^{p+q}_{\text{ét}}(X, \mathbb{G}_m)$$

Remark. *The way to remember this is to specialze to $p = q = 0$.*

Combine with:

$$H^0_{\text{ét}}(X_s, \mathbb{G}_m) = k_s^\times$$
$$H^1_{\text{ét}}(X_s, \mathbb{G}_m) = \text{Pic}(X_s)$$
$$H^2_{\text{ét}}(X_s) = \text{Br}X_s$$

and similarly for X. Moreover:

$$H^1(G, k_s^\times) = 0$$
$$H^2(G, k_s^\times) = \text{Br}k$$

Combining all these, we have a long exact sequence:

$$0 \to \text{Pic}X \to (\text{Pic}X_s)^G \to \text{Br}k \to \text{Br}_1X \to H^1(G, \text{Pic}X_s) \to H^3(G, k_s^\times)$$

where $\text{Br}_1X = \ker(\text{Br}X \to \text{Br}X_s)$ is the "algebraic part of $\text{Br}X$". If k is local / global field, then $H^3(G, k_s^\times) = 0$ (this is a by-product from CFT).

If X is normal integral noetherian scheme, I have an exact sequence:

$$1 \to \mathcal{O}_X^\times \to k(X)^\times \to \bigoplus_{x \text{ codim 1 points}} \mathbb{Z}$$

Theorem 4.21. *If X is a regulaar integral noetherian scheme, then there exists an exact sequence:*

$$0 \to \text{Br}X \to \text{Br}k(X) \xrightarrow{\text{res}} \bigoplus_{x \text{ codim 1 points}} H^1(k(x), \mathbb{Q}/\mathbb{Z})$$

except that one should exclude p-primary parts if some $k(x)$ is imperfect of characteristic p.

Corollary 4.22. *Let (U_i) be an Zariski open cover for X, then:*

$$\text{Br}X = \bigcap_i \text{Br}U_i$$

Theorem 4.23 (Purity for the Brauer group, conjectured by Grothendieck 1968, proved by Česnavičius 2019 using perfectoid methods)**.** *Let X be regular integral noetherian scheme. Let $Z \subset X$ of codimension ≥ 2, and $U = X \setminus Z$. Then:* $\text{Br}X \simeq \text{Br}U$.

Corollary 4.24. *Let X, Y be nice k-varieties, birational. Then* $\text{Br}X \simeq \text{Br}Y$.

Proof. There is $U \subset X$ with $\text{codim}(X - U) \geq 2$, such that there is a well-defined map $U \to Y$. So $\text{Br}Y \to \text{Br}U = \text{Br}X \hookrightarrow \text{Br}k(X) = \text{Br}k(Y)$. Thus $\text{Br}Y \subset \text{Br}X$. Similarly $\text{Br}X \subset \text{Br}Y$. □

Proposition 4.25. *Let k be a field. Then* $\text{Br}\mathbb{P}^1_k \simeq \text{Br}k$.

Proof.

$$0 \to \text{Pic}(\mathbb{P}^1_k) \to \text{Pic}(\mathbb{P}^1_{k_s})^G \to \text{Br}k \to (\ker \text{Br}\mathbb{P}^1_k \to \text{Br}\mathbb{P}^1_{k_s}) \to H^1(G, \text{Pic}\mathbb{P}^1_{k_s})$$

Note:

$$\text{Pic}(\mathbb{P}^1_k) \simeq \text{Pic}(\mathbb{P}^1_{k_s})^G \simeq \mathbb{Z}$$
$$\text{Br}\mathbb{P}^1_{k_s} = 0$$
$$H^1(G, \text{Pic}\mathbb{P}^1_{k_s}) = H^1(G, \mathbb{Z}) = \text{Hom}(G, \mathbb{Z}) = 0$$

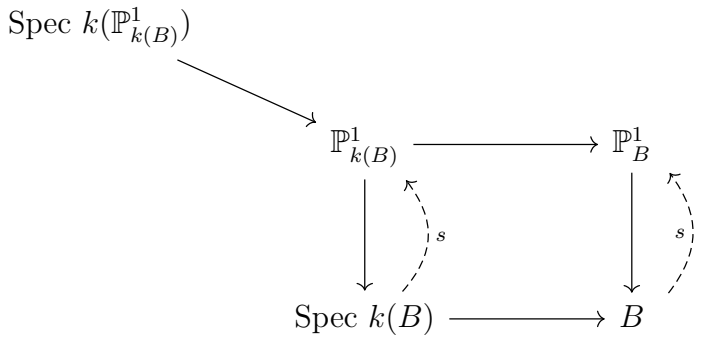
The result follows. □

Proposition 4.26. *Let B be a integral noetherian scheme. The natural map:*

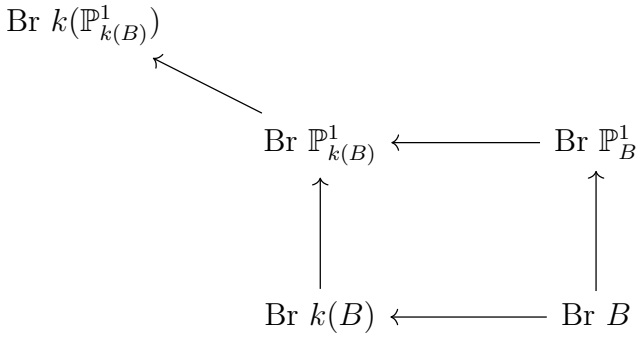
$$\mathrm{Br}(B) \to \mathrm{Br}(\mathbb{P}^1_B)$$

is an isomorphism.

Proof. Consider the diagram:



Apply Br, we get a diagram:



Note that $\mathrm{Br}(B) \to \mathrm{Br}(k(B))$ is injection, and $\mathrm{Br}(\mathbb{P}^1_B) \to \mathrm{Br}(\mathbb{P}^1_{k(B)}) \to \mathrm{Br}(k(\mathbb{P}^1_{k(B)}))$ is injection; and $\mathrm{Br}(k(B)) \to \mathrm{Br}(\mathbb{P}^1_{k(B)})$ is an isomorphism. This implies that $\mathrm{Br}(B) \to \mathrm{Br}(\mathbb{P}^1_B)$ is an injection. Conversely, the sections s will provide us with the map in the other direction (which is again injective). □

Corollary 4.27. *We have:*

$$\mathrm{Br}(\mathbb{P}^2_k) \simeq \mathrm{Br}(\mathbb{P}^1_k \times \mathbb{P}^1_k) \simeq \mathrm{Br}\mathbb{P}^1_k \simeq \mathrm{Br}k$$

Similarly $\mathrm{Br}\mathbb{P}^n_k \simeq \mathrm{Br}k$.

Let's now consider quadrics: degree 2 hypersurface in $\mathbb{P}^n_k$ for some $n \geq 2$.

Proposition 4.28. *Let X be a smooth quadric ove k. Then:*

$$\mathrm{Br}(k) \to \mathrm{Br}(X)$$

is surjective.

Proof. Note that $X_s \sim \mathbb{P}^1_{k_s}$, so:

$$\mathrm{Br}(X_s) \simeq \mathrm{Br}(\mathbb{P}^{n-1}_{k_s}) \simeq \mathrm{Br}(k_s) = 0$$

The Hochschild-Serre spectral sequence reads:

$$0 \to \mathrm{Pic}X \to (\mathrm{Pic}X_s)^G \to \mathrm{Br}k \to \ker(\mathrm{Br}X \to \mathrm{Br}X_s) = \mathrm{Br}X_s \to H^1(G, \mathrm{Pic}X_s)$$

Now:

$$\mathrm{Pic}X_s = \begin{cases} \mathbb{Z} \times \mathbb{Z}, & \text{if } \dim X = 2; \\ \mathbb{Z}, & \text{otherwise.} \end{cases}$$

There are not so many choices for the Galois action (it must preserve the ampleness of line bundles). It turns out that $H^1(G, \mathrm{Pic}X_s) = 0$. The result follows. □

We can genearlize this to quadratic bundles.

Proposition 4.29. *Let* $\mathrm{char}\ k = 0$. *Let* $\pi : X \to B$ *be a flat morphism of regular integral* k*-varieties. If:*

1. *Every fibre of* π *has an irreducible component of multiplicity one, that is geometrically irreducible.*
2. *The generic fibre* $X_{k(B)}$ *is a smooth quadric.*

Then $\mathrm{Br}(B) \to \mathrm{Br}(X)$ *is a surjection.*

Proof. Let's use:

$$0 \to \mathrm{Br}(B) \to \mathrm{Br}(k(B)) \xrightarrow{\mathrm{res}} \bigoplus_{b \in B^{(1)}} H^1(k(b), \mathbb{Q}/\mathbb{Z})$$

Compare with:

$$0 \to \mathrm{Br}(X) \to \mathrm{Br}(X_{k(B)}) \xrightarrow{\mathrm{res}} \bigoplus_{b \in B^{(1)}} \bigoplus_{x \in X^{(1)}, \pi(x) = b} H^1(k(x), \mathbb{Q}/\mathbb{Z})$$

For each $b \in B^{(1)}$, let x be the generic point of geometrically irreducible component of multiplicity one in X_b. Then $k(b)$ is separably closed in $k(x)$. Thus: $G_{k(x)} \to G_{k(b)}$ is surjective. Therefore:

$$\bigoplus_{b \in B^{(1)}} H^1(k(b), \mathbb{Q}/\mathbb{Z}) \hookrightarrow \bigoplus_{b \in B^{(1)}} \bigoplus_{x \in X^{(1)}, \pi(x) = b} H^1(k(x), \mathbb{Q}/\mathbb{Z})$$

$$\mathrm{Br}k(B) \to \mathrm{Br}X_{k(B)} \to 0$$

It follows that $\mathrm{Br}(B) \to \mathrm{Br}(X)$ is a surjection. □

4.5 Brauer-Manin Obstruction

Let k be a field, and X a k-variety. Let $\alpha \in \mathrm{Br}(X)$. If $x \in X(k)$, then:

$$x : \mathrm{Spec}\ k \to X$$

Apply Br:

$$\mathrm{Br}X \to \mathrm{Br}k, \alpha \mapsto \alpha(x) = \mathrm{ev}_\alpha(x)$$

In other word, we have a map:

$$\mathrm{ev}_\alpha : X(k) \to \mathrm{Br}k$$

Similarly for $x \in X(L)$ for a field extension L/k, we get $\alpha(x) = \mathrm{ev}_\alpha(x) \in \mathrm{Br}(L)$.

Now assume k to be a global field, X a nice k-variety, $\alpha \in \mathrm{Br}(X)$.

Proposition 4.30. *The evaluation:*

$$\mathrm{ev}_\alpha : X(k_v) \to \mathrm{Br}(k_v)$$

is zero for all but finitely many v.

Proof. Spread out X to a proper scheme $\mathfrak{X}$ over some $\mathcal{O}_{k,S}$, such that α comes from some $\mathfrak{a} \in \mathrm{Br}(\mathfrak{X})$. For places $v \notin S$, we have:

$$\begin{array}{ccc} \mathfrak{X}(\mathcal{O}_v) & \xrightarrow{\mathrm{ev}_\mathfrak{a}} & \mathrm{Br}\ \mathcal{O}_v = \mathrm{Br}\ \mathbb{F}_v = 0 \\ \downarrow \sim & & \downarrow \\ \mathfrak{X}(k_v) & \xrightarrow{\mathrm{ev}_\alpha} & \mathrm{Br}\ k_v \end{array}$$

□

Recall we have:

$$0 \to \mathrm{Br}(k) \to \bigoplus_v \mathrm{Br}(k_v) \xrightarrow{\sum} \mathbb{Q}/\mathbb{Z} \to 0$$

This fits into the diagram:

$$\begin{array}{ccccccccc} & & X(k) & \hookrightarrow & X(\mathbb{A}) = \prod_v X(k_v) & & & & \\ & & \downarrow{\scriptstyle \mathrm{ev}_\alpha} & & \downarrow{\scriptstyle \mathrm{ev}_\alpha} & \searrow^{\varphi_\alpha} & & & \\ 0 & \longrightarrow & \mathrm{Br}\ k & \longrightarrow & \bigoplus_v \mathrm{Br}(k_v) & \longrightarrow & \mathbb{Q}/\mathbb{Z} & \longrightarrow & 0 \end{array}$$

Now the commutativity of the diagram tells us:

$$X(k) \subset X(\mathbb{A})^\alpha \coloneqq \varphi_\alpha^{-1}(0)$$

We thus define the **Brauer set**:

$$X(\mathbb{A})^{\mathrm{Br}} \coloneqq \bigcap_{\alpha \in \mathrm{Br}(X)} X(\mathbb{A})^\alpha$$

Definition 4.31. We say "there is a Brauer-Manin obstruction to the local-global principle for X" means: $X(\mathbb{A}) \neq \emptyset$ (i.e. X has a k_v point for each place v), but $X(\mathbb{A})^{\mathrm{Br}} = \emptyset$. In this case $X(k) = \emptyset$, and local-global principle fails.

Example (Iskovskikh's conic bundle, 1971)**.** *Consider:*

$$U : y^2 + z^2 = (3 - x^2)(x^2 - 2) \subset \mathbb{A}^3_{\mathbb{Q}}$$

Let's construct a smooth projective model X. Let's start with some generalities:

1. *By a conic over k we mean the locus $\{s = 0\} \subset \mathbb{P}^2_k$ for some nonzero quadratic form s. We view $\mathbb{P}^2_k = \mathbb{P}E = \mathrm{Proj}\ \mathrm{Sym}\ E$ and thus $s \in \mathrm{Sym}^2 E$.*

2. *A conic bundle over a base B is the locus $\{s = 0\} \subset \mathbb{P}\mathcal{E}$ for some rank 3 vector bundle $\mathcal{E}$, and $s \in \Gamma(B, \mathrm{Sym}^2\mathcal{E})$ vanishing nowhere on B.*

3. *In the diagonal case, we can decompose:*

$$\mathcal{E} = \mathcal{L}_0 \oplus \mathcal{L}_1 \oplus \mathcal{L}_2$$

$$s = s_0 \oplus s_1 \oplus s_2, \quad s_i \in \Gamma(B, \mathcal{L}_i^{\otimes 2})$$

4. *Châtlet surface. Given $a \in k^\times$ with $\mathrm{char}(k) \neq 2$, and $P(x) \in k[x]$ separable of degree 4, we look at:*

$$U : y^2 - az^2 = P(x) \subset \mathbb{A}^3$$

Suppose $\widetilde{P}(w, x) = w^4 P(\frac{x}{w})$. Now take:

$$B = \mathbb{P}^1, \mathcal{L}_0 = \mathcal{L}_1 = \mathcal{O}, \mathcal{L}_2 = \mathcal{O}(2)$$

with $s_0 = 1, s_1 = -a, s_2 = -\widetilde{P}(w, x)$ to form $X \supset U$, a projective model for U.

Proposition 4.32. *The Iskovskikh's surface X has a $\mathbb{Q}_p$-point for all $p \leq \infty$*

Proof. Use Lang-Weil to get a smooth $\mathbb{F}_p$-point for p sufficiently large and use Hensel's lemma to lift it to a $\mathbb{Q}_p$ point. For example, let's fix $x = 1$, we get $y^2 = z^2 = -2$, which has a smooth $\mathbb{F}_p$ point for all $p \geq 3$. Now we only need:

$$(0, 1, \sqrt{-7}) \in X(\mathbb{Q}_2), (\sqrt{2}, 0, 0) \in X(\mathbb{R})$$

□

Remark. *Lang-Weil says the following. If X is a geometrically irreducible variety over $\mathbb{F}_q$ of dimension d. Then $\#X(\mathbb{F}_q) = q^d + O(q^{d-\frac{1}{2}})$, the big O term only depends on geometry of X.*

Let $K = k(X)$. We know $\mathrm{Br}(X) \subset \mathrm{Br}(K)$. Let $A \coloneqq (3 - x^2, -1) \in \mathrm{Br}(K)$. We claim that $A \in \mathrm{Br}(X)$. Consider:

$$B \coloneqq (x^2 - 2, -1) \in \mathrm{Br}(K)$$

$$C \coloneqq (\frac{3}{x^2} - 1, -1) \in \mathrm{Br}(K)$$

Then $A + B = ((3 - x^2)(x^2 - 2), -1) = (y^2 + z^2, -1) = 0$ since $y^2 + z^2$ is a norm of some element in a quadratic extension. Also, $A - C = (x^2, -1) = 0$. Since they're all 2-torsions, we have $A = B = C \in \mathrm{Br}(K)$. Now A extends to $U_A = X - X_\infty - X_{3-x^2}$, for X_∞ the fibre above $\infty \in \mathbb{P}^1$, X_{3-x^2} the fibre above the point defined by $3 - x^2 = 0$. Similarly:

$$U_B = X - X_\infty - X_{x^2-2}$$

$$U_C = X - X_0, X_{3-x^2}$$

Since U_A, U_B, U_C cover X, we see that $A \in \mathrm{Br}(X)$. Our next goal is to compute $X(\mathbb{A})^A$. We need to understand:

$$X(\mathbb{Q}_p) \to \mathrm{Br}(\mathbb{Q}_p) \hookrightarrow \mathbb{Q}/\mathbb{Z}$$

$$P \mapsto A(P) \mapsto \mathrm{inv}_p A(P)$$

For example, if $P \in U_A(\mathbb{Q}_p)$, then:

$$\mathrm{inv}_p A(P) = \begin{cases} 0, & \text{if } 3 - x(P)^2 \in N_{\mathbb{Q}_p(\sqrt{-1})/\mathbb{Q}_p} \mathbb{Q}_p(\sqrt{-1})^\times; \\ \frac{1}{2}, & \text{otherwise.} \end{cases}$$

Let $x = x(P) \in \mathbb{Q}_p(\infty)$.

1. If $p \notin \{2, \infty\}$. If $v(x) < 0$ or $x = \infty$, then $\frac{3}{x^2} - 1 \in \mathbb{Z}_p^\times$. If $v(x) \geq 0$, then either $3 - x^2$ or $x^2 - 2$ is in $\mathbb{Z}_p^\times$ (their sum is 1). So in either cases, we have: $A(P) = (u_1, u_2)$ with $u_1, u_2 \in \mathbb{Z}_p^\times$. So $A(P) \in \mathrm{Br}(\mathbb{Z}_p) = 0 \implies \mathrm{inv}_p A(P) = 0$.

2. If $P \in X(\mathbb{R})$, then $2 \leq x^2 \leq 3$. Thus either $3 - x^2$ or $x^2 - 2$ is positive, hence a square in $\mathbb{R}^\times$. Thus $\mathrm{inv}_\infty A(P) = 0$.

3. Consider $p = 2$. If $v_2(x) > 0$, then $3 - x^2 \equiv 3 \equiv -1 (\mod 4)$; if $v_2(x) = 0$, then $x^2 - 2 \equiv -1 (\mod 4)$; if $v_2(x) < 0$, then $\frac{3}{x^2} - 1 \equiv -1 (\mathrm{mod}\ 4)$. But an element in $\mathbb{Z}_2$ that is -1 mod 4 is not a norm from $\mathbb{Q}_2(\sqrt{-1})/\mathbb{Q}_2$. So $\mathrm{inv}_2 A(P) = \frac{1}{2}$

Therefore the map:

$$\varphi_A : X(\mathbb{A}) \to \mathbb{Q}/\mathbb{Z}$$

is the constant function $\frac{1}{2}$, so $X(\mathbb{A})^A = \emptyset$. Hence:

$$X(\mathbb{Q}) \subset X(\mathbb{A})^{\mathrm{Br}} = \emptyset$$

4.6 Descent Obstruction

Let k be a number field. Let G be a smooth affine group scheme of fintie type over k. We know the (right) G-torsors are classified by $H^1(X, G)$ (nonabelian Čech cohomology). Let's fix a G-torsor $f : Y \to X$, this defines a map:

$$\varphi : X(k) \to H^1(k, G)$$

Hence:

$$X(k) = \coprod_{[\sigma] \in H^1(k,G)} \{x \in X(k) \mid \varphi(x) = [\sigma]\}$$

Now we know:

$$\{x \in X(k) \mid \varphi(x) = 1\} = f(Y(k))$$

More generally, for $[\sigma] \in H^1(k, G)$, we get a twisted torsor $f^\sigma : Y^\sigma \to X$, we know:

$$\{x \in X(k) \mid \varphi(x) = [\sigma]\} = f^\sigma(Y^\sigma(k))$$

Therefore:

$$X(k) = \coprod_{[\sigma] \in H^1(k,G)} f^\sigma(Y^\sigma(k)) \subset \bigcup_{[\sigma]} f^\sigma(Y^\sigma(\mathbb{A}))$$

It turns out that $Y^\sigma(\mathbb{A})$ is nonempty for only finitely many $[\sigma]$. Let's denote:

$$X(\mathbb{A})^f = \bigcup_{[\sigma]} f^\sigma(Y^\sigma(\mathbb{A}))$$

and

$$X(k) \subset X(\mathbb{A})^{\text{descent}} := \bigcap_{G, f:Y \to X} X(\mathbb{A})^f$$

We can further consider:

$$X(\mathbb{A})^{\text{ét,Br}} := \bigcap_{\text{finite étale } G, f:Y \to X} \bigcup_{[\sigma] \in H^1(k,G)} f^\sigma(Y^\sigma(\mathbb{A})^{\text{Br}})$$

It turns out that:

$$X(k) \subset X(\mathbb{A})^{\text{descent}} = X(\mathbb{A})^{\text{ét,descent}} = X(\mathbb{A})^{\text{ét,Br}} \subset X(\mathbb{A})^{\text{Br}} \subset X(\mathbb{A})$$

Each inclusion is strict for some X.

Theorem 4.33 (Poonen 2010)**.** *There exists nice $\mathbb{Q}$-variety X such that $X(\mathbb{A})^{\text{ét,Br}} \neq \emptyset$, but $X(\mathbb{Q}) = \emptyset$.*

Proof. The proof breaks into several steps.

1. Let C be a nice $\mathbb{Q}$-curve such that $C(\mathbb{Q}) = \{c\}$ (for example, take a nice modle of $y^2 = x^3 - 3$). Choose $f : C \to \mathbb{P}^1$ étale at c. WLOG we may assume $f(c) = \infty$. Choose a simply connected open neighborhood U of $c \in C(\mathbb{R})$, mapping to a simply connected open neighborhood $f(U)$ of $\infty \in \mathbb{P}(\mathbb{R})$. We may also assume $1 \in f(U)$ such that f is étale above $0, 1 \in \mathbb{P}^1$.

2. Now glue
$$Y^{(t)} : t(t-1)x_0^2 + \cdots x_4^2 = 0 \in \mathbb{P}^4 \times \mathbb{A}^1_t$$
and
$$Y^{(T)} : (1-T)X_0^2 + x_1^2 + \cdots x_4^2 = 0 \in \mathbb{P}^4 \times \mathbb{A}^1_T$$
using $t = \frac{1}{T}$, $x_0 = X_0 T$ to obtain a quadric bundle $Y/\mathbb{P}^1$. This has two degenerate fibres, above 0 and 1.

3. Now consider the fibre product $X = Y \times_{\mathbb{P}^1} C$. This is a quadric bundle over X. Note $Y \to \mathbb{P}^1$ is projective with geometrically integrable fibres, so the same holds for $X \to C$. Since $Y \to \mathbb{P}^1$ has smooth fibres outside $0, 1$; but $C \to \mathbb{P}^1$ is smooth above $0, 1$. Therefore X is smooth over C or Y at each point. This implies X is smooth.

4. Let's now show that $X(\mathbb{Q}) = \emptyset$. Any $\mathbb{Q}$-point of X mapsto $c \in C(\mathbb{Q})$, and to ∞ at $\mathbb{P}^1$. But the fibre Y_∞ is:
$$Y_0^{(T)} : X_0^2 + x_1^2 + \cdots + x_4^2 = 0 \in \mathbb{P}^4$$
which has no $\mathbb{Q}$-points.

5. Every quadratic form of rank ≥ 5 over $\mathbb{Q}_p$ has a nontrivial zero, so choose $y_p \in Y_\infty(\mathbb{Q}_p)$ for each prime p. Let $x_p = (y_p, c) \in X(\mathbb{Q}_p)$. Let $y_\mathbb{R}$ be the point of $Y_1(\mathbb{R})$. Let $c_\mathbb{R} \in U$ be such that $f(c_\mathbb{R}) = 1$ in $\mathbb{P}^1(\mathbb{R})$. Let $x_\mathbb{R} = (y_\mathbb{R}, c_\mathbb{R}) \in X(\mathbb{R})$. Finally let's put:
$$x = (x_\mathbb{R}, x_2, x_3, \cdots) \in X(\mathbb{A})$$

6. We claim that $x \in X(\mathbb{A})^{\mathrm{Br}}$. Suppose that $A \in \mathrm{Br}(X)$. Since $\pi : X \to C$ is a quadric bundle, so $\mathrm{Br}(C) \to \mathrm{Br}(X)$ is surjective. Choose $A_c \in \mathrm{Br}(C)$ mapping to A. Then:
$$\sum_{p \leq \infty} \mathrm{inv}_p A(x_p) = \sum_{p \leq \infty} \mathrm{inv}_p A_c(\pi(x_p)) = \sum_{p \leq \infty} \mathrm{inv}_p A_C(c) = 0$$

7. Finally let's prove $x \in X(\mathbb{A})^{\text{ét,Br}}$. The fibres of $X \to C$ are simply connected. So every étale cover of X is the pullback of an étale cover of C. Let G be a finite étale group scheme over $\mathbb{Q}$. Let $X' \to X$ be a G-torsor. It is the pullback of a G-torsor $C' \to C$. Replace C' by a twist to assume $c \in C(\mathbb{Q})$ lifts to some $c'' \in C'(\mathbb{Q})$. Let C'' be the irreducible component of C' containing c''. Now C'' is smooth projective irreducible with a $\mathbb{Q}$-point (so C'' is a nice curve). It follows that $X'' = X \times_C C''$ is nice too. We claim x lifts to some $x'' \in X''(\mathbb{A})$. For each p, we choose $x''_p = (x_p, c'') \in X''(\mathbb{Q}_p)$. Since U is simply connected, its inverse image in $C''(\mathbb{R})$ is a disjoint union of copies of U. Therefore, we may let $c''_\mathbb{R} \in C''(\mathbb{R})$ to be the preimage of $c_\mathbb{R}$ in the same copy as c'' . Let $x''_\mathbb{R} = (x_\mathbb{R}, c''_\mathbb{R})$. So I get:
$$x'' = (x''_\mathbb{R}, x''_2, x''_3, \cdots) \in X''(\mathbb{A})$$
As in the previous step, we can show:
$$x'' \in X''(\mathbb{A})^{\mathrm{Br}} = X''(\mathbb{A}) \cap X'(\mathbb{A})^{\mathrm{Br}}$$
In particular $X'(\mathbb{A})^{\mathrm{Br}}$ has a point (namely, x'') mapping to x. This holds for all finite étale covers, so $x \in X(\mathbb{A})^{\text{ét,Br}}$.

□

Conjecture 4.34 (Colliot,Thélène, Kanevsky, Sansuc)**.** *If X is a ncie (geometrically) rationally connected over a number field, and $X(\mathbb{A})^{\mathrm{Br}} \neq \emptyset$, then $X(k) \neq \emptyset$.*

Remark. *It is proved for Châtelet surfaces.*

5 Cluster Algebras and Applications to Arithmetics

5.1 Origins and Basic Notions

Definition 5.1 (Classical frieze)**.** A frieze is the central section of an entablatune above Greek columns.

Definition 5.2 ("Classical" frieze)**.** A frieze pattern (or frieze) is an array $\{x_{i,j} \in \mathbb{N} \mid x_{i,j}x_{i,j+1} - x_{i-1,j+1}x_{i+1,j} = 1\}$ with positive integers, with $n \in \mathbb{N}$ rows, infinitly many columns, with an SL_2 unimodular condition.

Proposition 5.3 (Gauss, 1866)**.** *All friezes with 2 rows are periodic with period* 5*.*

Theorem 5.4 (Coxeter, 1971)**.** *All frieze with n rows are periodic with period $n+3$.*

Theorem 5.5 (Conway-Coxeter, 1973)**.** *Fix n, there are $\frac{1}{n+2}\binom{2n+2}{n+1}$ many friezes with n rows.*

Sketched. One should construct a bijection between friezes with n rows and triangulations of $(n+3)$-gons. For each $(n+3)$-gon, we count the number of edges connecting to each vertex, and use it to fill in the first row. □

Definition 5.6 (Dynkin Friezes)**.** Let Δ be a Dynkin type with generalized Cartan matrix C. A frieze of type Δ is an array:

$$\{x_{i,j} \in \mathbb{N} \mid x_{i,j}x_{i,j+1} - \prod_{k<i} x_{k,j+1}^{|C_{k,i}|} \prod_{k>i} x_{k,j}^{|C_{k,i}|} = 1\}$$

for $1 \leq i \leq n, j \in \mathbb{Z}$

Theorem 5.7. *A frieze of type Δ is periodic iff Δ is finite type; a frieze of type Δ has finitely many objects iff Δ is finite type. Moreover, we can explicitly enumerate these numbers. The number of frieze of type B_n equals:*

$$\sum_{m=1}^{[\sqrt{n+1}]} \binom{2n - m^2 + 1}{n}$$

The number of frieze of type C_n equals:

$$\binom{2n}{n}$$

The number of frieze of type D_n equals:

$$\sum_{m=1}^{n} d(m)\binom{2n-m-1}{n-m}$$

For exceptional types:

$$E_6 : 868; \quad E_7 : 4400; \quad E_8 : 26952; \quad F_4 : 112; \quad G_2 : 9.$$

Remark. *Most of these answers use cluster algebras; E_7, E_8 use Diophantine aspects of cluster varieties.*

Proposition 5.8. *We have a bijection between friezes of type Δ and a cluster algebra over $\mathbb{Z}$ of type Δ and a homomorphism into $\mathbb{Z}$, such that we send cluster variables into $\mathbb{N}$.*

Definition 5.9 (Simplified)**.** A cluster algebra $\mathcal{A}$ is a commutative ring in $\mathbb{Q}(x_1, \cdots, x_n)$ given by the data of a mutation matrix $B = (b_{ij})$ (this is a Cartan matrix), with generators (called cluster variables) constructed following certain combinatorial rules:

$$x_k x_k' = \prod_{j: b_{jk} < 0} x_j^{|b_{kj}|} + \prod_{j: b_{jk} > 0} x_j^{|b_{kj}|}$$

Example. *Let $F = \mathbb{Q}, x = (x_1, x_2), B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ We proceed in several steps.*

1. *We mutate $\{x_1, x_2\} \mapsto \{x_1', x_2\}$, for:*

$$x_1' \coloneqq \frac{x_2 + 1}{x_1}$$

and flip th sign of the mutation matrix in column 1 and column 2:

$$B' = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

2. *We mutate $\{x_1', x_2\} \mapsto \{x_1', x_2'\}$, for:*

$$x_2' \coloneqq \frac{x_1 + 1}{x_2}$$

$$B'' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

3. *Keep doing this:*

$$x_1'' = \frac{x_2' + 1}{x_1'}, x_2'' = \frac{x_1'' + 1}{x_2'}$$

Note that $x_2'' = x_1$.

4. *In this example, I get:*

$$\mathcal{A} = \mathbb{Z}[x_1, x_2, \frac{x_2 + 1}{x_1}, \frac{x_1 + x_2 + 1}{x_1 x_2}, \frac{x_1 + 1}{x_2}] \subset \mathbb{Q}(x_1, x_2)$$

This is a cluster algebra with 5 cluster variables, and has type A_2.

5.2 The Classification

Definition 5.10. Let $B \in \mathrm{Mat}_n(\mathbb{Z})$ be a skew-symmetric (or skew-symmetrizable, i.e. can be written as a product of diagonal matrix with positive entries and a skew-symmetric matrix). Given the data of:

1. a seed $x = (x_1, \cdots, x_n)$.
2. a mutation matrix $B \in \mathrm{Mat}_n(\mathbb{Z})$ skew-symmetrizable.

For each index $k \in \{1, 2, \cdots, n\}$, we can define the k-th mutation to be:

$$x_i' = \mu_k(x_i) = \begin{cases} x_i, & \text{if } i \neq k; \\ \dfrac{\prod_{j:b_{jk}>0} x_j^{|b_{jk}|} + \prod_{j:b_{jk}<0} x_j^{|b_{jk}|}}{x_k}, & \text{if } i = k. \end{cases}$$

$$b_{ij}' = \begin{cases} -b_{ij}, & \text{if } i = k \text{ or } j = k; \\ b_{ij} + \dfrac{|b_{ik}|b_{kj} + b_{ik}|b_{kj}|}{2}, & \text{otherwise.} \end{cases}$$

The **cluster algebra** $\mathcal{A}/\mathbb{Q}$ associated to (x, B) is the $\mathbb{Z}$-subalgebra of $\mathbb{Q}(x)$ generated by all possible cluster variables.

Definition 5.11. A cluster algebra is of **finite type** iff there are only finitely many distinct cluster variables.

Definition 5.12. A cluster algebra is **acyclic** if the directed graphs of all mutations of B have no oriented cycles. The directed graph is a graph obtained by drawing b_{ij} arrows from vertex i to vertex j whenever $b_{ij} > 0$.

Example. *Let* $B = \begin{pmatrix} 0 & 2 & -2 \\ -2 & 0 & 2 \\ 2 & -2 & 0 \end{pmatrix}$. *The mutations are Vieta jumps if we specialize to* $(x_1, x_2, x_3) \mapsto (1, 1, 1)$, *we get solutions to Markov equation:*

$$3xyz = x^2 + y^2 + z^2$$

Theorem 5.13 (Fomin-Zelevinsky, Laurent Phenomenon)**.** *Let* $\mathcal{A}$ *be a cluster algebra with initial seeds* (x, B). *Any cluster variable in* $\mathcal{A}$ *is a Laurent polynomial in* x.

The key ingredient of the proof is:

Lemma 5.14 (Intersection Lemma)**.** *Consider:*

$$x = (x_1, \cdots, x_n), \quad x' = (x_1, \cdots, x_{i-1}, y_i', x_{i+1}, \cdots, x_n)$$

and add the exchange relation:

$$x_i x_i' = P(x_1, \cdots, \widehat{x_i}, \cdots, x_n)$$

for P *a binomial in* $\mathbb{Z}[x_1, \cdots, \widehat{x_i}, \cdots, x_n]$ *not divisible by* x_j *for all* $j \neq i$. *Let* R *be the space of Laurent polynomials in variables excluding* x_i. *Then:*

$$R[x_i^{\pm 1}] \cap R[(x_i')^{\pm 1}] = R[x_i, x_i']$$

Theorem 5.15 (Positivity)**.** *Every cluster variable is a Laurent polynomial in $x_1, \cdots, x_n$ with non-negative coefficients.*

Remark. *The proof uses scattering diagrams from mirror symmetry.*

Definition 5.16. Let B be skew symmetric matrix. The Cartan companion of B is:

$$c_{ij} = \begin{cases} 2, & \text{if } i = j; \\ -|b_{ij}|, & \text{otherwise.} \end{cases}$$

Theorem 5.17 (Fomin-Zelevinsky, Classification)**.** *A cluster algebra $\mathcal{A}$ is of finite type iff some mutation of B has finite type Cartan companion.*

Proof. (Sketched). Let's start from direction $\implies$. Suppose we have infinite type Cartan companion. We can check by hand in the rank 2 case:

$$\begin{pmatrix} 0 & b \\ -c & 0 \end{pmatrix}$$

is finite type iff $|bc| \leq 3$. For higher ranks, in the indefinite case we can find a 2×2 submatrix which has infinite type. We then check by hands for all affine types. For the reverse direction, suppose B has finite type. Let Φ be the corresponding finite root system and $\Pi = \{\alpha_1, \cdots, \alpha_n\}$ simple roots. One can check that for every non-initial cluster variable z, we can write:

$$z = \frac{P(x_1, \cdots, x_n)}{x_1^{d_1} \cdots x_n^{d_n}}$$

for d_i not all zero, and $x_i \nmid P$. Now let's take the map:

$$z \mapsto \sum_{i=1}^{n} d_i \alpha_i$$

$$x_i \mapsto -\alpha_i$$

We have an inductive property for these d_k, and this will give us a bijection:

$$\{\text{cluster algebras of } \mathcal{A}\} \iff \Phi^+ \cup \{-\alpha_1, \cdots, -\alpha_n\}$$

□

5.3 Cluster Varieties

Definition 5.18. An (acyclic) cluster variety is an affine variety that is isomorphic to Spec $\mathcal{A}$ for some (acyclic) cluster algebra $\mathcal{A}$.

Remark. *More generally, a cluster variety is a collection of algebraic tori glued using exchange relations.*

Definition 5.19. Let $\mathcal{A}$ be a cluster algebra with initial seeds (x, B). The lower bound cluster algebra $\mathcal{L}$ of $\mathcal{A}$ is the subalgebra:

$$\mathbb{Z}[x_1, \cdots, x_n, y_1, \cdots, y_n] \subset \mathcal{A}$$

where $y_i = \mu_i(x)_i$ the first step mutation cluster variables. The upper bound cluster algebra $\mathcal{U}$ of $\mathcal{A}$ is the super-algebra:

$$\mathcal{A} \subset \bigcap_{\text{seeds in } \mathcal{A}} \mathbb{Z}[z_1^{\pm 1}, \cdots, z_n^{\pm 1}]$$

Proposition 5.20 (Bernstein-Fomin-Zelevinsky)**.** *There is a sandwich $\mathcal{L} \subset \mathcal{A} \subset \mathcal{U}$. If $\mathcal{A}$ is acyclic, then $\mathcal{L} = \mathcal{A} = \mathcal{U}$.*

Definition 5.21. Let C be an $n \times n$ generalized Cartan matrix. Let:

$$f_{C,i} \coloneqq x_i y_i - \prod_{j<i} x_j^{|c_{j,i}|} - \prod_{j>i} x_j^{|c_{j,i}|} \in \mathbb{Z}[x_1, \cdots, x_n, y_1, \cdots, y_n]$$

Let $Y_1^{\text{frieze}}(C) \coloneqq V(f_{C,1}, \cdots, f_{C,n})$ be an n-dimensional affine vareity in $\mathbb{A}^{2n}$.

Proposition 5.22. *Let $\mathcal{A}$ be acyclic of type C, then* $\operatorname{Spec} \mathcal{A} = Y_1^{\text{frieze}}(C)$.

Theorem 5.23. *There is a bijection:*

$$Y_1^{\text{frieze}}(C)(\mathbb{N}) \xrightarrow{\sim} \text{Frieze}(C, \mathbb{N})$$

Theorem 5.24. *Every cluster variety is log Calabi-Yau.*

Theorem 5.25. *There is a volume preserving automorphism*

$$\tau : Y_1^{\text{frieze}}(C) \to Y_1^{\text{frieze}}(C)$$

with period divides $h+2$ if C has finite type (where h is the Coxeter number); and the period is infinite otherwise.

Theorem 5.26. *$Y_1^{\text{frieze}}(C)(\mathbb{N})$ is finite type iff C is finite type. Furthermore: we can explicitly count $\#Y_1^{\text{frieze}}(C)(\mathbb{N})$ for each type C (details in the first section).*

Corollary 5.27. *Consider the equation:*

$$xyz = (x^a + 1)^b + y$$

It has $5, 6, 9$ positive integer solutions if $ab \leq 3$. It has infinitely many positive integer solutions otherwise.

Proof. We have a bijection:

$$Y_1^{\text{frieze}}(\begin{pmatrix} 2 & -a \\ -b & 2 \end{pmatrix})(\mathbb{N}) \simeq (xyz = (x^a + 1)^b + y)(\mathbb{N})$$

and use the previous theorem. □

6 Vector Bundles with Flat Connection

Let $k = \overline{k}$, X/k a smooth algebraic variety. A vector bundle with a flat connection is a pair $(\mathcal{E}, \nabla)$ for $\mathcal{E}$ a locally free finite rank $\mathcal{O}$-module, with $\nabla : \mathcal{E} \to \mathcal{E} \otimes \Omega^1$ with $\nabla \wedge \nabla = 0$.

Given a local section $v \in \mathrm{Vect}(X) = \mathrm{Der}(\mathcal{O}_X)$, we have $v \mapsto \nabla_v$ such that $[\nabla_v, \nabla_w] = \nabla_{[v,w]}$ (the connection is flat). Therefore, we get an action of the Lie algebra vector field on $\mathcal{E}$.

6.1 $\mathcal{D}$-modules over $\mathbb{C}$

If $k = \mathbb{C}$, we get a local system: a locally constant sheaf of ∇-parallel section of $\mathcal{E}$.

Example (Gauss-Manin Connection)**.** *Consider $f : Y \to X$ with f smooth projective. Then $R^i f_* \underline{\mathbb{C}}$ is a local system with fibre at x equals $H^i(Y_x, \mathbb{C})$. Note $H^i(Y_x)$ has pure Hodge structure:*

$$H^i(Y_x, \mathbb{C}) = \bigoplus_{p+q=i} H^q(\Omega^p_{Y_x})$$

This comes from a vector bundle with a flat connection, setting

$$H^i_{\leq n} = \bigoplus_{q \leq n, p+q=i} H^q(\Omega^p_{Y_x})$$

be fibres of an algebraic subbundle. Then we have:

$$\nabla : H^i_{\leq n} \to H^i_{\leq n+1}$$

which gives global action:

$$\nabla : (R^i f_* \underline{\mathbb{C}})_{\leq n} \to (R^i f_* \underline{\mathbb{C}})_{\leq n+1}$$

Remark. *Same defintion applies if $\mathcal{E}$ is a quasi-coherent sheaf. A quasi-coherent sheaf with a flat connection is a $\mathcal{D}$-module.*

Example. *Let $X = \mathbb{A}^n, \mathcal{D}_X = k\langle x_i, \partial_i \rangle / ([\partial_i, x_j] = \delta_{ij})$ is the Weyl algebra.*

Note that $\mathcal{D}$ comes with a filtration such that $\mathrm{gr}(\mathcal{D}_X) = \mathcal{O}(T^*X)$. Using Rees construction, can define $\mathcal{D}_\hbar(X)$ by imposing the relation: $vw - wv = \hbar[v, w]$, $v \cdot f - f \cdot v = \hbar \cdot v(f)$. This forms a flat family with $\mathcal{D}_{\hbar=1} = \mathcal{D}, \mathcal{D}_{\hbar=0} = \mathcal{O}(T^*X)$.

Remark. *Quasi-coherent sheaves on T^*X can be thought of a "Higges fields". Then we get:*

$$\overline{\nabla} : \mathcal{E} \to \mathcal{E} \otimes \Omega^1$$

a morphism of $\mathcal{O}$-modules, with $\overline{\nabla}^2 = 0$.

Example. *Suppose $(\mathcal{E}, \nabla)$ comes from $R^i f_* \mathbb{C}$ as before. Then:*

$$\mathrm{gr}(\mathcal{E}) = \bigoplus \mathcal{E}_{\leq n} / \mathcal{E}_{\leq n-1}$$

is a Higgs field. More generally, if $\mathcal{M}$ is a $\mathcal{D}$-module, given a filtration by submodules $\mathcal{M}_{\leq n}$ with:

$$v : \mathcal{M}_{\leq n} \to \mathcal{M}_{\leq n+1}$$

we thus get $\mathrm{gr}(\mathcal{M}) \in \mathsf{QCoh}(T^*X)$*. Ree's construction again tells me there exists* $\widetilde{\mathcal{M}}$ *such that* $\widetilde{\mathcal{M}}_1 = \mathcal{M}, \widetilde{\mathcal{M}}_0 = \mathrm{gr}(\mathcal{M})$*.*

We often assume $\bigcup \mathcal{M}_{\leq i} = \mathcal{M}$*, and* $\mathrm{gr}(\mathcal{M}) \in \mathrm{Coh}(T^*X)$*. This is called a good filtration.*

Let's discuss functoriality of $\mathcal{D}$-modules.

1. We can pull back $\mathcal{D}$-modules (in fact $\mathcal{D}_\hbar$-modules).
2. We cal also pushforward $\mathcal{D}$-modules. Look at

$$\begin{array}{ccc} Y & & T^*X \times_X Y \xrightarrow{\quad df \quad} T^*Y \\ \downarrow{\scriptstyle f} & & \downarrow \\ X & & T^*X \end{array}$$

assuming f smooth. This diagram quantizes. We get a $(f^{-1}\mathcal{D}_X, \mathcal{D}_Y)$-bimodule

$$\mathcal{D}_{Y\to X} = \mathcal{O}_X \otimes_{\mathcal{O}_Y} \mathcal{D}_Y$$

From this we get a family of functors:

(a) For $\hbar = 1$, $\mathcal{M} \mapsto R\pi_*(\mathcal{D}_{Y\to X} \otimes^{\mathbb{L}}_{\mathcal{D}_Y} \mathcal{M})$ we get a $f^{-1}\mathcal{D}_X$-module.

(b) For $\hbar = 0$, we get:

$$\mathrm{pr}_*(df^*) : D^b(\mathsf{QCoh}(T^*Y)) \to D^b(\mathsf{QCoh}(T^*X))$$

Remark. *The coherent sheaf* $\mathrm{gr}(\mathcal{E}) = H^i(\mathrm{pr}_* df^*(\mathcal{O}_Y))$*.*

6.2 $\mathcal{D}$-modules over Positive Characteristic

Let's now turn to the case when $\mathrm{char}(k) = p$. We have:

$$d(f^p) = 0, \quad v(f^p) = 0$$

Therefore if $\delta \in \mathrm{Der}(A)$, then $\delta^p \in \mathrm{Der}(A)$. Therefore we have:

$$\mathrm{Vect}(X) \to \mathrm{Vect}(X), v \mapsto v^{[p]}$$

Now $v^p - v^{[p]} \in \mathcal{D}_X$ acts by zero on $\mathcal{O}_X$. In fact $v^p - v^{[p]} \in Z(\mathcal{D}_X) \simeq \mathcal{O}(T^*X^{(1)})$. Moreover, $\mathcal{D}_X$ is an Azumaya algebra over $T^*X^{(1)}$ of rank $p^{2\dim X}$.

Example. *Suppose k is perfect, and consider:*

$$W_{\alpha,\beta} = k\langle x, \partial\rangle/([\partial, x] = 1, x^p = \alpha, \partial^p = \beta) \simeq \mathrm{Mat}_p(k)$$

Indeed we can reduce to the case $\alpha = \beta = 0$, and $W_{0,0}$ acts on $V = k[x]/x^p$.

If we consider $\mathcal{D}_X|_{X^{(1)}}$-action on $\mathcal{O}_X$, we obtain:

$$\mathcal{D}_X|_{X^{(1)}} \xrightarrow{\sim} \mathrm{End}_{\mathcal{O}_{X^{(1)}}}(\mathrm{Fr}_*\mathcal{O}_X)$$

Namely the Azumaya algebra splits on the zero section.

One can aslo see the splitting by considering:

$$m : H^0(\Omega^1) \to \mathrm{Br}[p]$$

and then $m(\theta) = [\mathcal{D}]$ for θ the Liouville form on T^*X.

Remark. *Ogus-Vologodsky: if X lifts to $W_2(k)$, then the Azumaya algebra splits on the p-neighborhood of zero section.*

If $\mathcal{M}$ is a a $\mathcal{D}_X$-module, then $\mathcal{M}$ lives on $T^*X^{(1)}$. If its support is on the zero section $X^{(1)}$, then $\mathcal{M} \simeq \mathrm{Fr}^*\mathcal{M}'$.

The action of $Z(\mathcal{D}_X) = \mathcal{O}_{T^*X^{(1)}}$ is related to *p-curvature*. Consider:

$$\mathrm{curv}_p(\nabla) : v \mapsto \nabla_v^p - \nabla_{v^{[p]}}$$

This is a section of $\mathrm{Fr}^*\Omega^1 \otimes \mathrm{End}(\mathcal{E})$. This section is flat in the canonical flat structure on $\mathrm{Fr}^*(\Omega^1)$. In this way we get a quasi-coherent sheaf on $T^*X^{(1)}$.

Suppose X is smooth projective defined over some number field K, with $(\mathcal{E}, \nabla)$ defined over K. Then $(X, (\mathcal{E}, \nabla))$ are defined over a finite localization of $\mathcal{O}_K$. For almost all prime $\mathfrak{p}$ with residue field $k_\mathfrak{p}$, we have $(X_{k_\mathfrak{p}}, (\mathcal{E}_{k_\mathfrak{p}}, \nabla_{k_\mathfrak{p}}))$. Grothendieck conjectured the following:

Conjecture 6.1. *If for almost all p, $(\mathcal{E}_{k_\mathfrak{p}}, \nabla_{k_\mathfrak{p}})$ has zero p-curvature, then $(\mathcal{E}_\mathbb{C}, \nabla_\mathbb{C})$ gives a local system with finite monodromy.*

Remark. *If $(\mathcal{E}_\mathbb{C}, \nabla_\mathbb{C})$ has regular singularity, then the converse is also true. The local system trivializes on a finite analytic cover, hence on an algebraic cover. Riemann-Hilbert tells me $(\mathcal{E}, \nabla)$ also trivializes.*

Remark. *The conjecture is known for rank 1 connections (Honda etc.) and Gauss-Manin connections (N. Katz 1972)*

Question: What can we say about p-support of $(\mathcal{E}_{k_\mathfrak{p}}, \nabla_{k_\mathfrak{p}})$ as p varies.

It is known that for almost all p, p-support is Lagrangian.

Lemma 6.2. *Let $\mathcal{E} = \mathcal{O}$, $\nabla = d + \alpha$ with $\alpha \in \Omega^1$. Then:*

$$\mathrm{curv}_p = \alpha - C(\alpha)$$

for C the Cartier isomorphism.

Recall the Cartier isomorphism is given by:

$$\Omega^1/d\mathcal{O} \simeq \Omega^1_{X^{(1)}}$$

Example. *Take $X = \mathbb{A}^1_t$, then $\Omega^1_{\text{cl}} = \langle t^i dt \rangle$. So that:*

$$\text{coker } (\mathcal{O} \xrightarrow{d} \Omega^1) = \langle t^{pi-1} dt \rangle$$

The Cartier isomorphism sends $t^{pi-1}dt \mapsto T^{i-1}dT$. The Cartier isomorphism has properties:

1. $C(f^p \alpha) = fC(\alpha)$.
2. $C(df) = 0$.
3. $C(\frac{df}{f}) = \frac{df}{f}$.

Example. *Let $X = \mathbb{A}^1$ and $\mathcal{E} = \mathcal{O}$, $\alpha = dt$. Then $C(\alpha) = 0$ and thus:*

$$\text{curv}_p(\mathcal{E}, \nabla) = dT$$

Example. *Let $X = \mathbb{G}_m$ and $\mathcal{E} = \mathcal{O}$, $\alpha = \lambda \frac{dt}{t}$. Then:*

$$\text{curv}_p(\mathcal{E}, \nabla) = (\lambda^p - \lambda)\frac{dT}{T}$$

Example. *If E is an elliptic curve over $\mathbb{Q}$ and ω is a one-form on E. Then:*

$$\omega_p - C(\omega_p) = 0 \iff p \mid |E(\mathbb{F}_p)|$$

Therefore for large p, $(\mathcal{O}, d+\omega_p)$ has zero p-curvature iff $E(\mathbb{F}_p) = p$ (use Hasse-Weil bound).

If $\text{rank}(\mathcal{E}) > 1$, only invariant polynomials of $\text{curv}_p(\mathcal{E}, \nabla)$ descend to $X^{(1)}$.

$$\text{Tr}(\text{curv}_p(\mathcal{E}, \nabla)^n) \in \Gamma(\text{Sym}^n(\Omega^1_{X^{(1)}}))$$

For a rank n vector bundle $\mathcal{E}$, these define:

$$T^*X^{(1)} \supset \widetilde{X}^{(1)} \to X^{(1)}$$

for $\text{supp}(\mathcal{E}, \nabla) \subset \widetilde{X}^{(1)}$ as a module for $Z(\mathcal{D}_X)$. This is defined to be the p-support.

Recall that we can view $\mathcal{D}_X$ as a Azumaya algebra on $T^*X^{(1)}$ whose class in $\text{Br}(T^*X^{(1)})$ comes from Liouville form θ. Suppose $\widetilde{X} \subset T^*X$ is smooth, then we have a bijection:

$$(\mathcal{E}, \nabla) \text{ with spectral cover } \widetilde{X} \leftrightarrow \text{ line bundles } (\mathcal{L}, \nabla_{\mathcal{L}}) \text{ on } \widetilde{X} \text{ with } \text{curv}(\nabla_{\mathcal{L}}) = \theta|_{\widetilde{X}}$$

Let X be a complete curve of genus $g > 1$. Let K be the canonical line bundle. We have a unique nonsplit extension:

$$0 \to K^{\frac{1}{2}} \to \mathcal{E} \to K^{-\frac{1}{2}} \to 0$$

Let $(\mathcal{E}, \nabla)$ have trivial dterminant. If we write ∇_0 for a fixed connection, and $\nabla = \nabla_0 + A$. Here:

$$A = \begin{pmatrix} 0 & B \\ 0 & 0 \end{pmatrix} \in H^0(\text{End}(\mathcal{E}) \otimes K), \quad B \in H^0(K^2)$$

Now $\widetilde{X} \subset T^*X$ is given by a section of K^2.

Now let's go back to general discussion about differential operators. Recall we have the diagram:

$$\begin{array}{ccc} Y & & T^*X \times_X Y \xrightarrow{df} T^*Y \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle \pi} \\ X & & T^*X \end{array}$$

and $\mathcal{D}$-module pushforward f^* is an $\hbar$-deformation of $\pi_* \circ df^*$. In characteristic p, there is another way to relate f_* to this diagram.

Lemma 6.3. *We have a canonical isomorphism:*

$$\mathrm{can} : df^{(1)*}(\mathcal{D}_Y) \sim \pi^{(1)*}\mathcal{D}_X$$

This gives a functor:

$$f_* = \pi_*^{(1)} \circ \mathrm{can} \circ df^{(1)*}$$

For proper f, "spreading works well".

Corollary 6.4. *(N. Katz) p-curvature of Gauss-Manin connection is nilpotent.*

Assume X lifts to $W_2(k)$, then $\mathcal{D}_X$ as an Azumaya algebra on $T^*X^{(1)}$ splits canonically on p-th infinitesimal neightborhood of the zero section. So for $n < p$, we can define the Cartier transform:

$$C : (\mathcal{E}, \nabla)_{\mathrm{nilp}} \to \mathrm{Coh}(T^*X^{(1)})$$

More generally we have:

$$C : \mathcal{D}\text{-mod}_{\mathrm{nilp}} \to \mathrm{Coh}(T^*X^{(1)})$$

One can check that Cartier transform is compatible with can. This implies

$$C(\nabla, \mathcal{E}_i) = \mathsf{H}^i(\pi_* df^*(\mathcal{O}_{Y^{(1)}}))$$

coincides with the formula for $\mathrm{gr}_{\mathrm{Hodge}}(\mathcal{E}_i)$, where ∇ is the Gauss-Manin connection.

Theorem 6.5. *(N. Katz) Grothendieck conjecture holds for Gauss Manin connection $\mathcal{E}_i$ for smooth proper $Y \to X$.*

6.3 Simpson's Conjecture and Non-abelian Hodge Theory

Let $X/\mathbb{C}$ be a smooth projective space, we have $\mathcal{M}_{\mathrm{Betti}}$ the (coarse) moduli space of local systems of fixed dimension n. A *rigid* local system corresponds to isolated points of $\mathcal{M}_{\mathrm{Betti}}$.

Example. *Let X be a complete curve of genus $g > 1$. Let's fix a rank 1 local system L, and consider $\mathcal{M}^L_{\mathrm{Betti},n}$ the moduli space of rank n local systems with determinant $= L$. Isolated points of $\mathcal{M}^L_{\mathrm{Betti}}$ are also called rigid.*

Conjecture 6.6 (Simpson)**.** *Assume $L^{\otimes d}$ is trivial for some d. Then a rigid local system of* $\det L$ *has geometric origin, i.e. it's a direct summand in Gauss-Manin connection.*

Corollary 6.7. *Let $(\mathcal{E}, \nabla) \in \mathcal{M}_{\mathrm{DR}}$ be a rigid flat connection, and assume everything defined over a number field. Then for almost all prime p,* $\mathrm{char}(k) = p$*, we have $(\mathcal{E}_k, \nabla_k)$ has nilpotent p-curvature.*

Remark. *The corollary of the conjecture is actually a theorem.*

Theorem 6.8. *Let $(\mathcal{E}, \nabla)$ be rigid and* $\mathrm{curv}_p(\mathcal{E}_k, \nabla_k) = 0$ *for almost all p, then the monodromy is unitary.*

Let X be smooth projective and lifts to W_2. Consider the Hodge filtration:

$$0 \to H^{0,1} = H^0(\Omega^1) \to H^1_{\mathrm{DR}}(X) \to H^{1,0} = H^1(\mathcal{O}) \to 0$$

In characteristic p, we have the Hodge filtratoin coming from Cartier isomorphism:

$$0 \to H^1(\mathcal{O}_{X^{(1)}}) \to H^1_{\mathrm{DR}}(X) \to H^0(\Omega^1_{X^{(1)}}) \to 0$$

There is a multiplicative version:

$$0 \to H^0(\Omega^1_{\mathrm{cl}}) \to H^1_{\mathrm{DR}}(\mathcal{O}^\times) \to H^1_{\mathrm{Zar}}(\mathcal{O}^\times) \to 0$$

where the second map is given by $(\mathcal{L}, \nabla) \mapsto \mathcal{L}$. In characteristic p, we also have a dual filtration:

$$1 \to H^1(\mathcal{O}^\times_{X^{(1)}}) \xrightarrow{\mathrm{Fr}^*} H^1_{\mathrm{DR}}(\mathcal{O}^\times) \to H^0(\Omega^1_{X^{(1)}}) \to 0$$

where the second map is given by $(\mathcal{L}, \nabla) = \mathrm{curv}_p(\mathcal{L}, \nabla)$.

Remark. *A vector space with a filtration is nothing but a $\mathbb{G}_m$-equivariant vector bundle on $\mathbb{A}^1$. So the filtered vector space $H^1_{\mathrm{DR}}(X)$ gives a bunlde on $\mathbb{A}$. One can see that the fibre of the vector bundle at* 0 *is the associated graded of our original vector bundle, the fibre at* 1 *is the original vector bundle. Over $\mathbb{C}$, we have: $H^1_{\mathrm{DR}} \simeq H^1(X(\mathbb{C}), \mathbb{C})$ and there is a complex conjugation on $X(\mathbb{C})$. The vector bundle thus extends to $\mathbb{CP}^1$.*

Note that $H^1_{\mathrm{DR}}(\mathcal{O}^\times) = \mathcal{M}_{\mathrm{DR},n=1}$. Thus there exists $\widetilde{\mathcal{M}}$ a $\mathbb{G}_m$ family over $\mathbb{A}^1$ such that:

$$\widetilde{\mathcal{M}}_1 = \mathcal{M}_{\mathrm{DR}}, \quad \widetilde{\mathcal{M}}_0 = \{(\mathcal{E}, \varphi) \mid \varphi : \mathcal{E} \to \mathcal{E} \otimes \Omega^1, \varphi \wedge \varphi = 0\} \subset \mathrm{Coh}(T^*X)$$

Over $\mathbb{C}$ it extends to a family over $\mathbb{CP}^1$, then $\widetilde{\mathcal{M}}_\infty = \overline{\widetilde{\mathcal{M}}_0}$.

Remark. *There is an analytic isomorphism:*

$$\mathcal{M}_{\mathrm{DR}} \simeq \mathcal{M}_{\mathrm{Betti}}$$

but not algebraically. For example if $n = 1$, then we have:

$$0 \to \Omega^1 \to \mathcal{M}_{\mathrm{DR}} \to \mathrm{Pic}^0 \to 0$$

while $\mathcal{M}_{\mathrm{Betti}} \simeq (\mathbb{C}^\times)^{2g}$. A little exercise is to show:

$$H^0_{\mathrm{alg}}(\mathcal{M}_{\mathrm{DR}}, \mathcal{O}) = \mathbb{C}$$

But in characteristic p, we have a map $\mathcal{M}_{\mathrm{DR}} \to \Omega^1_{X^{(1)}}$.

The core hard construction in non-abelian Hodge theory is C^∞ trivialization of twistor family. Given a semisimple local system $(\mathcal{E}, \nabla)$, we need (hard analysis) to find a harmonic metric $(\mathcal{E}, \varphi)$.

Theorem 6.9 (Simpson)**.** $\mathbb{C}^*$*-invariant sections of* $\widetilde{M}$ *corresponds to complex variation of Hodge structures.*

Remark. *The Nonabelian counterpart for Hodge filtration is:*

$$\mathcal{M}_{\mathrm{DR},n} \to \mathrm{Bun}_n$$

via $(\mathcal{E}, \nabla) \mapsto \mathcal{E}$*. For* $\dim X = 1$*, this realizes* $\mathcal{M}_{\mathrm{DR}}$ *as a torsor over* $T^*\mathrm{Bun}_n$ *(in stack setting). The conjugate filtration is given by:*

$$\mathcal{M}_{\mathrm{DR}} \to H^0(\Omega^1_{X^{(1)}}) \oplus H^0(\mathrm{Sym}^2\Omega^1_{X^{(1)}}) \oplus \cdots \oplus H^0(\mathrm{Sym}^n\Omega^1_{X^{(1)}})$$

via taking invariant polynomials of p-curvature.There is a twisted version for any characteristic:

$$\mathcal{M}_{\mathrm{Higgs}} \to B = H^0(\mathrm{Sym}^n\Omega^1_{X^{(1)}})$$

This is the Hitchin integrable system.

Proof. (of Corollary 6.7, sketched). By Simpson theory, over $\mathbb{C}$ flat connections correspond to rigid Higgs fields. In particular, the two (finite) sets have same cardinality. Rigid Higgs fields are nilpotent, otherwise dilation gives a nontrivial determinant. Now check that for almost all p, $(\mathcal{E}, \nabla) \to (\mathcal{E}_k, \nabla_k)$ gives a bijection between rigid connections over $\mathbb{C}$ and over k, and the same for Higgs fields.

For a rigid Higgs field $(\mathcal{E}_k, \varphi_k)$ on X_k and $p \gg 0$, we can apply Cartier transform to get a rigid flat connection on X_k. By counting, we get all rigid connections on X_k this way. □

Proof. (of Theorem 6.8, sketched) Let $(\mathcal{E}, \nabla)$ be a rigid connection, it comes from a complex VHS by Simpson's theoorem. As in Katz's theorem, we need to check that $\mathrm{gr}(\mathcal{E}, \nabla)$ lives on the zero section (i.e. the Kodaira-Spencer class is zero). For this we prove that for infinitely many primes p such that for k of characteristic p, we have:

$$C^{-1}(\mathcal{E}_k, \nabla_k) \simeq \mathrm{gr}(\mathcal{E}_K atz)$$

For this we use Riemann-Hilbert correspondence in characteristic p. A key ingredient is Cassel's embedding theorem:

Theorem. *For* $F = \mathbb{Q}(\alpha_1, \cdots, \alpha_n) \subset \mathbb{C}$ *for infinitely many p, there exists* $F \to \mathbb{Z}_p$ *sending* α_i *to a p-adic unit.*

□

A Hodge Theory

In this appendix, we review the classical Hodge theory.

A.1 Hodge Decomposition Theorem

Let X be a compact n-dimensional Riemannian manifold with a Riemannian metric g. Note that g induces an inner product on TX, so that g induces an inner product on cotangent bundle, and more generally on the bundle on m-forms:

$$\mathcal{E}_X^m \coloneqq \Lambda^m T^*X$$

Choose $\{e_1, \cdots, e_n\}$ be a local orthonormal frame for the cotangent bundle, then $\{e_{i_1} \wedge \cdots e_{i_m} \mid i_1 < i_2 < \cdots\}$ is an orthonormal frame for $\mathcal{E}_X^m$.

Assume X is orientable, we choose the local frames compatible with the orientation, we have the volume form:

$$\mathrm{vol}_g = e_1 \wedge e_2 \cdots \wedge e_n$$

The **Hodge $*$-operators** is defined to be:

$$* : \Lambda^m T_x^*X \to \Lambda^{n-m} T_x^*X$$

such that:

$$\alpha \wedge *\beta = g(\alpha, \beta)[\mathrm{vol}_g]_x$$

The **De Rham spaces** are defined to be:

$$E_{\mathrm{DR}}^m(X) = \Gamma(X, \mathcal{E}_X^m)$$

this space thus carry a global inner product:

$$(\alpha, \beta) = \int_X g(\alpha, \beta)\mathrm{vol}_g = \int_X \alpha \wedge *\beta$$

The De Rham cohomology group are:

$$H_{\mathrm{DR}}^k(X) \coloneqq H^k(E_{\mathrm{DR}}^\bullet(X), d)$$

We also define

$$d^* = (-1)^{nm+1} * d *$$

this is adjoint to d in the sense:

$$(d\alpha, \beta) = (\alpha, d^*\beta), \quad \alpha \in E_{\mathrm{DR}}^{m-1}(X), \beta \in E_{\mathrm{DR}}^m(X)$$

We have an associated **Laplacian**:

$$\Delta_d = dd^* + d^*d$$

We thus can define the d-harmonic:

$$\mathrm{Har}^m(X) = \{\alpha \in \Gamma(X, \mathcal{E}_X^m) \mid \Delta_d \alpha = 0\}$$

Theorem A.1 (Hodge Isomorphism Theorem)**.** *Let X be a compact Riemannian manifold. Then:*

1. $\dim \mathrm{Har}^m(X) < \infty$

2. *Let*

$$H : E^m_{\mathrm{DR}}(X) \to \mathrm{Har}^m(X)$$

be the orthogonal projection. In fact we have an orthogonal factorization:

$$E^m_{\mathrm{DR}}(X) = \mathrm{Har}^m(X) \oplus dE^{m-1}_{\mathrm{DR}}(X) + d^* E^{m+1}_{\mathrm{DR}}(X)$$

and H induces an isomorphism:

$$H^m_{\mathrm{DR}}(X) \simeq \mathrm{Har}^m(X)$$

Given a Riemannian metric g, we have an associated Levi-cevita connection, which in turn defines parallel transport for use. Now that we defined the **holonomy group** to be

$$\mathrm{Hol}_x \subset O(T_x X)$$

generated by $P_\gamma : T_x X \to T_x X$ for every closed loop γ based at x. It's easy to see this is independent of x up to conjugation. We can thus state:

Theorem A.2 (Chern's Theorem)**.** *Let (X, g) be a compact connected Riemannian manifold of dimension n. Let $A \in \mathrm{End}(\Lambda^k T^* X)$ be an operator which fibrewise commutes with the holonomy actions. Then the action of A on $E^m_{\mathrm{DR}}(X)$ commutes with the Laplacian, and hence preserve the subspace of harmonic forms.*

Now let's assume X is a complex manifold with a Hermitian metric h. In local coordinate:

$$h = \sum_{j,k} h_{jk} dz_j \otimes d\overline{z}_k$$

Note that $\Re(h)$ defines a Riemannian metric for the underlying real vector space, while $\omega_h = \Im(h)$ is a real-valued skew-form. The almost complex structure J on TX corresponding to multiplication by i preserves ω_h, this means ω_h is a $(1,1)$-form. So in local coordinate, we can write:

$$\omega_h = \frac{i}{2} \sum_{j,k} h_{jk} dz_j \wedge d\overline{z}_k$$

this defines a volume form:

$$\mathrm{vol} = \frac{\omega_h^n}{n!}$$

Now let $\mathcal{E}^{p,q}_X$ be the space complex-valued (p,q)-forms. The differential

$$d : \mathcal{E}^m_X(\mathbb{C}) \to \mathcal{E}^{m+1}_X(\mathbb{C})$$

splits as $d = \partial + \overline{\partial}$, with:

$$\partial : \mathcal{E}^{p,q}_X \to \mathcal{E}^{p+1,q}_X, \quad \overline{\partial} : \mathcal{E}^{p,q}_X \to \mathcal{E}^{p,q+1}_X$$

Now h gives a volume form, which then defines a **global inner product** on the space of (p,q)-forms, as well as the space complex valued smooth forms. With respect these metrics, we have an orthogonal splitting:

$$\Gamma(\mathcal{E}_X^m(\mathbb{C})) = \bigoplus_{p+q=m} \Gamma(\mathcal{E}_X^{p,q})$$

The volume form gives us a $\mathbb{C}$-linear Hodge star operator:

$$* : \mathcal{E}_X^{p,q} \to \mathcal{E}_X^{n-q,n-p}$$

satisfying

$$\alpha \wedge \overline{*\beta} = h(\alpha, \beta)\text{vol}$$

So the global inner product is again given by:

$$(\alpha, \beta) = \int_X \alpha \wedge \overline{*\beta}$$

More generally, we may consider forms with coefficients in a holomorphic vector bundel E equipped with a Hermitian metric h_E. Let's consider the bundle of differentiable E-forms of type (p,q):

$$\mathcal{E}_X^{p,q}(E) = \mathcal{E}_X^{p,q} \otimes E$$

we have well-defined opertor:

$$\overline{\partial}_E : \mathcal{E}_X^{p,q}(E) \to \mathcal{E}_X^{p,q+1}(E), \quad \alpha \otimes s \mapsto \overline{\partial}\alpha \otimes s$$

Note the naive definition for ∂_E will not work. Moreover, one may check $\overline{\partial}_E^2 = 0$.

Now that the Hermitian metric h_E induces a conjugate linear isomorphism:

$$\tau : E \to E^\vee$$

we can thus define:

$$\overline{*}_E : \mathcal{E}_X^{p,q}(E) \to \mathcal{E}_X^{n-p,n-q}(E^\vee), \quad \alpha \otimes e \mapsto \overline{*\alpha} \otimes \tau(e)$$

Thus we can define the global Hodge inner product on $\Gamma(\mathcal{E}_X^{p,q}(E))$ by:

$$(\alpha, \beta) = \int_X \alpha \wedge \overline{*}_E \beta$$

Now view $\overline{\partial}_E$ as a map on the level of global section, it has a formal adjoint $\overline{\partial}_E^*$ and the Laplacian:

$$\Delta_{\overline{\partial}_E} = \overline{\partial}_E \overline{\partial}_E^* + \overline{\partial}_E^* \overline{\partial}_E$$

and one has the associated Harmonic forms $\text{Har}^{p,q}(E)$.

Let

Theorem A.3 (Hodge Isomorphism Theorem, Second Version)**.** *Let X be a compact complex manifold with holomorphic vector bundle E . Let T_X and E both equip with a Hermitian metric. We have:*

1. $\dim \mathrm{Har}^{p,q}(E) < \infty$
2. *Let*
$$H : \Gamma(X, \mathcal{E}_X^{p,q}(E)) \to \mathrm{Har}^{p,q}(E)$$
be the orthogonal projection onto harmonic forms. We have a orthogonal decomposition:
$$\Gamma(\mathcal{E}_X^{p,q}((E))) = \mathrm{Har}^{p,q}(E) \oplus \overline{\partial}_E \Gamma(\mathcal{E}_X^{p,q-1}(E)) \oplus \overline{\partial}_E^* \Gamma(\mathcal{E}_X^{p,q+1}(E))$$
Hence H induces an isomorphism:
$$H_{\overline{\partial}}^{p,q}(E) \simeq \mathrm{Har}^{p,q}(E)$$

Remark. *One see the operator $\overline{*}_E : \mathrm{Har}^{p,q}(E) \simeq \mathrm{Har}^{n-p,n-q}(E^\vee)$ is a conjugate-linear isomorphism. Note $\mathrm{Har}^{n-p,n-q}(E^\vee)$ has the global Hodge inner product. So we have $\mathbb{C}$-linear isomorphism:*
$$\mathrm{Har}^{p,q}(E) \simeq \mathrm{Har}^{n-p,n-q}(E^\vee)^\vee$$
It follows that:
$$H_{\overline{\partial}}^{p,q}(E) \simeq H_{\overline{\partial}}^{n-p,n-q}(E^\vee)^\vee$$
is linearly isomorphic. A theorem of Dolbeault says:
$$H_{\overline{\partial}}^{p,q}(E) \simeq H^q(X, \Omega_X^p(E))$$
We thus obtain Serre duality:
$$H^q(X, \Omega_X^p(E)) \simeq H^{n-q}(X, \Omega_X^{n-p}(E^\vee))^\vee$$

Definition A.4. A Hermitian metric h on a complex manifold is called **Kähler** is the $(1,1)$-form ω_h is closed. In this case, this form is called the Kähler form.

Example. *We can put Fubini-study metric on $\mathbb{P}^n$:*
$$\omega = \frac{i}{2\pi} \partial\overline{\partial} \log \|Z\|^2$$
where $Z = (Z_0, Z_1, \cdots, Z_n)$ as coordinates on $\mathbb{P}^n$. This makes $\mathbb{P}^n$ a Kähler manifold. Note every submanifold of a Kähler manifold is Kähler, so every projective manifold is Kähler.

Theorem A.5 (Hodge Decomposition Theorem)**.** *Let X be a compact Kähler manifold. Let $H^{p,q}$ be the space of cohomology classes whose harmonic representative is of type (p,q). There is a direct sum decomposition:*
$$H_{\mathrm{DR}}^m(X) \otimes \mathbb{C} = \bigoplus_{p+q=m} H^{p,q}(X)$$
Moreover, $H^{p,q}(X) \simeq \overline{H^{q,p}(X)}$

Let's discuss a little bit on the ingredients of the proof. For a complex manifold X, look at the real tangent space T_xX at a point x. It carries a natural complex structure $J \in \mathrm{End}(T_xX)$. If in addition X is Hermitian, it can be shown that the metric is Kähler if and only if $\nabla J = 0$. This compatibility of the metric with the complex structure has nice consequences for the action of the holonomy group.

We can decompose the complexified cotangent space into $\pm i$-eigenspaces of J:

$$T_x^*X \otimes \mathbb{C} \simeq T_x^{*(1,0)}X \oplus T_x^{*(0,1)}X$$

One sees both components are preserved by holonomy. We now let

$$\Lambda^{p,q}(T_x^*X) = \Lambda^p(T_x^{*(1,0)}X) \otimes \Lambda^q(T_x^{*(0,1)X})$$

be the space of (p,q) covectors. The operator J acts by i^{p-q} on this space, and they're again preserved by holonomy. Moreover, in this case, the holonomy will act unitarily: $\mathrm{Hol}_x \subset U(T_xX)$. Moreover, we have:

Lemma A.6. *The group $U(n)$ acts on each $\Lambda^{p,q}(T_x^*X)$. The operator "multiplication by Kähler form" commutes with this action.*

Proof. (of Hodge Decomposition Theorem). In light of Hodge isomorphism theorems, one must check:

$$\mathrm{Har}^m(X) \otimes \mathbb{C} \simeq \bigoplus_{p+q=m} \mathrm{Har}^{p,q}(X)$$

It suffices to check the projection map:

$$\Lambda^{p+q}(T_x^*X \otimes \mathbb{C}) \to \Lambda^{p,q}(T_x^*X)$$

preserves harmonic forms. But by Chern's theorem, we only need to see this commutes with holnomy actions, which we discussed before. □

A striking fact is that the Hodge isomorphism doesn't depend on the Kähler metric we choose. This can be seen by comparing with **Bott-Chern** cohomology groups:

$$H_{\mathrm{BC}}^{p,q} \coloneqq \frac{d\text{-closed forms of type } (p,q)}{\partial\overline{\partial}\Gamma(\mathcal{E}_X^{p-1,q-1})}$$

Proposition A.7. *The natural morphism:*

$$H_{\mathrm{BC}}^{p,q} \to H_{\mathrm{DR}}^{p+q}(X) \otimes \mathbb{C}$$

is injective with image $H^{p,q}(X)$. Therefore, $H^{p,q}$ is independent of the Kähler metric.

A.2 Hodge Structures

Definition A.8. A **Hodge structure** of weight k on an abelian group V of finite rank is a direct sum decomposition:

$$V_{\mathbb{C}} \coloneqq V \otimes \mathbb{C} = \bigoplus_{p+q=k} V^{p,q}$$

with $V^{p,q} = \overline{V^{q,p}}$. The numbers $h^{p,q}(V) = \dim V^{p,q}$ are the **Hodge numbers** of the Hodge structure. It is clear what should be the right definition of a moprhism of Hodge structures.

We tautologically have:

Theorem A.9. *Let X be a compact Kähler manifold. The integral homology group $H^k(X)$ carries a weight k Hodge structure. If $f : X \to Y$ is holomorphic map between compact Kähler manifolds, then f^* is a morphism of Hodge structures.*

A basic example is the Hodge structure on the Tate $\mathbb{Z}(m)$:

$$\mathbb{Z}(m) = (2\pi i)^m \mathbb{Z} \subset \mathbb{C}, \mathbb{Z}(m) \otimes \mathbb{C} = [\mathbb{Z}(m) \otimes \mathbb{C}]^{-m,-m}$$

For each Hodge structure $V_{\mathbb{Z}}$ of weight k, we have the **Tate twist** $V(m)$ of weights $k - 2m$.

Now let X be a compact complex manifold of dimension n. Denote

$$Z^c(X) = Z_{n-c}(X)$$

be the group of algebraic cycles of codimension $n - c$. We wish to define a **cycle class map** to be the composition:

$$\mathrm{cl} : Z^c(X) \to H_{2n-2c}(X) \xrightarrow{\mathrm{PD}} H^{2c}(X)$$

An element in the image is called an **algebraic class**.

If we further embed $H^{2c}(X)$ into $H^{2c}_{\mathrm{DR}}(X)$, one get an alternative construction of this ma as following. Given a subvariety Y of codimension c. We have a map:

$$L_Y : \{\text{smooth closed } (2n-2k) \text{ forms}\} \to \mathbb{R}, \quad \eta \mapsto \int_Y \eta$$

Note this vanishes on exact forms since:

$$\int_Y d\lambda = \int_{\partial Y} \lambda = 0$$

So L_Y descends to:

$$L_Y : H^{(2n-2k)}_{\mathrm{DR}}(X) \to \mathbb{R}$$

By Poincare duality we know there must be a cohomology class $[\omega]$ such that:

$$\int_Y \eta = L_Y(\eta) = \int_X \omega \wedge \eta$$

We then put: $\mathrm{cl}(Y) = [\omega]$ and extend linearly.

We call a class is **pure** of type (p, q) if it has some representative pure with type (p, q). One sees that $\mathrm{cl}(Y)$ is pure of type (c, c). In particular, the twisted trace map is an isomorphism:

$$\mathrm{tr}_X : H^{2n}(X) \to \mathbb{Z}(-n), \quad [\alpha] \mapsto [\frac{1}{2\pi i}]^n \int_X \alpha$$

A theorem of Lefschetz shows that: for X projective and $c = 1$, the cycle class map surjects onto the integral $(1, 1)$-classes. But this is false in general.

We're now ready to state:

Conjecture A.10 (Hodge Conjecture)**.** *Let X be a smooth projective variety. Every (c,c)-class with rational coefficents is algebraic. Namely, the cycle class map:*

$$\mathrm{cl} : Z^c(X) \to H^{c,c}(X) \cap H^{2c}(X;\mathbb{Q})$$

is surjective.

Remark. *Hodge conjecture is false for compact Kähler manifold.*

Use the isomorphism

$$H_k(X,\mathbb{C}) \simeq \mathrm{Hom}(H^k(X),\mathbb{C})$$

we can define a Hodge decomposition on homology:

$$H_k(X,\mathbb{C})^{-p,-q} = \{\varphi : H^k(X,\mathbb{C}) \to \mathbb{C} \mid \varphi(H^{r,s}(X)) = 0, \text{whenever } (r,s) \neq (p,q)\}$$

So $H_k(X)$ has a Hodge structure of weight $-k$. There are a few nice consequences.

Lemma A.11. *Let X be a compact Kähler manifold. There is a natural Hodge structure of weight $k+\ell$ on $H^k(X) \otimes H^\ell(X)$, and a natural Hodge structure of weight $k-\ell$ on $H^k(X) \otimes H_\ell(X)$. We have:*

1. *The cup product map*
$$H^k(X) \otimes H^\ell(X) \to H^{k+\ell}(X)$$
is a morphism of Hodge structures.

2. *The cap product map*
$$H^k(X) \otimes H_\ell(X) \to H^{k-\ell}(X)$$
is a morphism of Hodge structures.

3. *The Poincaré dualtiy:*
$$H^{2n-k}(X)(n) \to H_k(X)$$
is an isomoprhism of Hodge structures.

A.3 Lefschetz Decomposition Theorem

Let X be an n-dimensional complex manifold with Hermitian metric h, with Kähler form ω_h. Let's consider operators:

$$F : \alpha \mapsto \omega_h \wedge \alpha, \quad E = F^* = \overline{*}^{-1} F \overline{*}, \quad H = [E,F]$$

on $\Lambda^\bullet(T_x^*X)$. One can see that this is a $\mathfrak{sl}_2(\mathbb{R})$-representation, with $\Lambda^m(T_x^*X)$ the weight space with weight $n-m$.

From $\mathfrak{sl}_2$-theory, we have "local" isomorphisms for $k \leq n, p+q \leq n$:

$$F^{n-k} : \Lambda^k(T_x^*X \otimes \mathbb{C}) \simeq \Lambda^{2n-k}(T_x^*X \otimes \mathbb{C})$$

and hence:

$$F^{n-p-q} : \Lambda^{p,q}(T_x^*X) \to \Lambda^{n-q,n-p}(T_x^*X)$$

Globally, we have an infinite-dimensional $\mathfrak{sl}_2(\mathbb{R})$-representation $E_{\mathrm{DR}}X$. But note this representation commutes with the holnomy action, Chern's theorem tells us this restricts to a finite dimensional representation on harmonic forms. We call a covector **primitive** if it is killed by E (i.e. a highest weight vector). We call a form primitive if it is fibrewise primitive. Hence from $\mathfrak{sl}_2$-theory, we again have a decomposition for harmonic forms:

$$\mathrm{Har}^{p,q}(X) = \bigoplus_{r \geq \max(p+q-n,0)} F^r(\mathrm{Har}^{p-r,q-r})_{\mathrm{prim}}(X)$$

The first Hodge isomorphism allows to transport this representation to cohomology. We immediately have:

Theorem A.12 (Hard Lefschetz Theorem)**.** *For any Kähler manifold (X, ω), let F be cup product with the Kähler class $[\omega]$, we have isomorphisms:*

$$F^{n-k} : H^k(X; \mathbb{C}) \to H^{2n-k}(X; \mathbb{C}), \quad k \leq n$$

$$F^{n-p-q} : H^{p,q}(X) \to H^{n-q,n-p}(X), \quad p + q \leq n$$

To state Lefschetz decomposition theorem, we need to identify the image of $(\mathrm{Har}^{p,q})_{\mathrm{prim}}$ under the isomorphism $\mathrm{Har}^{p,q}(X) \simeq H^{p,q}(X)$. Explicitly, the image is the so called "primitve cohomology":

$$\begin{aligned} H^{p,q}_{\mathrm{prim}}(X) &= \ker(E : H^{p,q}(X) \to H^{p-1,q-1}(X)) \\ &= \ker(F^{n-p-q+1} : H^{p,q}(X) \to H^{n-q+1,n-p+1}(X)) \end{aligned}$$

and of course:

$$H^m_{\mathrm{prim}}(X; \mathbb{C}) = \bigoplus_{p+q=m} H^{p,q}_{\mathrm{prim}}(X)$$

So we have:

Theorem A.13 (Lefschetz Decomposition)**.** *We have:*

$$H^m(X; \mathbb{C}) = \bigoplus_{r \geq \max(m-n,0)} F^r H^{m-2r}_{\mathrm{prim}}(X; \mathbb{C})$$

$$H^{p,q}(X) = \bigoplus_{r \geq \max(p+q-n,0)} F^r H^{p-r,q-r}_{\mathrm{prim}}(X)$$

The final major ingredients in classical Hodge theory are the **Hodge-Riemann bilinear relations**. We define the **Weil-operator** C on Dolbeault cohomology to be the operator coming from the complex structure J-action on $\Lambda^{p,q}(T^*_x X)$. Concretely, this is saying:

$$C|_{H^{p,q}(X)} = i^{p-q}$$

Theorem A.14 (Hodge Riemann Bilinear Relations)**.** *For $k \in \mathbb{Z}$, put $\varepsilon(k) = (-1)^{\frac{1}{2}k(k-1)}$. Let (X, ω) be a Kähler manifold of dimension n. The* **Hodge-Riemann form** *is:*

$$Q([\alpha], [\beta]) = \varepsilon(k) \int_X \alpha \wedge \beta \wedge \omega^{n-k}, \quad [\alpha], [\beta] \in H^k_{\mathrm{DR}}(X; \mathbb{C})$$

This is $(-1)^k$-symmetric. We have:

1. *If* $(r,s) \neq (q,p)$*, we have:*
$$Q(H^{p,q}, H^{r,s}) = 0$$

2. *For* $u \in H^{p,q}_{\text{prim}}$*, we have:*
$$Q(Cu, \overline{u}) = (u,u)$$
where RHS is the global Hodge inner product.

B Homework 1

Exercise B.1. *Let F be a field of characteristic 0. Consider an affine elliptic curve (we choose a, b so that it is smooth):*

$$X = \text{Spec } F[x,y]/(y^2 - (x^3 + ax + b)) \subset \mathbb{A}^2_F$$

Compute $H^1_{\text{dR}}(X)$ by presenting a basis.

Proof. First we compute:

$$d(y^2) = d(x^3 + ax + b) \implies 2ydy = (3x^2 + a)dx$$

Therefore, since $\dim X = 1$ so that all 1-forms are closed:

$$H^1_{\text{dR}}(X) = \{fdx + gdy \mid f, g \in \mathcal{O}(X)\}/\{\mathcal{O}(X) \cdot (2ydy - (3x^2 + a)dx), \text{exact forms}\}$$

as vector spaces. Note that as vector spaces, $\mathcal{O}(X)$ has a basis:

$$x^i, x^i y \quad i \geq 0$$

It follows that exact forms have a basis:

$$d(x^j) = jx^{j-1}dx \text{ for } j \geq 1, dy, d(x^j y) = x^j dy + jx^{j-1}ydx \text{ for } j \geq 1$$

So we compute:

$$\begin{aligned}
H^1_{\text{dR}}(X) \simeq & F\{x^i dx, x^i dy, x^i ydx, x^i ydy\}/ \\
& F\{x^i(2ydy - (3x^2 + a)dx), x^i y(2ydy - (3x^2 + a)dx), x^{j-1}dx, x^j dy + jx^{j-1}ydx, dy\} \\
\simeq & F\{x^i dy, x^i ydx, x^i ydy\}/F\{x^i ydy, x^i y^2 dy, x^j dy + jx^{j-1}ydx, dy\} \\
\simeq & F\{x^i dy, x^i ydx\}/F\{x^i(x^3 + ax + b)dy, x^j dy + jx^{j-1}ydx, dy\} \\
\simeq & F\{x^i dy\}/F\{(x^{3+i} + ax^{i+1} + bx^i)dy, dy\}
\end{aligned}$$

Therefore evidently it has a basis $xdy, x^2 dy$. We can identify

$$H^1_{\text{dR}}(X) \simeq F\{xdy, x^2 dy\}$$

□

Exercise B.2. *Compute $H^*_{\text{dR}}(\mathbb{P}^1_F)$, where F is a field of any characteristic.*

Proof. The spectral sequence starts at the E_0 page:

$$\begin{array}{ccc}
\mathcal{O}(U_1 \cap U_2) = F[t, t^{-1}] & \xrightarrow{\quad d \quad} & \Omega^1(U_1 \cap U_2) = F[t, t^{-1}]dt \\
\uparrow & & \uparrow \\
\mathcal{O}(U_1) \oplus \mathcal{O}(U_2) = F[t] \oplus F[t^{-1}] & \xrightarrow{\quad d \quad} & \Omega^1(U_1) \oplus \Omega^1(U_2) = F[t]dt \oplus F[t^{-1}]d(t^{-1})
\end{array}$$

The differentials on the E_0 page is just de Rham differential. In characteristic 0, it gives E_1 page:

$$\begin{array}{ccc} F & \xrightarrow{0} & F \cdot t^{-1}dt \\ \uparrow & & \uparrow \\ F \oplus F & \xrightarrow{0} & 0 \end{array}$$

The differntials d_1 comes from the Cech differentials (vertical maps), so that it gives E_2-page:

$$\begin{array}{cc} 0 & F \cdot t^{-1}dt \\ \\ F & 0 \end{array}$$

Note there will be no further differentials. Now if F has characteristic p, then the E_1-page will become:

$$\begin{array}{ccc} F[t^p, t^{-p}] & \xrightarrow{0} & \bigoplus_{i\in\mathbb{Z}} F \cdot t^{pi-1}dt \\ \uparrow & & \uparrow \\ F[t^p] \oplus F[t^{-p}] & \xrightarrow{0} & (\bigoplus_{i\geq 1} F \cdot t^{pi-1}dt) \oplus (\bigoplus_{i\geq 1} F \cdot t^{-pi-1}dt) \end{array}$$

Now again the differentials d_1 are the Cech differentials. One sees tht on the E_2-page we still get:

$$\begin{array}{cc} 0 & F \cdot t^{-1}dt \\ \\ F & 0 \end{array}$$

and there will be no further differentials. Therefore in any characteristic, we always have:

$$H^*_{\mathrm{dR}}(\mathbb{P}^1_F) = \begin{cases} F, & \text{if } * = 0; \\ F \cdot t^{-1}dt, & \text{if } * = 2; \\ 0, & \text{otherwise.} \end{cases}$$

□

Exercise B.3. *Let $X = \mathbb{P}^1_k$ be the projective line defined over a field k of characteristic p. Let $F : X \to X^{(1)}$ be the geometric Frobenius. Describe*

$$F_*\mathcal{O}_X, F_*\Omega^1_X$$

on $X^{(1)} \simeq \mathbb{P}^1_k$ as direct sum of line bundles. Compute:

$$d : F_*\mathcal{O}_X \to F_*\Omega^1_X$$

in terms of this description.

Proof. Let $U_0 = \text{Spec } k[t], U_1 = \text{Spec } k[s]$ be the affine charts for $\mathbb{P}^1$ glued via $t \mapsto s^{-1}$. We know: $F_*\mathcal{O}_X(U_0)$ corresponds to the $k[t^p]$-module $k[t]$, $F_*\mathcal{O}_X(U_1)$ corresponds to the $k[s^p]$-module $k[s]$. We have decomposition of $k[t^p]$-modules:

$$k[t] = k[t^p] \oplus tk[t^p] \oplus \cdots \oplus t^{p-1}k[t^p]$$

Similarly we have:

$$k[s] = k[s^p] \oplus sk[s^p] \oplus \cdots \oplus s^{p-1}k[s^p]$$

Note that $t^i k[t^p]$ localizes to $t^i k[t^p, t^{-p}]$; $s^j k[s^p]$ localizes to $s^j k[s^p, s^{-p}]$. They will glue iff $i + j = 0, p$. So there are two types of gluing:

1. $i = j = 0$. This will glue to $\mathcal{O}_X$.
2. $i + j = p$. The gluing map is:

$$\begin{array}{ccc} k[t^p, t^{-p}] & \xrightarrow{\quad t^p \quad} & k[s, s^{-1}] = k[t, t^{-1}] \\ & {\scriptstyle t^i} \searrow \quad \nearrow {\scriptstyle s^{-j}=t^j} & \\ & t^i k[t^p, t^{-p}] = s^j k[s^p, s^{-p}] & \end{array}$$

This corresponds to line bundle $\mathcal{O}(-1)$. It follows that:

$$F_*\mathcal{O}_X \simeq \mathcal{O}_X \oplus \mathcal{O}_X(-1)^{\oplus p-1}$$

Similarly, $F_*\Omega^1(U_0)$ corresponds to $k[t^p]$-module $k[t]dt$, $F_*\Omega^1(U_1)$ corresponds to $k[s^p]$-module $k[s]ds$. Note that:

$$s^j k[s^p, s^{-p}]ds = t^{-j}k[t^p, t^{-p}]t^{-2}dt = t^{-j-2}k[t^p, t^{-p}]$$

It follows that we will glue $t^i k[t]dt$ with $s^j k[s]ds$ for $i + j + 2 = p, 2p$. So there are two types of gluing:

1. $i = j = p - 1$. The gluing is:

$$\begin{array}{ccc} k[t^p, t^{-p}] & \xrightarrow{\quad t^{2p} \quad} & k[s, s^{-1}] = k[t, t^{-1}] \\ & {\scriptstyle t^{p-1}dt} \searrow \quad \swarrow {\scriptstyle s^{p-1}ds} & \\ & t^{p-1}k[t^p, t^{-p}]dt = s^{p-1}k[s^p, s^{-p}]ds & \end{array}$$

This gives me a copy of $\mathcal{O}(-2)$.

2. $i+j=p-2$. The gluing is:

$$\begin{array}{ccc} k[t^p,t^{-p}] & \xrightarrow{t^{i+j+2}=t^p} & k[s,s^{-1}]=k[t,t^{-1}] \\ & {\scriptstyle t^i dt}\searrow \quad \swarrow{\scriptstyle s^j ds} & \\ & t^i k[t^p,t^{-p}]dt = s^j k[s^p,s^{-p}]ds & \end{array}$$

This gives me a copy of $\mathcal{O}(-1)$.

Therefore:

$$F_*\Omega^1 = (\mathcal{O}(-1)^{\oplus p-1}) \oplus \mathcal{O}(-2)$$

The above descriptions respect the increasing order of i. To describe F_*d, we only describe:

$$F_*d : F_*\mathcal{O}_X(U_0) \to F_*\Omega^1_X(U_0)$$

It is induced via the diagram:

$$\begin{array}{ccc} k[t^p] & \xrightarrow{i} & k[t^p] \\ \downarrow{\scriptstyle t^i} & & \downarrow{\scriptstyle t^{i-1}dt} \\ t^i k[t^p] & \xrightarrow{d} & t^{i-1}k[t^p]dt \end{array}$$

In other word, the matrix is given by

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & 0 & p-1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

□

Exercise B.4. *Calculate $M \coloneqq H^1_{\mathrm{dR}}(\widehat{\mathbb{A}^1}_{W(k)})$. Is this module p-adically complete?*

Proof. It is easy to see:

$$W\langle t\rangle = \{a_0 + a_1 t + a_2 t^2 + \cdots \mid a_i \in W, v_p(a_i) \to \infty\} \subset W[[t]]$$

Now the two step complex:

$$W\langle t\rangle \xrightarrow{d} W\langle t\rangle dt$$

computes the de Rham cohomology of the formal affine line. One computes that:

$$\text{im } (d) = \{(a_0 + a_1 t + a_2 t^2 + \cdots)dt \mid a_i \in W, v_p(a_i) - v_p(i+1) \to \infty\}$$

Therefore explicitly, we have:

$$M \simeq \{(a_0, a_1, \cdots) \in W^{\mathbb{N}} \mid v_p(a_i) \to \infty\} / \{(a_0, a_1, \cdots) \in W^{\mathbb{N}} \mid 0 \leq v_p(a_i) - v_p(i+1) \to \infty\}$$

Now if we write $M = N_1/N_2$, then for each m:

$$M/p^m = N_1/(N_2 + p^m N_1)$$

Note each element in M/p^m can be represented by finitely many nonzero terms. Since when $v_p(a_i) > k$, we can adjust a_i by some multiple of p^k to make it has arbitrarily large valuation. Therefore:

$$M/p^m = \bigcup_{j=1}^{\infty} \bigoplus_{i=0}^{j} \mathbb{Z}/p^{\min(m, v_p(i+1))} = \bigoplus_{i=0}^{\infty} \mathbb{Z}/p^{\min(m, v_p(i+1))}$$

It follows that M has p-adic completion:

$$\hat{M} = \prod_{i=0}^{\infty} \mathbb{Z}/p^{v_p(i+1)}$$

But then $M \to \hat{M}$ is not injective, since for example $1 + pt^{p-1} + p^2 t^{p^2-1} + \cdots$ is a non-zero element in M but is sent to zero in $\hat{M}$. $\square$